

Continuous-Time Fourier Transform

5.1 INTRODUCTION

We saw in Chapter 3 that the Fourier series is a powerful tool in treating various problems involving periodic signals. We were also able to decompose any periodic signal with period T in terms of infinitely many harmonically related complex exponentials of the form $e^{jn\omega_0 t}$. All such harmonics have the common period $T = 2\pi/\omega_0$.

In this chapter, we consider another powerful mathematical technique, called the Fourier transform, for describing both periodic and nonperiodic signals for which no Fourier series exists. Like the Fourier series coefficients, the Fourier transform specifies the spectral content of a signal, thus providing a frequency domain description of the signal. Besides being useful in analytically representing aperiodic signals, the Fourier transform is a valuable tool in the analysis of LTI systems.

An aperiodic signal can be viewed as a periodic signal with an infinite period. More precisely, in the Fourier series representation of a periodic signal, as the period increases the fundamental frequency decreases and the harmonically related components become closer in frequency. As the period becomes infinite, the frequency components form a continuum and the Fourier series sum becomes an integral. The resulting spectrum of coefficients in this representation is called the Fourier transform, and the synthesis integral itself, which uses these coefficients to represent the signal as a linear combination of complex exponentials, is called the inverse Fourier transform.

5.2 FOURIER TRANSFORM REPRESENTATION OF APERIODIC SIGNALS

The generalization of the Fourier series to aperiodic signals was suggested by Fourier himself and can be deduced from an examination of the structure of the Fourier series for periodic signals as the period $T \rightarrow \infty$. After taking the limit, we will find that the magnitude spectrum of an aperiodic signal is not a line spectrum (as with a periodic signal) but instead occupies a continuum of frequencies. The same is true of the corresponding phase spectrum.

To clarify how the change from discrete to continuous spectra takes place, consider a periodic signal $\tilde{x}(t)$ shown in Fig. 5.1. Now think of keeping the waveform of one period of $\tilde{x}(t)$ unchanged, but carefully and intentionally increase T . In the limit as $T \rightarrow \infty$, or a single pulse remains because the nearest neighbours have been moved to infinity.

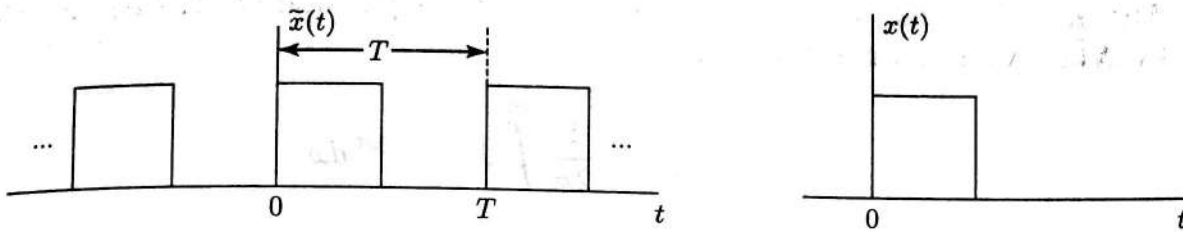


Fig. 5.1 Allowing the period T of $\tilde{x}(t)$ to increase to obtain the aperiodic signal $x(t)$.

saw in Chapter 3 (Section 3.4.5, time-scaling property) that as the period T increases the fundamental frequency $\omega_0 = 2\pi/T$ decreases and the harmonically related components become closer in frequency. As $T \rightarrow \infty$, the spacing between lines approaches zero. This means that the spectral lines move closer, eventually becoming a continuum. The overall shapes of the magnitude and phase spectra are determined by the shape of the single pulse that remains in the new signal $x(t)$, which is aperiodic.

To investigate what happens mathematically, we use the exponential form of the Fourier series representation for $\tilde{x}(t)$, i.e.,

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} \quad (5.1)$$

where

$$X_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt \quad (5.2)$$

In the limit as $T \rightarrow \infty$, we see that $\omega_0 = 2\pi/T$ becomes an infinitesimally small quantity, $d\omega$, so that

$$\frac{1}{T} \rightarrow \frac{d\omega}{2\pi}$$

We argue that in the limit, $n\omega_0$ should be a continuous variable. Then, from Eq. (5.2), the Fourier coefficients per unit frequency interval are

$$\frac{X_n}{d\omega} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{x}(t) e^{-j\omega t} dt \quad (5.3)$$

Substituting Eq. (5.3) into Eq. (5.1), and recognizing that in the limit the sum becomes an integral and $\tilde{x}(t)$ approaches $x(t)$, we obtain

$$x(t) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right) e^{j\omega t} \frac{d\omega}{2\pi} \quad (5.4)$$

The inner integral, in brackets, is a function of only ω and not t . Denoting the integral by $X(\omega)$, we can write Eq. (5.4) as

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (5.5)$$

where

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (5.6)$$

Equations (5.5) and (5.6) are referred to as the *Fourier transform pair*, with the function $X(\omega)$ referred to as the *Fourier transform* or Fourier integral of $x(t)$ and Eq. (5.5) as the *inverse Fourier transform* equation. $X(\omega)$ plays the same role for aperiodic signals that X_n plays for periodic signals. Thus, $X(\omega)$ is the spectrum of $x(t)$ and is a continuous function defined for all values of ω , whereas X_n is defined only for discrete frequencies. Therefore, an aperiodic signal has a continuous spectrum rather than a line spectrum.

We call $X(\omega)$ the Fourier transform of $x(t)$, and $x(t)$ the inverse Fourier transform of $X(\omega)$. The same information is conveyed by the statement that $x(t)$ and $X(\omega)$ are a Fourier transform pair. Symbolically, this statement is expressed as

$$X(\omega) = \mathcal{F}[x(t)] \quad \text{and} \quad x(t) = \mathcal{F}^{-1}[X(\omega)]$$

or

$$x(t) \longleftrightarrow X(\omega)$$

$X(\omega)$, in general, is a complex function of the variable ω . Thus, it can be written as

$$\begin{aligned} X(\omega) &= X_R(\omega) + jX_I(\omega) \\ X(\omega) &= |X(\omega)| e^{j\angle X(\omega)} \end{aligned} \quad (5.7)$$

where $|X(\omega)| = \sqrt{X_R^2(\omega) + X_I^2(\omega)}$ is the magnitude and $\angle X(\omega) = \tan^{-1} \left[\frac{X_I(\omega)}{X_R(\omega)} \right]$ is the angle (or phase) of $X(\omega)$.

The $|X(\omega)|$ plotted against ω is called the *magnitude spectrum* of $x(t)$, and the $\angle X(\omega)$ plotted against ω is called the *phase spectrum*. For real $x(t)$, the magnitude spectrum $|X(\omega)|$ is an even function, and the phase spectrum $\angle X(\omega)$ is an odd function of ω , i.e.,

$$|X(-\omega)| = |X(\omega)| \quad (5.8)$$

and

$$\angle X(-\omega) = -\angle X(\omega) \quad (5.9)$$

These results were derived earlier for the line spectrum of periodic signal (Section 3.2.8).

It is important at this point to comment on conventional notation. In communication system literature, Fourier optics, and image processing, the functional form $X(f)$ is

usually used for the transform of $x(t)$. In control system literature, one can find the use of both $X(\omega)$ and $X(j\omega)$. The Fourier transform in terms of cyclic frequency f is written as

$$X(f) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt \quad (5.10)$$

The inverse Fourier transform in terms of cyclic frequency f is written as

$$x(t) = \mathcal{F}^{-1}[X(f)] = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df \quad (5.11)$$

The first definition [Eqs (5.5) and (5.6)] is written in terms of the radian frequency variable ω instead of cyclic frequency f . Radian frequency has a somewhat more direct relationship to the time constants and resonant frequencies of real systems, and, as a result, the transform of some system functions are somewhat simpler using this form. Either definition can be converted to the other by using the relationship $\omega = 2\pi f$. It is natural to wonder at this point what the physical significance of $X(f)$ is. One way to understand it is to find the units of $X(f)$, which depend on what the units of $x(t)$ are. To make the idea concrete, suppose for the moment that the units of $x(t)$ are volts (V). The transformation process begins by multiplying $x(t)$ by the complex exponential $e^{-j2\pi ft}$. The exponent of $x(t)$ consists of three dimensionless numbers, $-j$, 2 , and π along with f and t , which are frequency and time, respectively. Frequency has units of hertz or 1/seconds, and time has units of seconds. Therefore, the exponent of e is dimensionless and so is $e^{-j2\pi ft}$. Then we multiply by dt which has units of seconds. Thus the integration process accumulates the area under the product of $x(t)$ and $e^{-j2\pi ft}$. This area has units of volt seconds. Therefore, $X(f)$ has units of volt seconds or volts/hertz. Similarly, $X(\omega)$ would have units of volts/radian/seconds.

The function $X(\omega)$ or $X(f)$ is sometimes called the *amplitude spectral density* or just the *spectrum* of $x(t)$. It expresses the variation of complex sinusoids with frequency which, when added, form $x(t)$. The word *spectral* refers to the variation with respect to frequency. The word *density* comes from the units, volts/hertz.

5.3 CONVERGENCE OF FOURIER TRANSFORM

Dirichlet showed that if $x(t)$ satisfies certain conditions (*Dirichlet conditions*), its Fourier transform is guaranteed to converge pointwise at all points where $x(t)$ is continuous. These conditions are as follows:

1. $x(t)$ be *absolutely integrable*, i.e.,

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty \quad (5.12)$$

This guarantees that $X(\omega)$ will be finite since

$$|X(\omega)| = \left| \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right|$$

$$|X(\omega)| \leq \int_{-\infty}^{\infty} |x(t) e^{-j\omega t}| dt = \int_{-\infty}^{\infty} |x(t)| dt$$

So

$$\text{if } \int_{-\infty}^{\infty} |x(t)| dt < \infty, \text{ then } |X(\omega)| < \infty$$

2. In any finite interval of time, $x(t)$ is of bounded variation, i.e., $x(t)$ have a finite number of maxima and minima.
3. In any finite interval of time, there are only a finite number of discontinuities. Furthermore, each of these discontinuities is finite.

Therefore, absolutely integrable signals that are continuous or that have a finite number of discontinuities have Fourier transform. At a point of discontinuity, t_0 , the inverse Fourier transform in Eq. (5.5) converges to $1/2[x(t_0^+) + x(t_0^-)]$.

The conditions just given for the existence of the Fourier transform of $x(t)$ are sufficient conditions. This means that there are signals that violate either one or both conditions and yet possess a Fourier transform. Examples are power signals (unit step signal, periodic signals, etc.) that are neither absolutely integrable nor square integrable over an infinite interval, but still have Fourier transforms.

Example 5.1 Find the Fourier transform (FT) of the continuous-time signal

$$x(t) = e^{-at}u(t), \quad a > 0$$

shown in Fig. 5.2. Plot the magnitude and phase spectrum of $x(t)$.

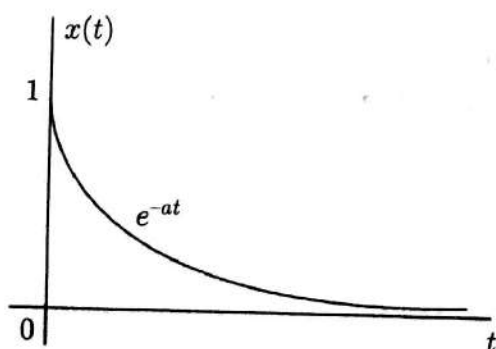


Fig. 5.2 Real-time domain exponential signal

Solution

By definition [Eq. (5.6)],

$$\begin{aligned}
 X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt \\
 &= \int_0^{\infty} e^{-(a+j\omega)t} dt \\
 &= -\frac{1}{a+j\omega} e^{-(a+j\omega)t} \Big|_0^{\infty} \\
 X(\omega) &= -\frac{1}{a+j\omega} (0 - 1) = \frac{1}{a+j\omega} \\
 e^{-at} u(t) &\longleftrightarrow \frac{1}{a+j\omega} \tag{5.13}
 \end{aligned}$$

Since this Fourier transform is complex valued, to plot it as a function of ω , we express $X(\omega)$ in terms of its magnitude and phase:

$$\begin{aligned}
 |X(\omega)| &= \frac{1}{\sqrt{a^2 + \omega^2}} \\
 \angle X(\omega) &= -\tan^{-1}\left(\frac{\omega}{a}\right)
 \end{aligned}$$

The magnitude spectrum $|X(\omega)|$ and phase spectrum $\angle X(\omega)$ are depicted in Fig. 5.3. Observe that $|X(\omega)|$ is an even function of ω , and $\angle X(\omega)$ is an odd function of ω , as expected.

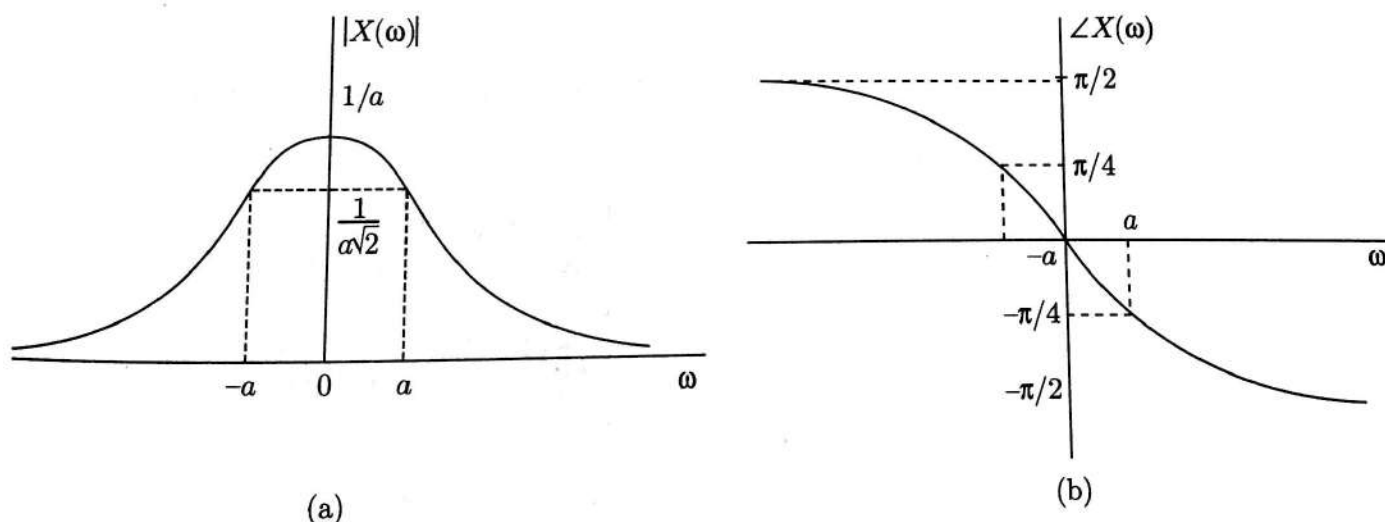


Fig. 5.3 (a) Magnitude spectrum and (b) phase spectrum

Example 5.2 Define the continuous-time signal [as shown in Fig. 5.4(a)]

$$x(t) = e^{-a|t|}, \quad a > 0$$

Find its Fourier transform.

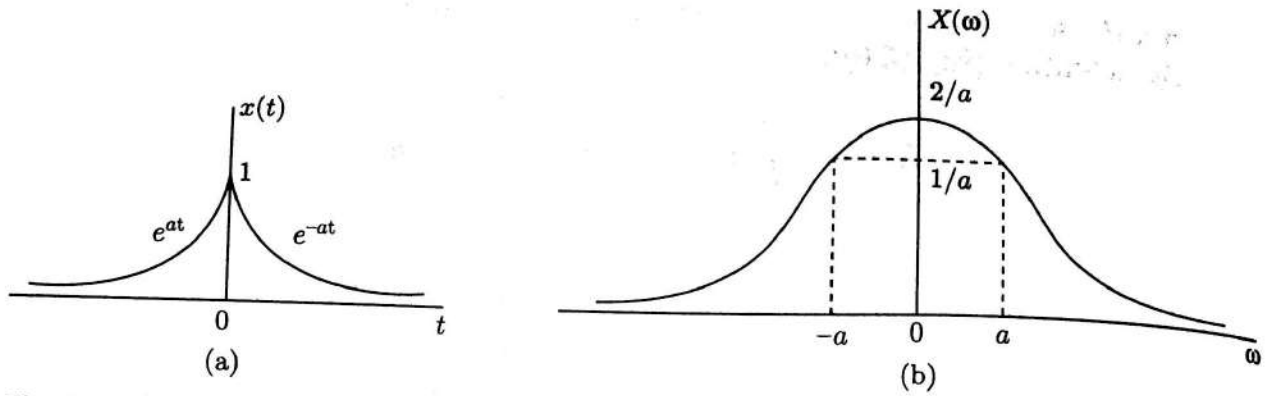


Fig. 5.4 (a) Time-domain signal and (b) its spectrum

Solution

The signal $x(t)$ is defined as

$$\begin{aligned} x(t) &= e^{-a|t|} = \begin{cases} e^{at}, & t < 0 \\ e^{-at}, & t \geq 0 \end{cases} \\ &= e^{at}u(-t) + e^{-at}u(t) \end{aligned}$$

By definition, we have

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} [e^{at}u(-t) + e^{-at}u(t)] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} e^{at}u(-t) e^{-j\omega t} dt + \int_{-\infty}^{\infty} e^{-at}u(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{-(a+j\omega)t} dt \\ &= \frac{1}{a-j\omega} e^{(a-j\omega)t} \Big|_{-\infty}^0 - \frac{1}{a+j\omega} e^{-(a+j\omega)t} \Big|_0^{\infty} \\ &= \frac{1}{a-j\omega} (1-0) - \frac{1}{a+j\omega} (0-1) \\ X(\omega) &= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} \\ X(\omega) &= \frac{2a}{a^2 + \omega^2} \end{aligned}$$

$$e^{-a|t|} \longleftrightarrow \frac{2a}{a^2 + \omega^2}$$

$X(\omega)$ is real and it is illustrated in Fig. 5.4(b).

Example 5.3 Determine the Fourier transform of the unit impulse

$$x(t) = \delta(t)$$

Solution

By definition [Eq. (5.6)],

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = e^{-j\omega t} \Big|_{t=0} = 1$$

$$\delta(t) \longleftrightarrow 1 \quad (5.15)$$

That is, the unit impulse has a Fourier transform consisting of equal contributions at all frequencies. The unit impulse function and its Fourier transform are depicted in Fig. 5.5.

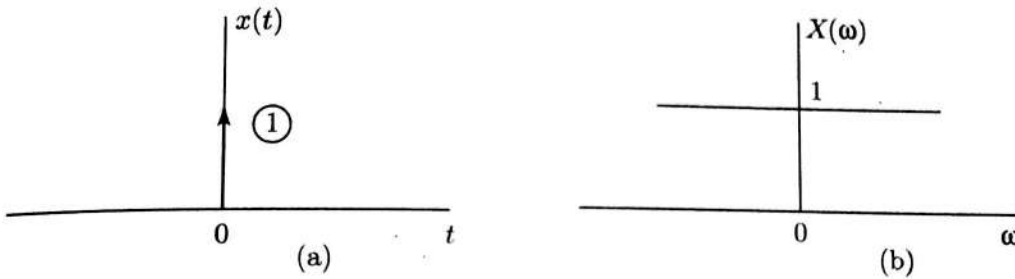


Fig. 5.5 (a) Unit impulse function and (b) its Fourier transform

Example 5.4 Find the inverse Fourier transform of $X(\omega) = \delta(\omega)$.

Solution

From Eq. (5.5), we have

$$x(t) = \mathcal{F}^{-1}[X(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$\mathcal{F}^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega$$

From the sampling property of the impulse function, we have

$$\mathcal{F}^{-1}[\delta(\omega)] = \frac{1}{2\pi} e^{j\omega t} \Big|_{\omega=0} = \frac{1}{2\pi}$$

or

$$\delta(\omega) = \mathcal{F} \left[\frac{1}{2\pi} \right]$$

Therefore,

$$\frac{1}{2\pi} \longleftrightarrow \delta(\omega) \quad (5.16)$$

or

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

(5.17)

This result shows that the spectrum of a constant signal $x(t) = 1$ is an impulse $2\pi\delta(\omega)$, as illustrated in Fig. 5.6.

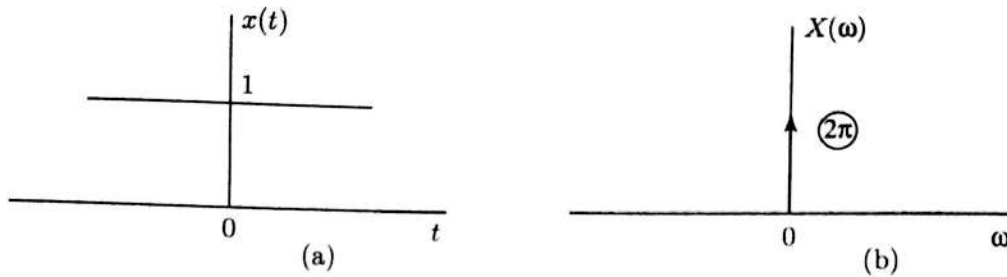


Fig. 5.6 (a) Constant signal and (b) its Fourier transform

Example 5.5 Find the inverse Fourier transform of $X(\omega) = \delta(\omega - \omega_0)$.

Solution

From Eq. (5.5), we have

$$x(t) = \mathcal{F}^{-1}[X(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$\mathcal{F}^{-1}[\delta(\omega - \omega_0)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega$$

From the sampling property of the impulse function, we have

$$\mathcal{F}^{-1}[\delta(\omega - \omega_0)] = \frac{1}{2\pi} e^{j\omega_0 t} \Big|_{\omega=\omega_0} = \frac{1}{2\pi} e^{j\omega_0 t}$$

or

$$\delta(\omega - \omega_0) = \mathcal{F} \left[\frac{1}{2\pi} e^{j\omega_0 t} \right]$$

Therefore,

$$\frac{1}{2\pi} e^{j\omega_0 t} \longleftrightarrow \delta(\omega - \omega_0) \quad (5.18)$$

or

$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0) \quad (5.19)$$

Similarly, we have

$$e^{-j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega + \omega_0) \quad (5.20)$$

Example 5.6 Consider the rectangular pulse (gate pulse) signal depicted in Fig. 5.7. and defined as

$$x(t) = A \operatorname{rect}\left(\frac{t}{2T_0}\right) = A \Pi\left(\frac{t}{2T_0}\right) = \begin{cases} A, & |t| < T_0 \\ 0, & |t| > T_0 \end{cases}$$

Find the Fourier transform of $x(t)$.

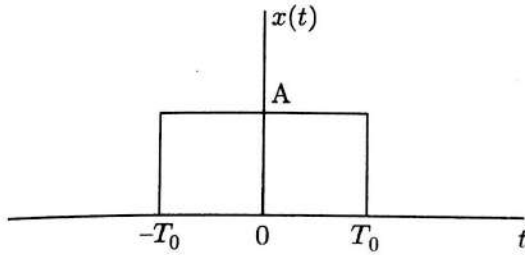


Fig. 5.7 Rectangular pulse or gate pulse

Solution

By definition [Eq. (5.6)],

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-T_0}^{T_0} A e^{-j\omega t} dt \\ &= A \frac{1}{-j\omega} e^{-j\omega t} \Big|_{-T_0}^{T_0} \\ &= \frac{A}{-j\omega} (e^{-j\omega T_0} - e^{j\omega T_0}) \\ &= \frac{2A}{\omega} \left(\frac{e^{j\omega T_0} - e^{-j\omega T_0}}{2j} \right) \\ &= \frac{2A}{\omega} \sin(\omega T_0) \end{aligned}$$

$$X(\omega) = 2AT_0 \frac{\sin(\pi\omega T_0/\pi)}{(\pi\omega T_0/\pi)}$$

$$X(\omega) = 2AT_0 \operatorname{sinc}\left(\frac{\omega T_0}{\pi}\right)$$

$$A \operatorname{rect}\left(\frac{t}{2T_0}\right) \longleftrightarrow 2AT_0 \operatorname{sinc}\left(\frac{\omega T_0}{\pi}\right) \quad (5.21)$$

$\operatorname{sinc}(\omega T_0/\pi) = 0$ when $\sin(\omega T_0) = 0$ except at $\omega T_0 = 0$, where it appears to be indeterminate. This means that $\operatorname{sinc}(\omega T_0/\pi) = 0$ for $\omega T_0 = \pm n\pi$, $n = 1, 2, 3, \dots$. The Fourier transform $X(\omega)$ shown in Fig. 5.8(a) exhibits positive and negative values.

A negative amplitude can be considered to be a positive amplitude with a phase of $-\pi$ or π . We use this observation to plot the magnitude spectrum $|X(\omega)| = |2AT_0 \text{sinc}(\omega T_0/\pi)|$ [Fig. 5.8(b)] and the phase spectrum $\angle X(\omega)$ [Fig. 5.8(c)].

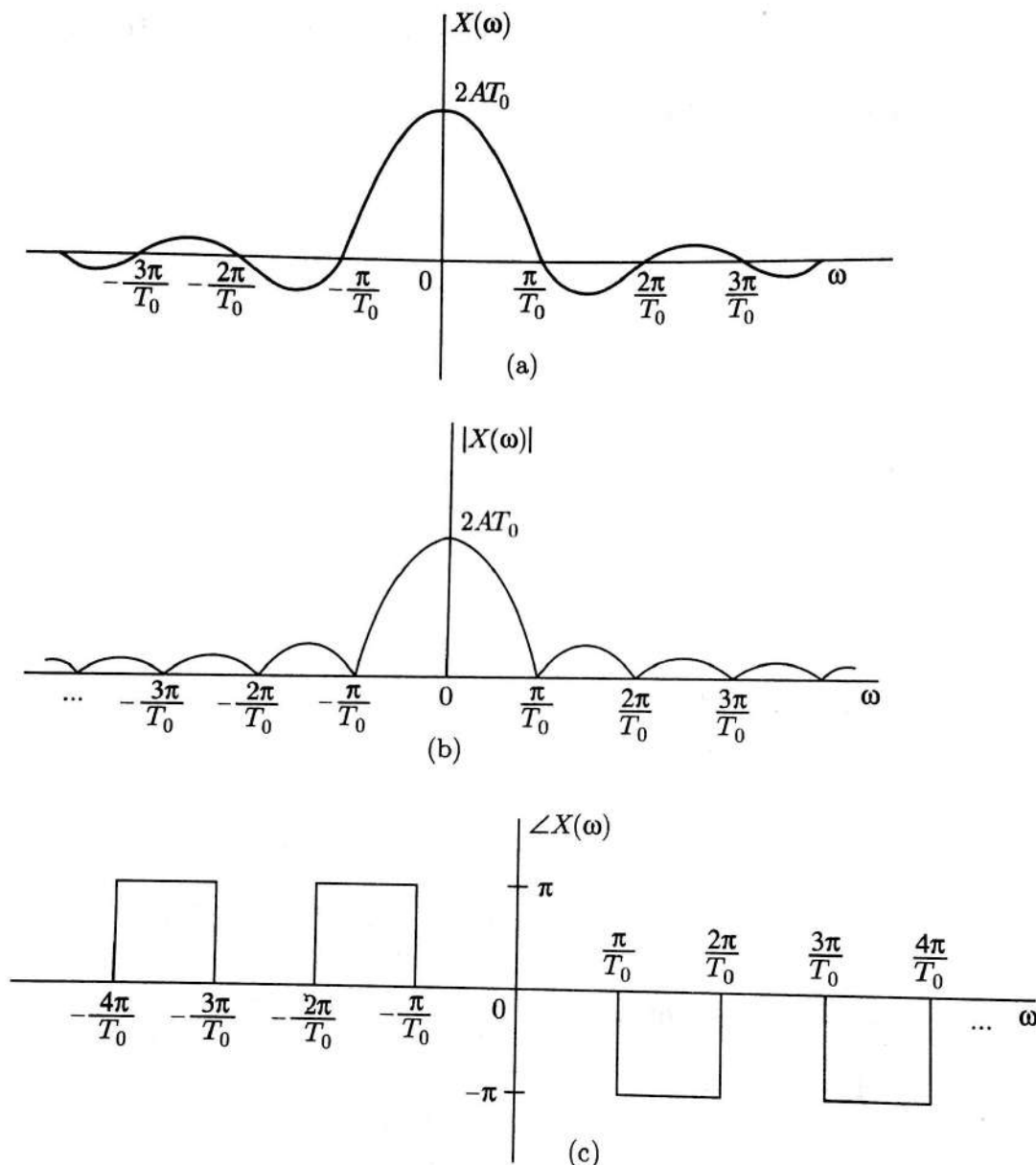


Fig. 5.8 (a) Fourier transform of a gate pulse, (b) its magnitude spectrum, and (c) its phase spectrum

Example 5.7 Consider the triangular pulse signal depicted in Fig. 5.9(a) and defined as

$$x(t) = A \Delta \left(\frac{t}{T} \right) = \begin{cases} A \left(1 - \frac{|t|}{T} \right), & |t| \leq T \\ 0, & |t| \geq T \end{cases} = \begin{cases} A \left(1 + \frac{t}{T} \right), & -T \leq t < 0 \\ A \left(1 - \frac{t}{T} \right), & 0 \leq t \leq T \\ 0, & |t| \geq T \end{cases}$$

Find the Fourier transform of $x(t)$.

Solution

By definition [Eq. (5.6)],

$$\begin{aligned}
X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-T}^0 A \left(1 + \frac{t}{T}\right) e^{-j\omega t} dt + \int_0^T A \left(1 - \frac{t}{T}\right) e^{-j\omega t} dt \\
&= \int_0^T A \left(1 - \frac{t}{T}\right) e^{j\omega t} dt + \int_0^T A \left(1 - \frac{t}{T}\right) e^{-j\omega t} dt \\
&= 2A \int_0^T \left(1 - \frac{t}{T}\right) \frac{e^{j\omega t} + e^{-j\omega t}}{2} dt \\
&= 2A \int_0^T \left(1 - \frac{t}{T}\right) \cos(\omega t) dt \\
&= \left[2A \left(1 - \frac{t}{T}\right) \frac{\sin(\omega t)}{\omega} \right]_0^T + \frac{2A}{\omega T} \int_0^T \sin(\omega t) dt \\
&= 0 - \frac{2A}{\omega^2 T} \cos(\omega t) \Big|_0^T \\
&= \frac{2A}{\omega^2 T} (1 - \cos(\omega T)) \\
X(\omega) &= \frac{4A}{\omega^2 T} \sin^2 \left(\frac{\omega T}{2} \right) \\
&= AT \left(\frac{\sin(\omega T/2)}{(\omega T/2)} \right)^2 \\
&= AT \left(\frac{\sin(\pi \omega T/2\pi)}{(\pi \omega T/2\pi)} \right)^2 \\
X(\omega) &= AT \operatorname{sinc}^2 \left(\frac{\omega T}{2\pi} \right)
\end{aligned}$$

$$A \triangle \left(\frac{t}{T} \right) \longleftrightarrow AT \operatorname{sinc}^2 \left(\frac{\omega T}{2\pi} \right) \quad (5.22)$$

$\operatorname{sinc}^2(\omega T/2\pi) = 0$ when $\sin(\omega T/2) = 0$ except at $\omega T/2 = 0$, where it appears to be indeterminate. This means that $\operatorname{sinc}^2(\omega T/2\pi) = 0$ for $\omega T/2 = \pm n\pi$, $n = 1, 2, 3, \dots$. The Fourier transform $X(\omega)$ is depicted in Fig. 5.9(b).

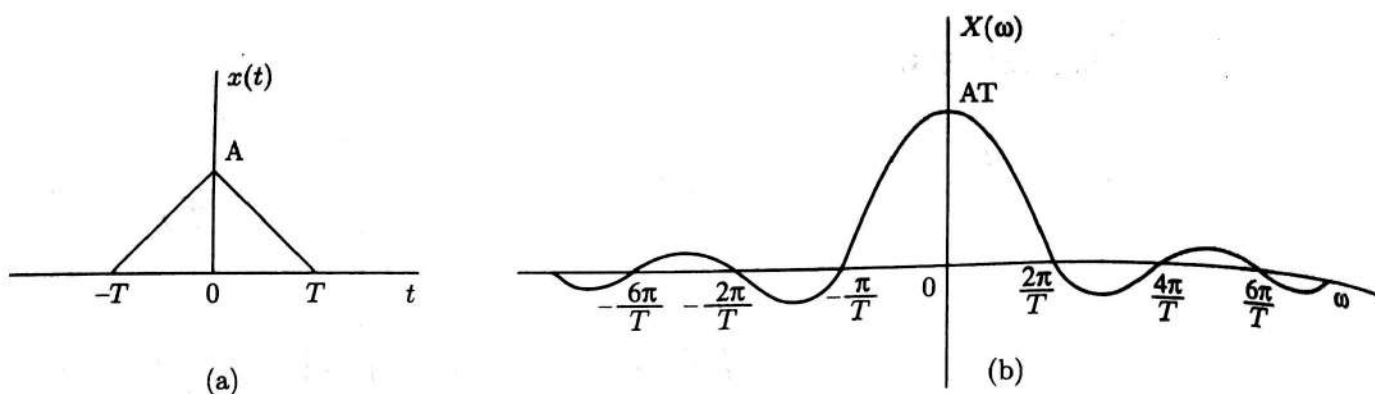


Fig. 5.9 (a) Triangular pulse and (b) its Fourier transform

Example 5.8 Consider the Gaussian pulse signal depicted in Fig. 5.10(a) and defined as

$$x(t) = e^{-\pi t^2}$$

Find the Fourier transform of $x(t)$.

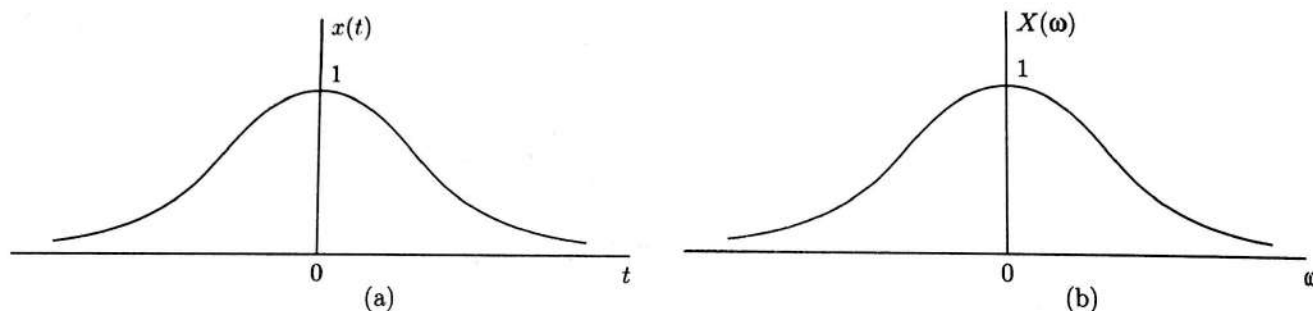


Fig. 5.10 (a) Gaussian pulse and (b) its Fourier transform

Solution

By definition [Eq. (5.6)],

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-\pi t^2} e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-(\pi t^2 + j\omega t)} dt$$

Substituting $\pi t^2 + j\omega t = (\sqrt{\pi}t + j\omega/2\sqrt{\pi})^2 + \omega^2/4\pi$ gives

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} e^{-(\sqrt{\pi}t + j\omega/2\sqrt{\pi})^2} e^{-\omega^2/4\pi} dt \\ &= e^{-\omega^2/4\pi} \int_{-\infty}^{\infty} e^{-(\sqrt{\pi}t + j\omega/2\sqrt{\pi})^2} dt \end{aligned}$$

A change of variables is performed by letting $u = \sqrt{\pi}t + j\omega/2\sqrt{\pi}$, which also yields $dt = du/\sqrt{\pi}$, $u \rightarrow -\infty$ as $t \rightarrow -\infty$, and $u \rightarrow \infty$ as $t \rightarrow \infty$. Therefore,

$$\begin{aligned} X(\omega) &= e^{-\omega^2/4\pi} \int_{-\infty}^{\infty} e^{-u^2} \frac{du}{\sqrt{\pi}} \\ &= e^{-\omega^2/4\pi} \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-u^2} du \end{aligned}$$

Since $\int_0^{\infty} e^{-u^2} du = \sqrt{\pi}/2$, we have

$$\begin{aligned} X(\omega) &= e^{-\omega^2/4\pi} \frac{2}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} \\ X(\omega) &= e^{-\omega^2/4\pi} \\ e^{-\pi t^2} &\longleftrightarrow e^{-\omega^2/4\pi} \end{aligned} \tag{5.23}$$

or

$$e^{-\pi t^2} \longleftrightarrow e^{-\pi f^2} \tag{5.24}$$

The Fourier transform $X(\omega)$ of a Gaussian pulse is a Gaussian pulse [as shown in Fig. 5.10(b)].

Example 5.9 Find the Fourier transform of $x(t) = \cos(\omega_0 t)$.

Solution

We know that (Euler's formula)

$$x(t) = \cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

Taking the Fourier transform, we have

$$\mathcal{F}[x(t)] = X(\omega) = \mathcal{F}\left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}\right] = \frac{1}{2}(\mathcal{F}[e^{j\omega_0 t}] + \mathcal{F}[e^{-j\omega_0 t}])$$

Using Eqs (5.19) and (5.20), we have

$$\begin{aligned} X(\omega) &= \frac{1}{2}[2\pi\delta(\omega - \omega_0) + 2\pi\delta(\omega + \omega_0)] \\ X(\omega) &= \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \\ \cos(\omega_0 t) &\longleftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \end{aligned} \tag{5.25}$$

The Fourier transform of $\cos(\omega_0 t)$ consists of two impulses at ω_0 and $-\omega_0$, as shown in Fig. 5.11.

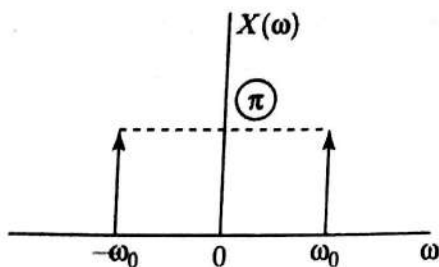


Fig. 5.11

Example 5.10 Find the Fourier transform of $x(t) = \sin(\omega_0 t)$.

Solution

We know that (Euler formula)

$$x(t) = \sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

By taking the Fourier transform, we have

$$\mathcal{F}[x(t)] = X(\omega) = \mathcal{F}\left[\frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}\right] = \frac{1}{2j} (\mathcal{F}[e^{j\omega_0 t}] - \mathcal{F}[e^{-j\omega_0 t}])$$

Using Eqs (5.19) and (5.20), we have

$$X(\omega) = \frac{1}{2j} [2\pi\delta(\omega - \omega_0) - 2\pi\delta(\omega + \omega_0)]$$

$$X(\omega) = \frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$\sin(\omega_0 t) \longleftrightarrow \frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] \quad (5.26)$$

Example 5.11 Find the Fourier transform of the signum function $x(t) = \text{sgn}(t)$.

$$\text{sgn}(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

Solution

The given signal $x(t) = \text{sgn}(t)$ can be rewritten as

$$\text{sgn}(t) = u(t) - u(-t)$$

This signal is not absolutely integrable. So we approach this problem by considering $\text{sgn}(t)$ to be a sum of exponentials $e^{-at}u(t) - e^{at}u(-t)$ in the limit as $a \rightarrow 0$ (Fig. 5.12).

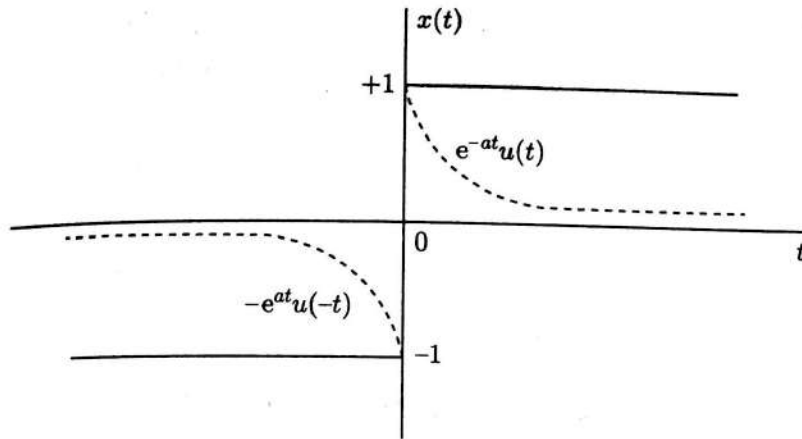


Fig. 5.12 Signum function

Thus,

$$\text{sgn}(t) = \lim_{a \rightarrow 0} [e^{-at}u(t) - e^{at}u(-t)]$$

and

$$\begin{aligned} X(\omega) &= \mathcal{F}[\text{sgn}(t)] = \lim_{a \rightarrow 0} \mathcal{F}[e^{-at}u(t) - e^{at}u(-t)] \\ &= \lim_{a \rightarrow 0} (\mathcal{F}[e^{-at}u(t)] - \mathcal{F}[e^{at}u(-t)]) \\ &= \lim_{a \rightarrow 0} \left(\frac{1}{a + j\omega} - \frac{1}{a - j\omega} \right) \\ &= \lim_{a \rightarrow 0} \left(\frac{-2j\omega}{a^2 + \omega^2} \right) \\ &= \frac{-2j\omega}{\omega^2} \\ X(\omega) &= \mathcal{F}[\text{sgn}(t)] = \frac{2}{j\omega} \end{aligned}$$

$$\text{sgn}(t) \longleftrightarrow \frac{2}{j\omega} \quad (5.27)$$

The magnitude spectrum is

$$|X(\omega)| = \left| \frac{2}{j\omega} \right|$$

and the phase spectrum is

$$\angle X(\omega) = -\tan^{-1} \left(\frac{\omega}{0} \right) = \begin{cases} -\frac{\pi}{2}, & \omega > 0 \\ \frac{\pi}{2}, & \omega < 0 \end{cases}$$

Figure 5.13 shows the magnitude and phase spectrum.

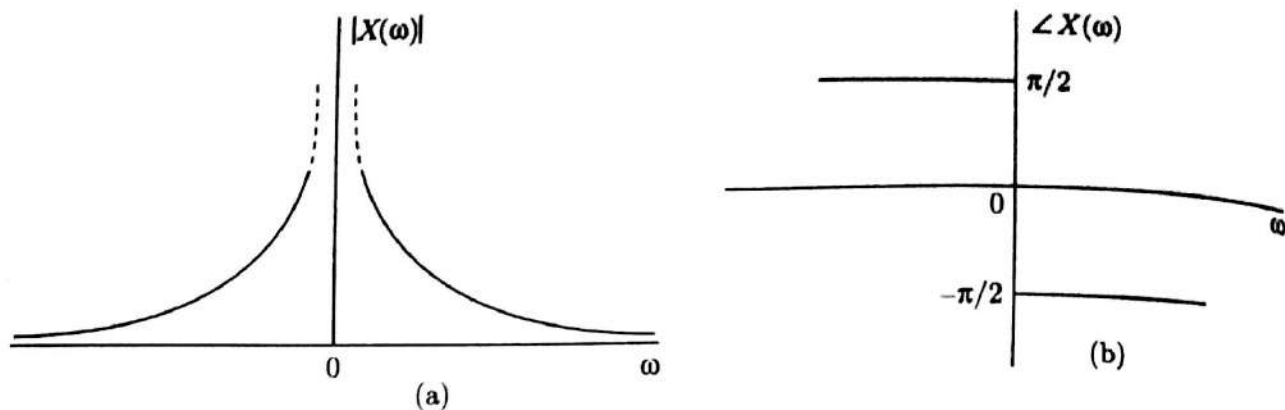


Fig. 5.13 (a) Magnitude spectrum and (b) phase spectrum of the signum function

Example 5.12 Find the Fourier transform of the unit step function $u(t)$.

Solution

Method 1: The unit step function can be expressed as

$$2u(t) = 1 + \text{sgn}(t)$$

$$u(t) = \frac{1}{2} + \frac{1}{2}\text{sgn}(t)$$

$$\mathcal{F}[u(t)] = U(\omega) = \mathcal{F}\left[\frac{1}{2} + \frac{1}{2}\text{sgn}(t)\right]$$

$$U(\omega) = \frac{1}{2}\mathcal{F}[1] + \frac{1}{2}\mathcal{F}[\text{sgn}(t)]$$

$$U(\omega) = \frac{1}{2}2\pi\delta(\omega) + \frac{1}{2}\frac{2}{j\omega}$$

$$U(\omega) = \pi\delta(\omega) + \frac{1}{j\omega}$$

$$u(t) \longleftrightarrow \pi\delta(\omega) + \frac{1}{j\omega} \quad (5.28)$$

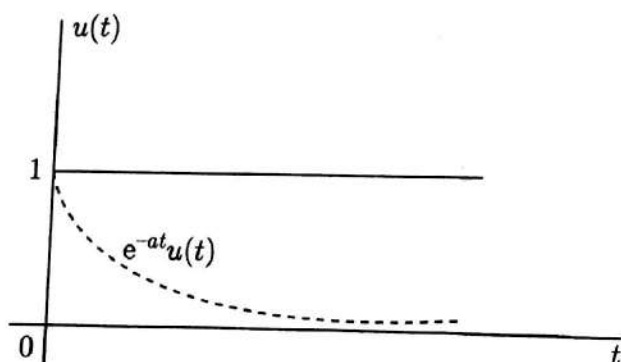


Fig. 5.14 Unit step function

Method 2: The unit step signal is not absolutely integrable. So we approach this problem by considering $u(t)$ to be a decaying exponential $e^{-at}u(t)$ in the limit as $a \rightarrow 0$

(Fig. 5.14). Thus,

$$u(t) = \lim_{a \rightarrow 0} e^{-at} u(t)$$

$$\mathcal{F}[u(t)] = U(\omega) = \mathcal{F}[\lim_{a \rightarrow 0} e^{-at} u(t)] = \lim_{a \rightarrow 0} \mathcal{F}[e^{-at} u(t)] = \lim_{a \rightarrow 0} \frac{1}{a + j\omega}$$

Expressing the RHS in terms of its real and imaginary parts yields

$$U(\omega) = \lim_{a \rightarrow 0} \left[\frac{a}{a^2 + \omega^2} - j \frac{\omega}{a^2 + \omega^2} \right] = \lim_{a \rightarrow 0} \left[\frac{a}{a^2 + \omega^2} \right] + \frac{1}{j\omega}$$

The function $a/a^2 + \omega^2$ has interesting properties. First, the area under this function is π regardless of the value of a :

$$\int_{-\infty}^{\infty} \frac{a}{a^2 + \omega^2} d\omega = \left[\tan^{-1} \frac{\omega}{a} \right]_{-\infty}^{\infty} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

Second, when $a \rightarrow 0$, this function approaches zero for all $\omega \neq 0$, and all its area (π) is concentrated at a single point $\omega = 0$. Clearly, this function approaches as impulse of strength π , i.e.,

$$\lim_{a \rightarrow 0} \left[\frac{a}{a^2 + \omega^2} \right] = \pi \delta(\omega)$$

Thus,

$$U(\omega) = \pi \delta(\omega) + \frac{1}{j\omega}$$

5.4 PROPERTIES OF FOURIER TRANSFORM

Fourier transform possesses a number of important properties that are useful for developing conceptual insights into the transform and into the relationship between the time-domain and frequency-domain descriptions of a signal. In addition, they can also help to reduce the complexity of the evaluation of the Fourier transform of many signals.

We will use a shorthand notation

$$x(t) \longleftrightarrow X(\omega)$$

to indicate the relationship between a time-domain signal $x(t)$ and its Fourier transform $X(\omega)$.

5.4.1 Linearity

If

$$x_1(t) \longleftrightarrow X_1(\omega) \quad \text{and} \quad x_2(t) \longleftrightarrow X_2(\omega)$$

then

$$ax_1(t) + bx_2(t) \longleftrightarrow aX_1(\omega) + bX_2(\omega) \quad (5.29)$$

Proof The Fourier transform of $ax_1(t) + bx_2(t)$ is given by

$$\mathcal{F}[ax_1(t) + bx_2(t)] = \int_{-\infty}^{\infty} [ax_1(t) + bx_2(t)] e^{-j\omega t} dt$$

$$\mathcal{F}[ax_1(t) + bx_2(t)] = a \underbrace{\int_{-\infty}^{\infty} x_1(t) e^{-j\omega t} dt}_{X_1(\omega)} + b \underbrace{\int_{-\infty}^{\infty} x_2(t) e^{-j\omega t} dt}_{X_2(\omega)} = aX_1(\omega) + bX_2(\omega)$$

5.4.2 Time Shifting

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$x(t - t_0) \longleftrightarrow X(\omega) e^{-j\omega t_0} \quad (5.30)$$

Proof The Fourier transform of $x(t - t_0)$ is given by

$$\mathcal{F}[x(t - t_0)] = \int_{-\infty}^{\infty} x(t - t_0) e^{-j\omega t} dt$$

A change of variables is performed by letting $\tau = (t - t_0)$, which also yields $d\tau = dt$, $\tau \rightarrow -\infty$ as $t \rightarrow -\infty$, and $\tau \rightarrow \infty$ as $t \rightarrow \infty$. Therefore,

$$\begin{aligned} \mathcal{F}[x(t - t_0)] &= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega(\tau + t_0)} d\tau \\ &= e^{-j\omega t_0} \int_{-\infty}^{\infty} x(\tau) e^{-j\omega \tau} d\tau \\ \mathcal{F}[x(t - t_0)] &= X(\omega) e^{-j\omega t_0} \end{aligned}$$

One consequence of this property is that when a signal is shifted in time, the *magnitudes* of its Fourier transform remain unaltered. That is, if we express $X(\omega)$ in polar form as

$$\mathcal{F}[x(t)] = X(\omega) = |X(\omega)| e^{j\angle X(\omega)}$$

then

$$\mathcal{F}[x(t - t_0)] = X(\omega) e^{-j\omega t_0} = |X(\omega)| e^{j(\angle X(\omega) - \omega t_0)}$$

Thus, the effect of a time shift on a signal is to introduce into its Fourier transform a phase shift, ωt_0 , which is a linear function of ω .

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5.4.3 Frequency Shifting

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$x(t) e^{j\omega_0 t} \longleftrightarrow X(\omega - \omega_0) \quad (5.31)$$

Proof The Fourier transform of $x(t) e^{j\omega_0 t}$ is given by

$$\begin{aligned} \mathcal{F}[x(t) e^{j\omega_0 t}] &= \int_{-\infty}^{\infty} [x(t) e^{j\omega_0 t}] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(t) e^{-j(\omega - \omega_0)t} dt \end{aligned}$$

$$\mathcal{F}[x(t) e^{j\omega_0 t}] = X(\omega - \omega_0)$$

Hence, a frequency shift corresponds to multiplication in time domain by a complex sinusoid whose frequency is equal to the time shift. ■

5.4.4 Time and Frequency Scaling

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$x(at) \longleftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right) \quad (5.32)$$

Proof The Fourier transform of $x(at)$ is given by

$$\mathcal{F}[x(at)] = \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt$$

Case I For a positive real constant a ,

$$\mathcal{F}[x(at)] = \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt$$

A change of variables is performed by letting $\tau = at$, which also yields $d\tau = a dt$, $\tau \rightarrow -\infty$ as $t \rightarrow -\infty$ and $\tau \rightarrow \infty$ as $t \rightarrow \infty$. Therefore,

$$\mathcal{F}[x(at)] = \frac{1}{a} \int_{-\infty}^{\infty} x(\tau) e^{-j(\omega/a)\tau} d\tau = \frac{1}{a} X\left(\frac{\omega}{a}\right)$$

Case II For a negative real constant $-a$,

$$\mathcal{F}[x(-at)] = \int_{-\infty}^{\infty} x(-at) e^{-j\omega t} dt$$

A change of variables is performed by letting $\tau = -at$, which also yields $d\tau = -a dt$, $\tau \rightarrow \infty$ as $t \rightarrow -\infty$ and $\tau \rightarrow -\infty$ as $t \rightarrow \infty$. Therefore,

$$\begin{aligned}\mathcal{F}[x(-at)] &= -\frac{1}{a} \int_{\infty}^{-\infty} x(\tau) e^{j(\omega/a)\tau} d\tau \\ &= \frac{1}{a} \int_{-\infty}^{\infty} x(\tau) e^{-j(-\omega/a)\tau} d\tau \\ &= \frac{1}{a} X\left(-\frac{\omega}{a}\right)\end{aligned}$$

Combining the two cases, we have

$$\mathcal{F}[x(at)] = \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

The scaling property states that time compression of a signal results in its spectral expansion, and time expansion of the signal results in its spectral compression. The relation between compression in one domain and expansion in the other is the basis for an idea called the *uncertainty principle* of Fourier analysis.

5.4.5 Time Reversal

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$x(-t) \longleftrightarrow X(-\omega) \quad (5.33)$$

Proof Substituting $a = -1$ in Eq. (5.32), we have

$$\begin{aligned}\mathcal{F}[x(-t)] &= \frac{1}{|-1|} X\left(\frac{\omega}{-1}\right) \\ \mathcal{F}[x(-t)] &= X(-\omega)\end{aligned}$$

The time-reversal property states that reversing a signal in time also reverses its Fourier transform. An interesting consequence of the time-reversal property is that if $x(t)$ is even, then its Fourier transform is also even, i.e.,

$$\text{if } x(-t) = x(t), \quad \text{then } X(-\omega) = X(\omega) \quad (5.34)$$

Similarly, if $x(t)$ is odd, then so is its Fourier transform, i.e.,

$$\text{if } x(-t) = -x(t), \quad \text{then } X(-\omega) = -X(\omega) \quad (5.35)$$

5.4.6 Differentiation in Time Domain

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$\frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega) \quad (5.36)$$

Proof From Eq. (5.5), we have

$$x(t) = \mathcal{F}^{-1}[X(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Differentiating both sides gives

$$\frac{dx(t)}{dt} = \frac{1}{2\pi} \int_{-\infty}^{\infty} [j\omega X(\omega)] e^{j\omega t} d\omega$$

$$\frac{dx(t)}{dt} = \mathcal{F}^{-1}[j\omega X(\omega)]$$

Therefore,

$$\frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega)$$

Similarly, the differentiation property can be extended to yield

$$\frac{d^2x(t)}{dt^2} \longleftrightarrow (j\omega)^2 X(\omega) \quad (5.37)$$

■

Example 5.13 Consider the triangular pulse signal depicted in Fig. 5.15(a) and defined as

$$x(t) = A \Delta \left(\frac{t}{T} \right) = \begin{cases} A \left(1 - \frac{|t|}{T} \right), & |t| \leq T \\ 0, & |t| \geq T \end{cases} = \begin{cases} A \left(1 + \frac{t}{T} \right), & -T \leq t < 0 \\ A \left(1 - \frac{t}{T} \right), & 0 \leq t \leq T \\ 0, & |t| \geq T \end{cases}$$

Find the Fourier transform of $x(t)$ using the differentiation property.

Solution

Using the ramp function $r(t)$, the given triangular pulse can be written as

$$x(t) = \frac{A}{T} r(t+T) - \frac{2A}{T} r(t) + \frac{A}{T} r(t-T)$$

The first derivative of the triangular pulse is [Fig. 5.15(b)]

$$\frac{dx(t)}{dt} = \frac{A}{T} u(t+T) - \frac{2A}{T} u(t) + \frac{A}{T} u(t-T)$$

and the second derivative is [Fig. 5.15(c)]

$$\frac{d^2x(t)}{dt^2} = \frac{A}{T} \delta(t+T) - \frac{2A}{T} \delta(t) + \frac{A}{T} \delta(t-T)$$

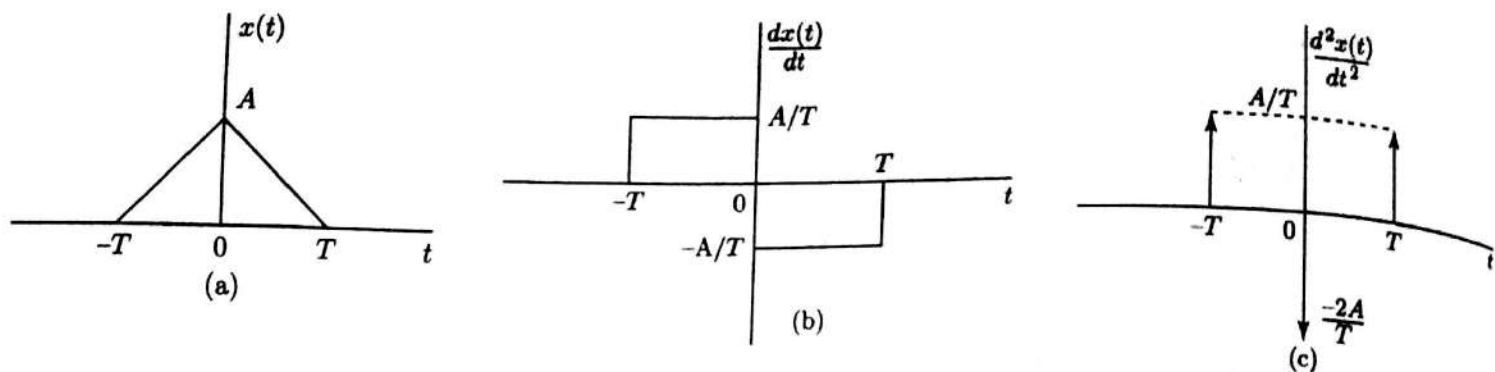


Fig. 5.15 (a) A triangular pulse, (b) its first derivative, and (c) its second derivative

The Fourier transform of the second derivative is

$$\begin{aligned}\mathcal{F}\left[\frac{d^2x(t)}{dt^2}\right] &= \mathcal{F}\left[\frac{A}{T}\delta(t+T) - \frac{2A}{T}\delta(t) + \frac{A}{T}\delta(t-T)\right] \\ &= \frac{A}{T}\mathcal{F}[\delta(t+T)] - \frac{2A}{T}\mathcal{F}[\delta(t)] + \frac{A}{T}\mathcal{F}[\delta(t-T)]\end{aligned}$$

Now, using the differentiation and the time-shifting property, we get

$$\begin{aligned}(j\omega)^2 X(\omega) &= \frac{A}{T} e^{j\omega T} - \frac{2A}{T} + \frac{A}{T} e^{-j\omega T} \\ &= \frac{2A}{T} \left(\frac{e^{j\omega T} + e^{-j\omega T}}{2} \right) - \frac{2A}{T} \\ (j\omega)^2 X(\omega) &= \frac{2A}{T} [\cos(\omega T) - 1] \\ &= \frac{2A}{(j\omega)^2 T} [\cos(\omega T) - 1] = \frac{2A}{\omega^2 T} [1 - \cos(\omega T)] \\ X(\omega) &= \frac{4A}{\omega^2 T} \sin^2\left(\frac{\omega T}{2}\right) = AT \left(\frac{\sin(\omega T/2)}{(\omega T/2)} \right)^2 \\ X(\omega) &= AT \left(\frac{\sin(\pi\omega T/2\pi)}{(\pi\omega T/2\pi)} \right)^2 = AT \operatorname{sinc}^2\left(\frac{\omega T}{2\pi}\right)\end{aligned}$$

5.4.7 Convolution Property

If

$$x_1(t) \longleftrightarrow X_1(\omega) \quad \text{and} \quad x_2(t) \longleftrightarrow X_2(\omega)$$

then

$$x_1(t) * x_2(t) \longleftrightarrow X_1(\omega) X_2(\omega) \quad (5.38)$$

The Fourier transform maps the convolution of two signals into the product of their Fourier transforms.

Proof The Fourier transform of $x_1(t) * x_2(t)$ is given by

$$\begin{aligned}\mathcal{F}[x_1(t) * x_2(t)] &= \int_{-\infty}^{\infty} [x_1(t) * x_2(t)] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) d\tau \right) e^{-j\omega t} dt\end{aligned}$$

Interchanging the order of integration and noting that $x_1(\tau)$ does not depend on t gives

$$\mathcal{F}[x_1(t) * x_2(t)] = \int_{-\infty}^{\infty} x_1(\tau) \left(\int_{-\infty}^{\infty} x_2(t - \tau) e^{-j\omega t} dt \right) d\tau$$

By the time-shifting property, Eq. (5.30), the bracketed term is $X_2(\omega) e^{-j\omega\tau}$. Substituting this into the above equation yields

$$\begin{aligned}\mathcal{F}[x_1(t) * x_2(t)] &= \int_{-\infty}^{\infty} x_1(\tau) (X_2(\omega) e^{-j\omega\tau}) d\tau \\ &= X_2(\omega) \int_{-\infty}^{\infty} x_1(\tau) e^{-j\omega\tau} d\tau\end{aligned}$$

$$\mathcal{F}[x_1(t) * x_2(t)] = X_2(\omega) X_1(\omega) = X_1(\omega) X_2(\omega)$$

Therefore,

$$x_1(t) * x_2(t) \longleftrightarrow X_1(\omega) X_2(\omega)$$

The convolution property states that convolution in the time domain corresponds to multiplication in the frequency domain. ■

5.4.8 Multiplication (or Modulation) Property

If

$$x_1(t) \longleftrightarrow X_1(\omega) \quad \text{and} \quad x_2(t) \longleftrightarrow X_2(\omega)$$

then

$$x_1(t)x_2(t) \longleftrightarrow \frac{1}{2\pi} [X_1(\omega) * X_2(\omega)] \quad (5.39)$$

The Fourier transform maps the multiplication of two signals into the convolution of their Fourier transforms.

Proof The Fourier transform of $x_1(t)x_2(t)$ is

$$\begin{aligned}\mathcal{F}[x_1(t)x_2(t)] &= \int_{-\infty}^{\infty} [x_1(t)x_2(t)] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\theta) e^{j\theta t} d\theta \right) x_2(t) e^{-j\omega t} dt\end{aligned}$$

Interchanging the order of integration and noting that $X_1(\theta)$ does not depend on t yields

$$\mathcal{F}[x_1(t)x_2(t)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\theta) \left(\int_{-\infty}^{\infty} [x_2(t) e^{j\theta t}] e^{-j\omega t} dt \right) d\theta$$

By the frequency-shifting property, Eq. (5.31), the bracketed term is $X_2(\omega - \theta)$. Substituting this into the above equation yields

$$\begin{aligned} \mathcal{F}[x_1(t)x_2(t)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\theta) X_2(\omega - \theta) d\theta \\ &= \frac{1}{2\pi} [X_1(\omega) * X_2(\omega)] \end{aligned}$$

Therefore,

$$x_1(t)x_2(t) \longleftrightarrow \frac{1}{2\pi} [X_1(\omega) * X_2(\omega)]$$

■

Example 5.14 Consider the pulse signal $g(t)$ shown in Fig. 5.16(a), which consists of a sinusoidal wave of amplitude A and frequency ω_c extending in duration from $t = -T/2$ to $t = T/2$. This signal is sometimes referred to as an *RF pulse* when the frequency ω_c falls in the radiofrequency band. Find the Fourier transform of the RF pulse $g(t)$.

Solution

The signal $g(t)$ of Fig. 5.16(a) may be expressed mathematically as the multiplication of two time-domain signals: rectangle signal of duration T and a sinusoidal signal of frequency ω_c :

$$\begin{aligned} g(t) &= A \operatorname{rect}\left(\frac{t}{T}\right) \cos(\omega_c t) \\ &= x(t) \cos(\omega_c t) \end{aligned}$$

where $x(t) = A \operatorname{rect}(t/T)$. The Fourier transform of the rectangle signal $x(t)$ is given by [Eq. (5.21)]:

$$X(\omega) = AT \frac{\sin(\omega T/2)}{(\omega T/2)} = AT \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right)$$

Now, consider the RF pulse $g(t)$:

$$g(t) = x(t) \cos(\omega_c t)$$

Applying the multiplication property gives

$$\mathcal{F}[g(t)] = \frac{1}{2\pi} \left(\mathcal{F}[x(t)] * \mathcal{F}[\cos(\omega_c t)] \right)$$

$$G(\omega) = \frac{1}{2\pi} \left(X(\omega) * \pi[\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] \right)$$

$$G(\omega) = \frac{1}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)]$$

Substituting $X(\omega) = AT \operatorname{sinc}(\omega T/2\pi)$ into the above equation yields

$$G(\omega) = \frac{1}{2} \left[AT \operatorname{sinc} \left(\frac{(\omega - \omega_c)T}{2\pi} \right) + AT \operatorname{sinc} \left(\frac{(\omega + \omega_c)T}{2\pi} \right) \right]$$

$$G(\omega) = \frac{AT}{2} \left[\operatorname{sinc} \left(\frac{(\omega - \omega_c)T}{2\pi} \right) + \operatorname{sinc} \left(\frac{(\omega + \omega_c)T}{2\pi} \right) \right]$$

The magnitude spectrum of the RF pulse is shown in Fig. 5.16(b).

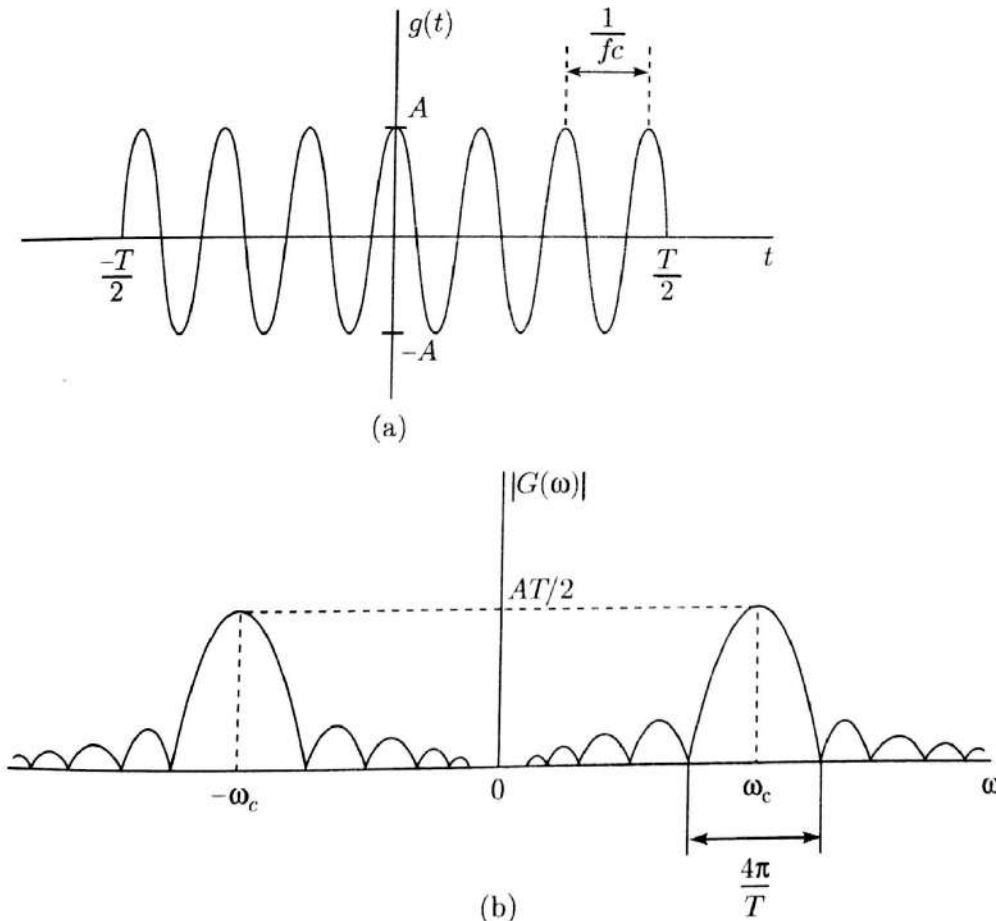


Fig. 5.16 (a) RF pulse and (b) its magnitude spectrum

Example 5.15 Consider an exponentially damped sinusoidal wave defined by (Fig. 5.17)

$$g(t) = e^{-t} \sin(\omega_c t) u(t)$$

Find the Fourier transform of $g(t)$.

Solution

The given exponentially damped sinusoidal wave is a multiplication of two time-domain signals: an exponentially decaying signal and a sinusoidal signal of frequency ω_c .

$$g(t) = e^{-t} \sin(\omega_c t) u(t) = e^{-t} u(t) \sin(\omega_c t) = x(t) \sin(\omega_c t)$$

where $x(t) = e^{-t} u(t)$. The Fourier transform of $x(t)$ is given by [substitute $a = 1$ into Eq. (5.13)]

$$X(\omega) = \frac{1}{1 + j\omega}$$

Now, consider the exponentially damped sinusoidal wave $g(t)$:

$$g(t) = x(t) \sin(\omega_c t)$$

Applying the multiplication property gives

$$\mathcal{F}[g(t)] = \frac{1}{2\pi} (\mathcal{F}[x(t)] * \mathcal{F}[\sin(\omega_c t)])$$

$$G(\omega) = \frac{1}{2\pi} \left(X(\omega) * \frac{\pi}{j} [\delta(\omega - \omega_c) - \delta(\omega + \omega_c)] \right)$$

$$G(\omega) = \frac{1}{2j} [X(\omega - \omega_c) - X(\omega + \omega_c)]$$

Substituting $X(\omega) = 1/(1 + j\omega)$ into the above equation yields

$$G(\omega) = \frac{1}{2j} \left[\frac{1}{1 + j(\omega - \omega_c)} - \frac{1}{1 + j(\omega + \omega_c)} \right]$$

$$G(\omega) = \frac{\omega_c}{(1 + j\omega)^2 + \omega_c^2}$$

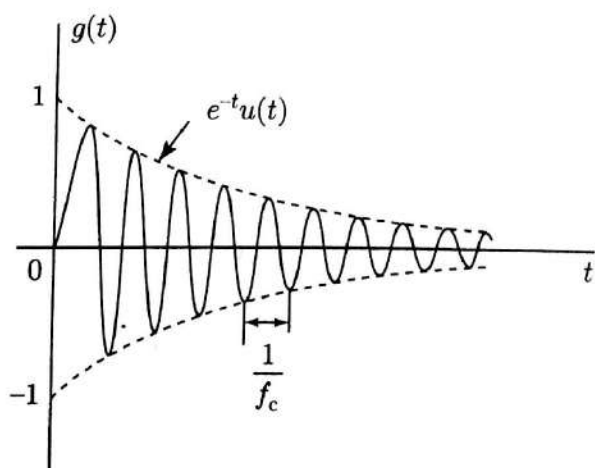


Fig. 5.17 Damped sinusoidal wave

Example 5.16 Consider an exponentially damped sinusoidal wave defined by

$$g(t) = e^{-at} \cos(\omega_c t) u(t)$$

Find the Fourier transform of $g(t)$.

Solution

The given exponentially damped sinusoidal wave is a multiplication of two time-domain signals: an exponentially decaying signal and a sinusoidal signal of frequency ω_c .

$$g(t) = e^{-at} \cos(\omega_c t) u(t) = e^{-at} u(t) \cos(\omega_c t) = x(t) \cos(\omega_c t)$$

where $x(t) = e^{-at} u(t)$. The Fourier transform of $x(t)$ is given by [Eq. (5.13)]

$$X(\omega) = \frac{1}{a + j\omega}$$

Now, consider the exponentially damped sinusoidal wave $g(t)$:

$$g(t) = x(t) \cos(\omega_c t)$$

Applying the multiplication property gives

$$\begin{aligned} \mathcal{F}[g(t)] &= \frac{1}{2\pi} \left(\mathcal{F}[x(t)] * \mathcal{F}[\cos(\omega_c t)] \right) \\ G(\omega) &= \frac{1}{2\pi} \left(X(\omega) * \pi[\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] \right) \\ G(\omega) &= \frac{1}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)] \end{aligned}$$

Substituting $X(\omega) = 1/(a + j\omega)$ into the above equation yields

$$\begin{aligned} G(\omega) &= \frac{1}{2} \left[\frac{1}{a + j(\omega - \omega_c)} + \frac{1}{a + j(\omega + \omega_c)} \right] \\ G(\omega) &= \frac{a + j\omega}{(a + j\omega)^2 + \omega_c^2} \end{aligned}$$

Example 5.17 Find the Fourier transform of $x(t) = \cos(\omega_c t) u(t)$.

Solution

Consider the given signal

$$x(t) = \cos(\omega_c t) u(t)$$

Applying the multiplication property gives

$$\begin{aligned} \mathcal{F}[x(t)] &= \frac{1}{2\pi} \left(\mathcal{F}[\cos(\omega_c t)] * \mathcal{F}[u(t)] \right) \\ X(\omega) &= \frac{1}{2\pi} \left(\pi[\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] * \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] \right) \\ X(\omega) &= \frac{1}{2\pi} \pi^2 [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] + \frac{1}{2\pi} \pi \left[\frac{1}{j(\omega - \omega_c)} + \frac{1}{j(\omega + \omega_c)} \right] \\ X(\omega) &= \frac{\pi}{2} [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] + \left[\frac{j\omega}{\omega_c^2 - \omega^2} \right] \end{aligned}$$

5.4.9 Differentiation in Frequency Domain

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$-jtx(t) \longleftrightarrow \frac{dX(\omega)}{d\omega} \quad (5.40)$$

or

$$tx(t) \longleftrightarrow j \frac{dX(\omega)}{d\omega} \quad (5.41)$$

Proof From Eq. (5.6), we have

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Differentiating both sides w.r.t. ω gives

$$\frac{dX(\omega)}{d\omega} = \int_{-\infty}^{\infty} [-jtx(t)] e^{-j\omega t} dt$$

$$\frac{dX(\omega)}{d\omega} = \mathcal{F}[-jtx(t)]$$

Therefore,

$$-jtx(t) \longleftrightarrow \frac{dX(\omega)}{d\omega}$$

or

$$tx(t) \longleftrightarrow j \frac{dX(\omega)}{d\omega} \quad (5.42)$$

Example 5.18 Find the Fourier transform of the signal $g(t)$ defined by

$$g(t) = t e^{-at} u(t)$$

Solution

Consider the given signal

$$g(t) = t e^{-at} u(t)$$

$$g(t) = tx(t)$$

From Eq. (5.42), we have

$$\mathcal{F}[g(t)] = \mathcal{F}[tx(t)] = j \frac{dX(\omega)}{d\omega}$$

Here, $x(t) = e^{-at}u(t)$ and its Fourier transform $X(\omega) = 1/(a + j\omega)$.

$$G(\omega) = \mathcal{F}[te^{-at}u(t)] = j \frac{d}{d\omega} \left(\frac{1}{a + j\omega} \right)$$

$$G(\omega) = \frac{1}{(a + j\omega)^2}$$

Therefore,

$$te^{-at}u(t) \longleftrightarrow \frac{1}{(a + j\omega)^2} \quad (5.43)$$

5.4.10 Integration

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow \frac{1}{j\omega} X(\omega) + \pi X(0)\delta(\omega) \quad (5.44)$$

Proof The convolution of a signal $x(t)$ with a unit step function $u(t)$ is given by

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau)u(t - \tau) d\tau$$

Since

$$u(t - \tau) = \begin{cases} 1, & (t - \tau) > 0 \longrightarrow \tau < t \\ 0, & (t - \tau) < 0 \longrightarrow \tau > t \end{cases}$$

we get

$$x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$$

The convolution of a signal with a unit step is the same as the cumulative integral of the signal. Now we can prove the integration property of the Fourier transform.

$$\begin{aligned} \int_{-\infty}^t x(\tau) d\tau &= x(t) * u(t) \\ \mathcal{F} \left[\int_{-\infty}^t x(\tau) d\tau \right] &= \mathcal{F}[x(t) * u(t)] \end{aligned}$$

Using the convolution property [Eq. (5.38)] and Eq. (5.28), we get

$$\begin{aligned}\mathcal{F}\left[\int_{-\infty}^t x(\tau) d\tau\right] &= X(\omega)U(\omega) = X(\omega)\left(\frac{1}{j\omega} + \pi\delta(\omega)\right) \\ &= \frac{1}{j\omega}X(\omega) + \pi X(\omega)\delta(\omega) \\ &= \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)\end{aligned}$$

Therefore,

$$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

5.4.11 Duality

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$X(t) \longleftrightarrow 2\pi x(-\omega) \quad (5.45)$$

Proof By definition

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$2\pi x(t) = \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

replacing t with $-t$ in the above equation gives

$$2\pi x(-t) = \int_{-\infty}^{\infty} X(\omega) e^{-j\omega t} d\omega$$

Now interchanging the variables t and ω yields

$$2\pi x(-\omega) = \int_{-\infty}^{\infty} X(t) e^{-j\omega t} dt$$

$$2\pi x(-\omega) = \mathcal{F}[X(t)]$$

Therefore,

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

Example 5.19 Find the Fourier transform $G(\omega)$ of the signal

$$g(t) = \frac{1}{1 + jt}$$

Solution

Define $X(\omega) = 1/(1 + j\omega)$ by replacing t with ω in the expression of $g(t)$. From Eq. (5.13), we have

$$e^{-at}u(t) \longleftrightarrow \frac{1}{a + j\omega}$$

By substituting $a = 1$, we get

$$e^{-t}u(t) \longleftrightarrow \frac{1}{1 + j\omega}$$

$$x(t) \longleftrightarrow X(\omega)$$

$$x(t) = e^{-t}u(t) \quad \text{and} \quad X(\omega) = \frac{1}{1 + j\omega}$$

$$x(-\omega) = e^{\omega}u(-\omega) \quad \text{and} \quad X(t) = \frac{1}{1 + jt}$$

The duality property of the Fourier transform states that if

$$x(t) \longleftrightarrow X(\omega)$$

then

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

$$\frac{1}{1 + jt} \longleftrightarrow 2\pi e^{\omega}u(-\omega)$$

Therefore,

$$\mathcal{F}\left[\frac{1}{1 + jt}\right] = 2\pi e^{\omega}u(-\omega) \quad (5.46)$$

Example 5.20 Find the Fourier transform $G(\omega)$ of the signal [Fig. 5.18(a)]

$$g(t) = \frac{1}{1 + t^2}$$

Solution

Define $X(\omega) = 1/(1 + \omega^2)$ by replacing t with ω in the expression of $g(t)$. From Eq. (5.14), we have

$$e^{-a|t|} \longleftrightarrow \frac{2a}{a^2 + \omega^2}$$

By substituting $a = 1$, we get

$$e^{-|t|} \longleftrightarrow \frac{2}{1 + \omega^2}$$

$$\frac{1}{2} e^{-|t|} \longleftrightarrow \frac{1}{1 + \omega^2}$$

$$x(t) \longleftrightarrow X(\omega)$$

$$x(t) = \frac{1}{2} e^{-|t|} \quad \text{and} \quad X(\omega) = \frac{1}{1 + \omega^2}$$

$$x(-\omega) = \frac{1}{2} e^{-|-\omega|} = \frac{1}{2} e^{-|\omega|} \quad \text{and} \quad X(t) = \frac{1}{1 + t^2}$$

The duality property of the Fourier transform states that if

$$x(t) \longleftrightarrow X(\omega)$$

then

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

$$\frac{1}{1 + t^2} \longleftrightarrow 2\pi \frac{1}{2} e^{-|\omega|}$$

$$g(t) = \frac{1}{1 + t^2} \longleftrightarrow \pi e^{-|\omega|} = G(\omega)$$

Therefore,

$$\mathcal{F} \left[\frac{1}{1 + t^2} \right] = \pi e^{-|\omega|} \quad (5.47)$$

The Fourier transform $G(\omega)$ is shown in Fig. 5.18(b).

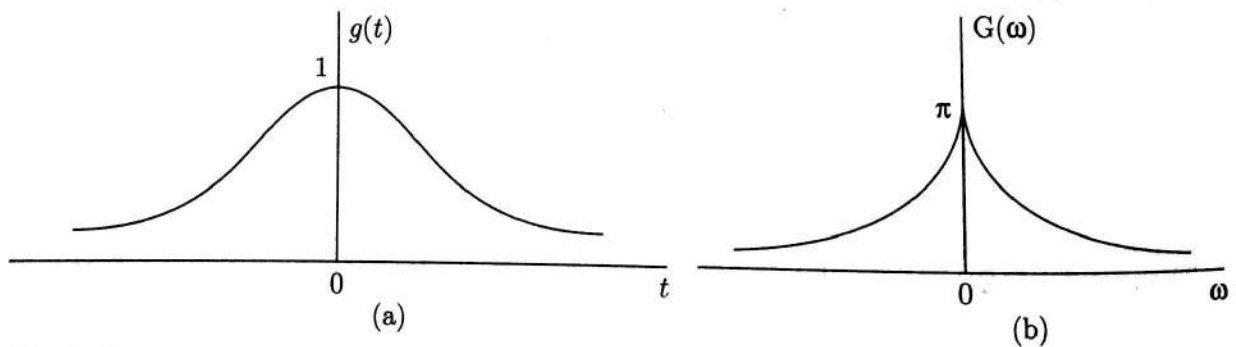


Fig. 5.18

Example 5.21 Find the Fourier transform $G(\omega)$ of the signal [Fig. 5.19(a)]

$$g(t) = \frac{1}{\pi t}$$

Solution

Define $X(\omega) = (1/\pi\omega)$ by replacing t with ω in the expression of $g(t)$. From Eq. (5.27), we have

$$\text{sgn}(t) \longleftrightarrow \frac{2}{j\omega}$$

$$\frac{j}{2\pi} \text{sgn}(t) \longleftrightarrow \frac{1}{\pi\omega}$$

$$x(t) \longleftrightarrow X(\omega)$$

$$x(t) = \frac{j}{2\pi} \text{sgn}(t) \quad \text{and} \quad X(\omega) = \frac{1}{\pi\omega}$$

$$x(-\omega) = \frac{j}{2\pi} \text{sgn}(-\omega) = -\frac{j}{2\pi} \text{sgn}(\omega) \quad \text{and} \quad X(t) = \frac{1}{\pi t}$$

The duality property of the Fourier transform states that if

$$x(t) \longleftrightarrow X(\omega)$$

then

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

$$\frac{1}{\pi t} \longleftrightarrow -2\pi \frac{j}{2\pi} \text{sgn}(\omega)$$

$$\frac{1}{\pi t} \longleftrightarrow -j \text{sgn}(\omega)$$

Therefore,

$$\mathcal{F}\left[\frac{1}{\pi t}\right] = -j \text{sgn}(\omega) = \begin{cases} -j, & \omega > 0 \\ j, & \omega < 0 \end{cases} \quad (5.48)$$

The Fourier transform $G(\omega)$ is shown in Fig. 5.19(b).

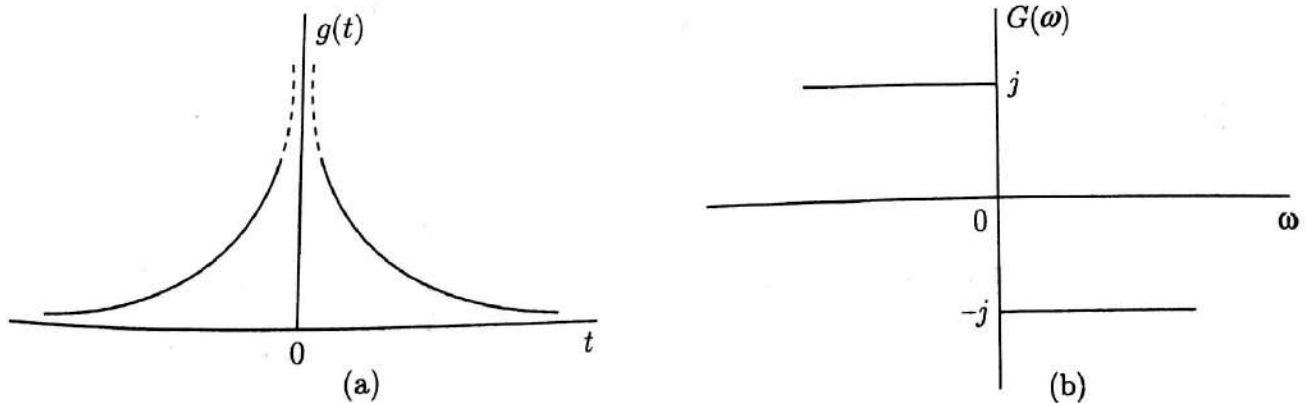


Fig. 5.19

Example 5.22 Find the Fourier transform $G(\omega)$ of the signal [Fig. 5.20(a)]

$$g(t) = A \operatorname{sinc} \left(\frac{t}{2\pi} \right)$$

Solution

Define $X(\omega) = A \operatorname{sinc}(\omega/2\pi)$ by replacing t with ω in the expression of $g(t)$. From Eq. (5.21), we have

$$A \operatorname{rect} \left(\frac{t}{2T_0} \right) \longleftrightarrow 2AT_0 \operatorname{sinc} \left(\frac{\omega T_0}{\pi} \right)$$

Substituting $2T_0 = 1$ yields

$$A \operatorname{rect}(t) \longleftrightarrow A \operatorname{sinc} \left(\frac{\omega}{2\pi} \right)$$

$$x(t) \longleftrightarrow X(\omega)$$

$$x(t) = A \operatorname{rect}(t) \quad \text{and} \quad X(\omega) = A \operatorname{sinc} \left(\frac{\omega}{2\pi} \right)$$

$$x(-\omega) = A \operatorname{rect}(-\omega) = A \operatorname{rect}(\omega) \quad \text{and} \quad X(t) = A \operatorname{sinc} \left(\frac{t}{2\pi} \right)$$

The duality property of the Fourier transform states that if

$$x(t) \longleftrightarrow X(\omega)$$

then

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

$$A \operatorname{sinc} \left(\frac{t}{2\pi} \right) \longleftrightarrow 2\pi A \operatorname{rect}(\omega)$$

Therefore,

$$\mathcal{F} \left[A \operatorname{sinc} \left(\frac{t}{2\pi} \right) \right] = 2\pi A \operatorname{rect}(\omega) = \begin{cases} 2\pi A, & -\frac{1}{2} < \omega < \frac{1}{2} \\ 0, & \text{otherwise} \end{cases} \quad (5.49)$$

The Fourier transform $G(\omega)$ is shown in Fig. 5.20(b).

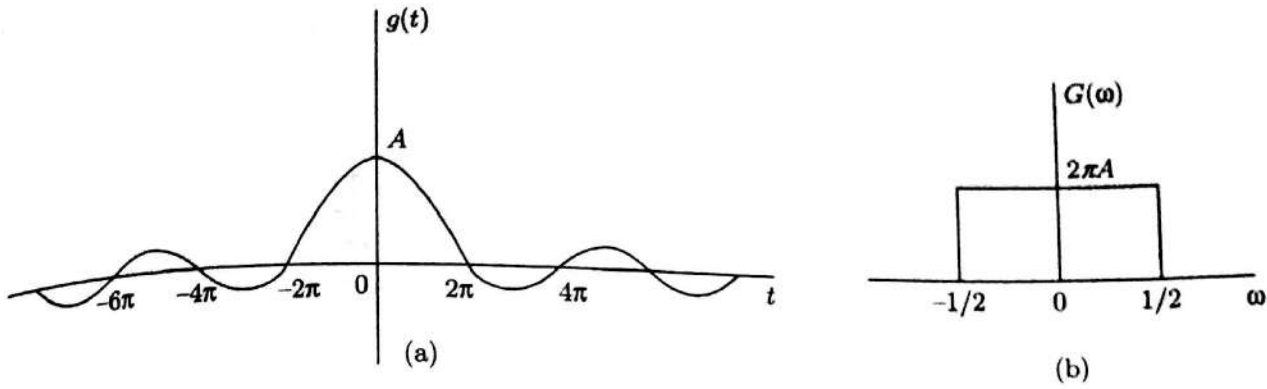


Fig. 5.20

Example 5.23 Find the Fourier transform $G(\omega)$ of the signal [Fig. 5.21(a)]

$$g(t) = 1, \quad \text{for all } t$$

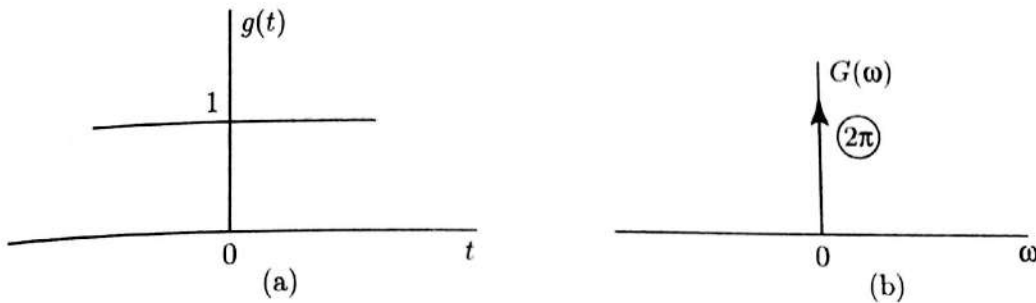


Fig. 5.21

Solution

Define $X(\omega) = 1$ for all ω , by replacing t with ω in the expression of $g(t)$. From Eq. (5.15), we have

$$\delta(t) \longleftrightarrow 1$$

$$x(t) \longleftrightarrow X(\omega)$$

$$x(t) = \delta(t) \quad \text{and} \quad X(\omega) = 1$$

$$x(-\omega) = \delta(-\omega) = \delta(\omega) \quad \text{and} \quad X(t) = 1$$

The duality property of the Fourier transform states that if

$$x(t) \longleftrightarrow X(\omega)$$

then

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

Therefore,

$$\mathcal{F}[1] = 2\pi\delta(\omega) \quad (5.50)$$

The Fourier transform $G(\omega)$ is shown in Fig. 5.21(b).

Example 5.24 Find the inverse Fourier transform of $u(\omega)$.

Solution

From Eq. (5.28), we have

$$u(t) \longleftrightarrow \pi\delta(\omega) + \frac{1}{j\omega}$$

Using the time-reversal property of the Fourier transform, we have

$$u(-t) \longleftrightarrow \pi\delta(-\omega) - \frac{1}{j\omega}$$

$$u(-t) \longleftrightarrow \pi\delta(\omega) - \frac{1}{j\omega}$$

$$x(t) \longleftrightarrow X(\omega)$$

$$x(t) = u(-t) \quad \text{and} \quad X(\omega) = \pi\delta(\omega) - \frac{1}{j\omega}$$

$$x(-\omega) = u(\omega) \quad \text{and} \quad X(t) = \pi\delta(t) - \frac{1}{jt}$$

The duality property of the Fourier transform states that if

$$x(t) \longleftrightarrow X(\omega)$$

then

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

$$\pi\delta(t) - \frac{1}{jt} \longleftrightarrow 2\pi u(\omega)$$

$$\frac{1}{2}\delta(t) - \frac{1}{j2\pi t} \longleftrightarrow u(\omega)$$

Therefore,

$$\mathcal{F}\left[\frac{1}{2}\delta(t) - \frac{1}{j2\pi t}\right] = u(\omega)$$

or

$$\mathcal{F}^{-1}[u(\omega)] = \frac{1}{2}\delta(t) - \frac{1}{j2\pi t} \quad (5.51)$$

5.4.12 Conjugation and Conjugate Symmetry

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$x^*(t) \longleftrightarrow X^*(-\omega) \quad (5.52)$$

Proof The Fourier transform of $x^*(t)$ is given by

$$\begin{aligned}\mathcal{F}[x^*(t)] &= \int_{-\infty}^{\infty} x^*(t) e^{-j\omega t} dt = \left[\int_{-\infty}^{\infty} x(t) e^{j\omega t} dt \right]^* \\ &= \left[\int_{-\infty}^{\infty} x(t) e^{-j(-\omega)t} dt \right]^* \\ &= [X(-\omega)]^* \\ \mathcal{F}[x^*(t)] &= X^*(-\omega)\end{aligned}$$

Case I If $x(t)$ is real, i.e., if

$$x^*(t) = x(t)$$

then

$$\begin{aligned}\mathcal{F}[x^*(t)] &= \mathcal{F}[x(t)] \\ X^*(-\omega) &= X(\omega)\end{aligned}$$

$$X(-\omega) = X^*(\omega) \quad (5.53)$$

the Fourier transform will be *conjugate symmetric*. Also if $x(t)$ is real and even, then from Eqs (5.34) and (5.53), we have

$$X(-\omega) = X^*(\omega) = X(\omega) \quad (5.54)$$

That is, if $x(t)$ is real and even, then so is its Fourier transform.

Case II If $x(t)$ is real and odd, then its Fourier transform is purely imaginary and odd. Using Eqs (5.35) and (5.53), we get

$$X(-\omega) = X^*(\omega) = -X(\omega) \quad (5.55)$$

Case III *Even and odd decomposition of real signals:* If $x(t)$ is real and

$$x(t) \longleftrightarrow X(\omega)$$

then

1.

$$x_e(t) = \mathcal{E}\{x(t)\} \longleftrightarrow \operatorname{Re}\{X(\omega)\} \quad (5.56)$$

The even part of a signal $x(t)$ is defined as

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

Taking the Fourier transform, we get

$$\begin{aligned}\mathcal{F}[x_e(t)] &= \mathcal{F}\left[\frac{1}{2}[x(t) + x(-t)]\right] = \frac{1}{2}(\mathcal{F}[x(t)] + \mathcal{F}[x(-t)]) = \frac{1}{2}(X(\omega) + X^*(\omega)) \\ &= \frac{1}{2}2\text{Re}\{X(\omega)\}\end{aligned}$$

$$\mathcal{F}[x_e(t)] = \text{Re}\{X(\omega)\}$$

Therefore,

$$x_e(t) = \mathcal{E}\{x(t)\} \longleftrightarrow \text{Re}\{X(\omega)\}$$

and

2.

$$x_o(t) = \mathcal{O}\{x(t)\} \longleftrightarrow j\text{Im}\{X(\omega)\} \quad (5.57)$$

The odd part of a signal $x(t)$ is defined as

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

Taking the Fourier transform, we get

$$\begin{aligned}\mathcal{F}[x_o(t)] &= \mathcal{F}\left[\frac{1}{2}[x(t) - x(-t)]\right] = \frac{1}{2}(\mathcal{F}[x(t)] - \mathcal{F}[x(-t)]) \\ &= \frac{1}{2}(X(\omega) - X^*(\omega)) \\ &= \frac{1}{2}2j\text{Im}\{X(\omega)\} \\ \mathcal{F}[x_o(t)] &= j\text{Im}\{X(\omega)\}\end{aligned}$$

Therefore,

$$x_o(t) = \mathcal{O}\{x(t)\} \longleftrightarrow j\text{Im}\{X(\omega)\}$$

5.4.13 Area Under $x(t)$

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$\int_{-\infty}^{\infty} x(t) dt = X(0) \quad (5.58)$$

That is, the area of a signal $x(t)$ is equal to the value of its Fourier transform $X(\omega)$ at $\omega = 0$.

Proof By definition

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X(\omega) \Big|_{\omega=0} = X(0) = \int_{-\infty}^{\infty} x(t) dt$$

5.4.14 Area Under $X(\omega)$

If

$$x(t) \longleftrightarrow X(\omega)$$

then

$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) d\omega \quad (5.59)$$

That is, the value of the function $x(t)$ at $t = 0$ is equal to the area under its Fourier transform $X(\omega)/2\pi$.

Proof From Eq. (5.5), we have

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$x(t) \Big|_{t=0} = x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) d\omega$$

Example 5.25 Find the total area under the function

$$x(t) = 10 \operatorname{sinc} \left(\frac{t+4}{7} \right)$$

Solution

From Eq. (5.49), we have

$$\mathcal{F} \left[A \operatorname{sinc} \left(\frac{t}{2\pi} \right) \right] = 2\pi A \operatorname{rect}(\omega)$$

$$A \operatorname{sinc} \left(\frac{t}{2\pi} \right) \longleftrightarrow 2\pi A \operatorname{rect}(\omega)$$

Using the time-scaling property (choose time-scaling constant $a = 2\pi$) Eq. (5.32), we get

$$A \operatorname{sinc}(t) \longleftrightarrow A \operatorname{rect}\left(\frac{\omega}{2\pi}\right)$$

Substitute $A = 10$ in the above equation:

$$10 \operatorname{sinc}(t) \longleftrightarrow 10 \operatorname{rect}\left(\frac{\omega}{2\pi}\right)$$

Using the time-shifting property, we get

$$10 \operatorname{sinc}\left(t + \frac{4}{7}\right) \longleftrightarrow 10 \operatorname{rect}\left(\frac{\omega}{2\pi}\right) e^{j\frac{4}{7}\omega}$$

Again using the time-scaling property ($a = \frac{1}{7}$), we get

$$10 \operatorname{sinc}\left(\frac{1}{7}t + \frac{4}{7}\right) \longleftrightarrow 70 \operatorname{rect}\left(\frac{7\omega}{2\pi}\right) e^{j4\omega}$$

$$10 \operatorname{sinc}\left(\frac{t+4}{7}\right) \longleftrightarrow 70 \operatorname{rect}\left(\frac{7\omega}{2\pi}\right) e^{j4\omega}$$

From Eq. (5.58), we have

$$\int_{-\infty}^{\infty} 10 \operatorname{sinc}\left(\frac{t+4}{7}\right) dt = \left[70 \operatorname{rect}\left(\frac{7\omega}{2\pi}\right) e^{j4\omega} \right] \Big|_{\omega=0} = 70$$

5.4.15 Parseval's Relation

Let $x(t)$ be an energy signal and if

$$x(t) \longleftrightarrow X(\omega)$$

then

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega \quad (5.60)$$

Parseval's theorem states that the signal energies of an energy signal and its Fourier transform are equal.

proof Consider the LHS of Eq. (5.60).

$$\begin{aligned}
 E_x &= \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t)x^*(t) dt \\
 &= \int_{-\infty}^{\infty} x(t) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \right)^* dt \\
 &= \int_{-\infty}^{\infty} x(t) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega \right) dt \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) \left(\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right) d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) X(\omega) d\omega \\
 \int_{-\infty}^{\infty} |x(t)|^2 dt &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega
 \end{aligned}$$

■

5.5 FOURIER TRANSFORM FOR PERIODIC SIGNALS

We can construct the Fourier transform of a periodic signal directly from its Fourier series representation. The resulting transform consists of a train of impulses in the frequency domain, with the areas of the impulses proportional to the Fourier series coefficients.

Consider the periodic signal $x(t)$ with period T and fundamental frequency $\omega_0 = (2\pi/T)$. The signal $x(t)$ has the Fourier series representation

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

Taking the Fourier transform on both sides yields

$$\begin{aligned}
 \mathcal{F}[x(t)] &= \mathcal{F} \left[\sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} \right] \\
 X(\omega) &= \sum_{n=-\infty}^{\infty} X_n \mathcal{F}[e^{jn\omega_0 t}]
 \end{aligned}$$

Using Eq. (5.19), we have $\mathcal{F}[e^{jn\omega_0 t}] = 2\pi\delta(\omega - n\omega_0)$, and therefore

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} X_n [2\pi\delta(\omega - n\omega_0)] \\ &= 2\pi \sum_{n=-\infty}^{\infty} X_n \delta(\omega - n\omega_0) \\ X(\omega) &= 2\pi \sum_{n=-\infty}^{\infty} X_n \delta(\omega - n\omega_0) \end{aligned} \quad (5.61)$$

Thus, the Fourier transform of a periodic signal is simply an impulse train with impulses located at $\omega = n\omega_0$, each of which has a strength $2\pi X_n$, and all impulses are separated from each other by ω_0 . Note that because the signal $x(t)$ is periodic, the magnitude spectrum $|X(\omega)|$ is a train of impulses of strength $2\pi|X_n|$, whereas the spectrum obtained through the use of the Fourier series is a line spectrum with lines of finite magnitude $|X_n|$.

Example 5.26 Find and sketch the Fourier transform of the impulse train $x(t) = \delta_{T_0}(t)$ depicted in Fig. 5.22(a).

Solution

The impulse train $x(t)$ is periodic with period $T = T_0$ and fundamental frequency $\omega_0 = (2\pi/T_0)$.

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0) \quad (5.62)$$

The impulse train for one period may be written as

$$x(t) = \delta(t), \quad -T_0/2 < t < T_0/2$$

The exponential Fourier series coefficients are

$$\begin{aligned} X_n &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \\ X_n &= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt \end{aligned}$$

$$= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \delta(t) e^{-jn\omega_0 t} dt$$

$$X_n = \frac{1}{T_0}$$

From Eq. (5.61), the Fourier transform of a periodic signal $x(t)$ is

$$X(\omega) = 2\pi \sum_{n=-\infty}^{\infty} X_n \delta(\omega - n\omega_0)$$

Substituting $X_n = (1/T_0)$ into the above equation yields

$$X(\omega) = 2\pi \sum_{n=-\infty}^{\infty} \frac{1}{T_0} \delta(\omega - n\omega_0)$$

$$= \frac{2\pi}{T_0} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

$$X(\omega) = \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

Thus, the Fourier transform of a periodic impulse in the time domain with period T_0 is a periodic impulse train in the frequency domain with period $\omega_0 = (2\pi/T)$, as sketched in Fig. 5.22(b).

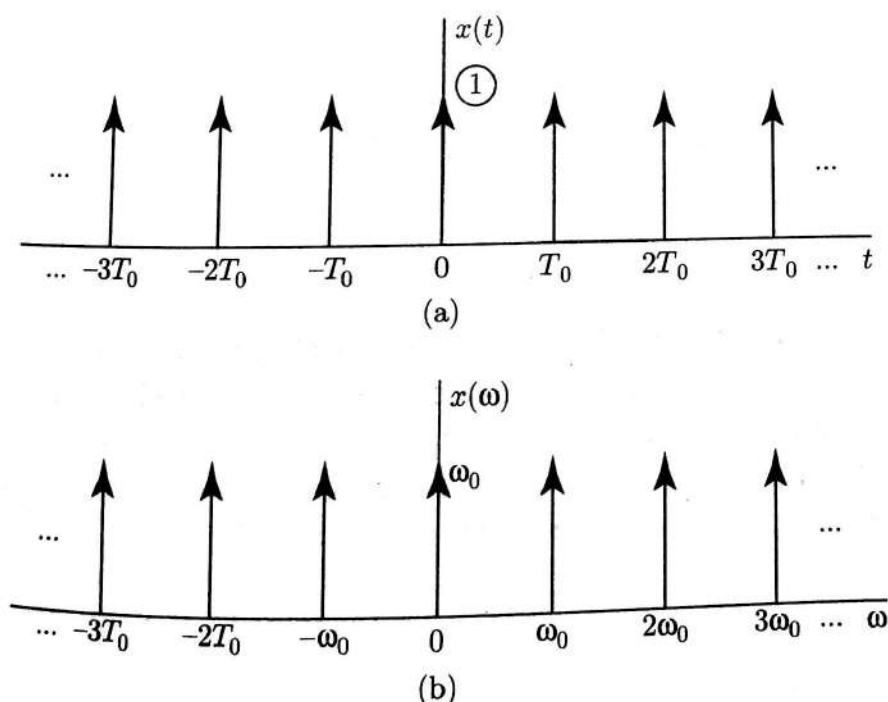


Fig. 5.22 (a) An impulse train and (b) its Fourier transform

5.6 SIGNAL TRANSMISSION THROUGH LTI SYSTEMS

If $x(t)$ and $y(t)$ are the input and output of an LTI system with impulse response $h(t)$, then

$$y(t) = x(t) * h(t)$$

Application of the time convolution property to the above equation yields [assuming that both $x(t)$ and $h(t)$ are Fourier transformable]

$$\mathcal{F}[y(t)] = \mathcal{F}[x(t)] \mathcal{F}[h(t)]$$

$$Y(\omega) = X(\omega)H(\omega) \quad (5.63)$$

or

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \quad (5.64)$$

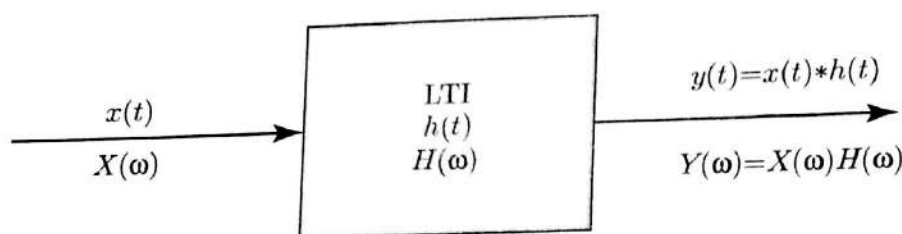


Fig. 5.23 Convolution property of LTI system response

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The use of the convolution property for LTI systems is demonstrated in Fig. 5.23. Equation (5.63) can be expressed in polar form as

$$|Y(\omega)| e^{j\angle Y(\omega)} = \left(|X(\omega)| e^{j\angle X(\omega)} \right) \left(|H(\omega)| e^{j\angle H(\omega)} \right)$$

$$|Y(\omega)| e^{j\angle Y(\omega)} = |X(\omega)| |H(\omega)| e^{j[\angle X(\omega) + \angle H(\omega)]}$$

Therefore,

$$|Y(\omega)| = |X(\omega)| |H(\omega)| \quad (5.65)$$

and

$$\angle Y(\omega) = \angle X(\omega) + \angle H(\omega) \quad (5.66)$$

During transmission, the input signal magnitude spectrum $|X(\omega)|$ is changed to $|X(\omega)| |H(\omega)|$. Similarly, the input signal phase spectrum $\angle X(\omega)$ is changed to $\angle X(\omega) + \angle H(\omega)$. An input signal spectral component of frequency ω is modified in amplitude by a factor $|H(\omega)|$ and is shifted in phase by an angle $\angle H(\omega)$. Clearly, $|H(\omega)|$ is the magnitude response, and $\angle H(\omega)$ is the phase response of the system. $H(\omega)$ is called the *frequency response* of the system.

Example 5.27 Consider an LTI system with impulse response

$$h(t) = e^{-at} u(t)$$

Find the response $y(t)$ of this system when the input is the unit step function, i.e., $x(t) = u(t)$.

Solution

From Eq. (5.63), we have

$$\begin{aligned} Y(\omega) &= X(\omega)H(\omega) = \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] \left(\frac{1}{a + j\omega} \right) \\ &= \frac{\pi}{a}\delta(\omega) + \frac{1}{j\omega(a + j\omega)} \\ &= \frac{\pi}{a}\delta(\omega) + \frac{1}{aj\omega} - \frac{1}{a(a + j\omega)} \\ Y(\omega) &= \frac{1}{a} \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] - \frac{1}{a} \frac{1}{a + j\omega} \end{aligned}$$

Taking the inverse Fourier transform of both sides results in

$$y(t) = \frac{1}{a}u(t) - \frac{1}{a}e^{-at}u(t) = \frac{1}{a}[1 - e^{-at}]u(t)$$

Example 5.28 The output of an LTI system in response to an input $x(t) = e^{-2t}u(t)$ is $y(t) = e^{-t}u(t)$. Find the frequency response and impulse response of this system.

Solution

We take the Fourier transform of $x(t)$ and $y(t)$, obtaining

$$X(\omega) = \frac{1}{2 + j\omega} \quad \text{and} \quad Y(\omega) = \frac{1}{1 + j\omega}$$

Now, we use Eq. (5.64) to obtain the frequency response $H(\omega)$:

$$\begin{aligned} H(\omega) &= \frac{Y(\omega)}{X(\omega)} = \frac{2 + j\omega}{1 + j\omega} = \frac{1 + 1 + j\omega}{1 + j\omega} \\ &= \frac{1 + j\omega}{1 + j\omega} + \frac{1}{1 + j\omega} \\ H(\omega) &= 1 + \frac{1}{1 + j\omega} \end{aligned}$$

The inverse Fourier transform of the above equation yields the impulse response $h(t)$ of the system:

$$h(t) = \delta(t) + e^{-t}u(t)$$

Example 5.29 Consider a stable LTI system characterized by the differential equation

$$\frac{dy(t)}{dt} + ay(t) = x(t), \quad a > 0$$

Find the frequency response $H(\omega)$ and the impulse response $h(t)$ of the system.

Solution

Given that

$$\frac{dy(t)}{dt} + ay(t) = x(t), \quad a > 0$$

Taking the Fourier transform of both sides of the above equation yields

$$j\omega Y(\omega) + aY(\omega) = X(\omega)$$

$$Y(\omega)(a + j\omega) = X(\omega)$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{a + j\omega}$$

Now take the inverse Fourier transform to obtain the impulse response of the system:

$$h(t) = e^{-at}u(t)$$

Example 5.30 Consider a stable LTI system characterized by the differential equation

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t)$$

- (a) Find the frequency response $H(\omega)$ and the impulse response $h(t)$ of the system.
 (b) What is the response of this system if the input $x(t) = e^{-t}u(t)$?

Solution

(a) Given that

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t)$$

Taking the Fourier transform on both sides yields

$$(j\omega)^2 Y(\omega) + 4j\omega Y(\omega) + 3Y(\omega) = j\omega X(\omega) + 2X(\omega)$$

$$Y(\omega)((j\omega)^2 + 4j\omega + 3) = X(\omega)(2 + j\omega)$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{2 + j\omega}{(j\omega)^2 + 4j\omega + 3} = \frac{2 + j\omega}{(1 + j\omega)(3 + j\omega)}$$

Using partial fraction expansion, we get

$$H(\omega) = \frac{1}{2} \frac{1}{1 + j\omega} + \frac{1}{2} \frac{1}{3 + j\omega}$$

The inverse Fourier transform of the above equation yields the impulse response $h(t)$ of the system:

$$\mathcal{F}^{-1}[H(\omega)] = h(t) = \frac{1}{2} e^{-t}u(t) + \frac{1}{2} e^{-3t}u(t)$$

(b) From Eq. (5.63), we have

$$Y(\omega) = X(\omega)H(\omega) = \left[\frac{1}{1+j\omega} \right] \left[\frac{2+j\omega}{(1+j\omega)(3+j\omega)} \right] = \frac{2+j\omega}{(1+j\omega)^2(3+j\omega)}$$

Using partial fraction expansion, we get

$$Y(\omega) = \frac{1}{4} \frac{1}{1+j\omega} + \frac{1}{2} \frac{1}{(1+j\omega)^2} - \frac{1}{4} \frac{1}{3+j\omega}$$

The inverse Fourier transform for each term in the above equation can be obtained by inspection. The first and third terms are of the same type that we have encountered in the previous part. To obtain the inverse Fourier transform of the second term, put $a = 1$ into the Eq. (5.43). Therefore, the response of the system is given by

$$\begin{aligned} \mathcal{F}^{-1}[Y(\omega)] &= y(t) = \frac{1}{4} \mathcal{F}^{-1} \left[\frac{1}{1+j\omega} \right] + \frac{1}{2} \mathcal{F}^{-1} \left[\frac{1}{(1+j\omega)^2} \right] - \frac{1}{4} \mathcal{F}^{-1} \left[\frac{1}{3+j\omega} \right] \\ y(t) &= \left[\frac{1}{4} e^{-t} + \frac{1}{2} t e^{-t} - \frac{1}{4} e^{-3t} \right] u(t) \end{aligned}$$

Example 5.31 Consider a causal LTI system with frequency response

$$H(\omega) = \frac{1}{3+j\omega}$$

For a particular input $x(t)$, this system is observed to produce the output

$$y(t) = e^{-3t}u(t) - e^{-4t}u(t)$$

Determine $x(t)$.

Solution

Consider the given output

$$y(t) = e^{-3t}u(t) - e^{-4t}u(t)$$

Taking the Fourier transform gives

$$Y(\omega) = \frac{1}{3+j\omega} - \frac{1}{4+j\omega} = \frac{1}{(3+j\omega)(4+j\omega)}$$

We know that the frequency response of an LTI system is given by

$$\begin{aligned} H(\omega) &= \frac{Y(\omega)}{X(\omega)} \\ X(\omega) &= \frac{Y(\omega)}{H(\omega)} = \frac{(3+j\omega)}{(3+j\omega)(4+j\omega)} = \frac{1}{4+j\omega} \end{aligned}$$

Taking the inverse Fourier transform yields

$$x(t) = e^{-4t}u(t)$$

Example 5.32 A causal and stable LTI system S has the frequency response

$$H(\omega) = \frac{4 + j\omega}{6 - \omega^2 + 5j\omega}$$

- (a) Determine a differential equation relating the input $x(t)$ and output $y(t)$ of S .
 (b) Determine the impulse response $h(t)$ of S .
 (c) What is the output of S when the input is

$$x(t) = e^{-4t}u(t) - te^{-4t}u(t)$$

Solution

- (a) Consider the given frequency response

$$H(\omega) = \frac{4 + j\omega}{6 - \omega^2 + 5j\omega}$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{4 + j\omega}{6 + (j\omega)^2 + 5j\omega}$$

$$(j\omega)^2 Y(\omega) + 5j\omega Y(\omega) + 6Y(\omega) = j\omega X(\omega) + 4X(\omega)$$

Taking the inverse Fourier transform to obtain the differential equation relating the input $x(t)$ and output $y(t)$ of S :

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + 4x(t)$$

- (b) Given that

$$H(\omega) = \frac{4 + j\omega}{6 + (j\omega)^2 + 5j\omega} = \frac{4 + j\omega}{(2 + j\omega)(3 + j\omega)}$$

Using partial fraction expansion, we get

$$H(\omega) = \frac{2}{2 + j\omega} - \frac{1}{3 + j\omega}$$

The inverse Fourier transform yields the impulse response $h(t)$ of the system:

$$h(t) = 2e^{-2t}u(t) - e^{-3t}u(t)$$

- (c) Consider the given input

$$x(t) = e^{-4t}u(t) - te^{-4t}u(t)$$

Taking the Fourier transform yields

$$X(\omega) = \frac{1}{4 + j\omega} - \frac{1}{(4 + j\omega)^2} = \frac{3 + j\omega}{(4 + j\omega)^2}$$

From Eq. (5.63), we have

$$Y(\omega) = X(\omega)H(\omega) = \frac{1}{(2+j\omega)(4+j\omega)} = \frac{1}{2} \frac{1}{2+j\omega} - \frac{1}{2} \frac{1}{4+j\omega}$$

Taking the inverse Fourier transform yields

$$\begin{aligned} y(t) &= \frac{1}{2} e^{-2t} u(t) - \frac{1}{2} e^{-4t} u(t) \\ &= \frac{1}{2} (e^{-2t} - e^{-4t}) u(t) \end{aligned}$$

5.6.1 Linear and Nonlinear Phase

Consider a continuous-time LTI system with impulse response $h(t)$ and frequency response $H(\omega)$. Let a signal $x(t)$ with Fourier transform $X(\omega)$ be applied to the input of the system. Let a signal $y(t)$ with Fourier transform $Y(\omega)$ denote the output of the system. In several applications, such as signal amplification or message signal transmission over a communication channel, we require that the output waveform be a replica of the input waveform, i.e., distortionless transmission. Transmission is said to be distortionless if the input and the output have identical waveshapes within a multiplicative constant and a constant time delay. Thus, in distortionless transmission, the input $x(t)$ and the output $y(t)$ satisfy the condition (Fig. 5.24):

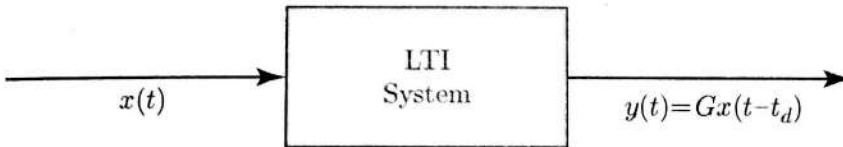


Fig. 5.24 Time-domain condition for distortionless transmission of a signal through an LTI system

$$y(t) = Gx(t - t_d) \quad (5.67)$$

where the constant G accounts for a change in amplitude and the constant t_d accounts for a delay in transmission. Taking the Fourier transform of the above equation yields

$$Y(\omega) = GX(\omega) e^{-j\omega t_d}$$

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = G e^{-j\omega t_d}$$

This is the frequency response required of a system for distortionless transmission. From this equation it follows that

1. the magnitude response $|H(\omega)|$ must be a constant, i.e., we must have

$$|H(\omega)| = G \quad (5.68)$$

for some constant G .

2. the phase response $\angle H(\omega)$ must be a linear function of ω with slope $-t_d$ and intercept zero, i.e., we must have

$$\angle H(\omega) = -\omega t_d \quad (5.69)$$

These two conditions are illustrated in Figs 5.25(a) and (b), respectively.

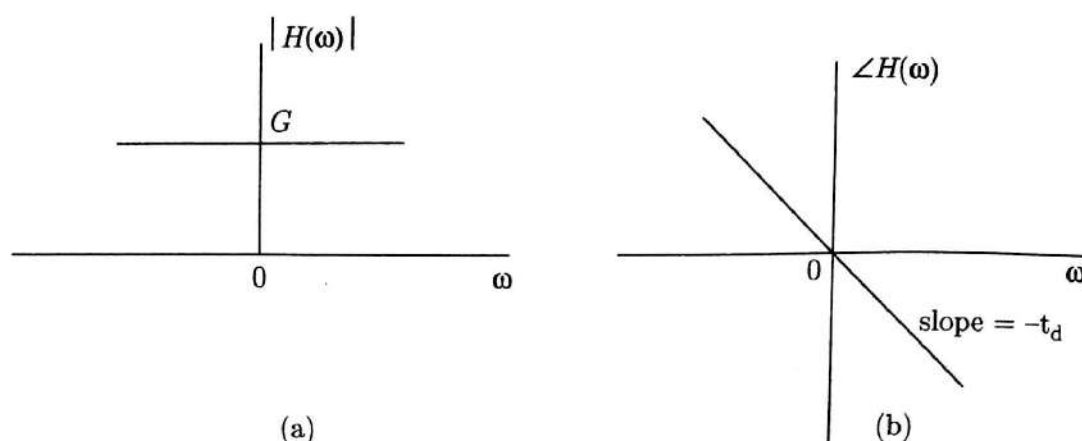


Fig. 5.25 Frequency response for distortionless transmission through an LTI system: (a) magnitude response and (b) phase response

The gain $|H(\omega)| = G$ means that every spectral component is multiplied by a constant G . A linear phase $\angle H(\omega) = -\omega t_d$ means that every spectral component is delayed by t_d seconds. For distortionless transmission, we require a linear phase characteristic. The phase is not only a linear function of ω , but it should also pass through the origin $\omega = 0$. Because each spectral component is attenuated by the same factor (G) and delayed by exactly the same amount (t_d), the output signal is an exact replica of the input (except for attenuating factor G and delay t_d).

5.6.2 Phase Delay and Group Delay

The two important parameters that characterize the form of the response $y(t)$ of an LTI system excited by an input signal $x(t)$ composed of a weighted sum of sinusoidal signals are phase delay and group delay. These two parameters are associated with the frequency response $H(\omega)$ of the system. For a linear phase system (whose phase varies linearly with frequency), both the phase delay and the group delay are constant.

Phase Delay

The time delay experienced by a single-frequency signal (i.e., a sinusoidal signal) when the signal passes through a system is referred to as the system phase delay. Now, let us assume that the input signal is the single-frequency signal

$$x(t) = A \cos(\omega_0 t + \theta)$$

$$x(t) = \frac{A}{2} e^{j(\omega_0 t + \theta)} + \frac{A}{2} e^{-j(\omega_0 t + \theta)}$$

and the system frequency response is

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$$

Then, the system output signal is [Eq. (3.64)]

$$y(t) = \frac{A}{2} |H(\omega_0)| e^{j(\omega_0 t + \theta + \angle H(\omega_0))} + \frac{A}{2} |H(-\omega_0)| e^{-j(\omega_0 t + \theta - \angle H(-\omega_0))}$$

Since $|H(\omega_0)| = |H(-\omega_0)|$ and $\angle H(\omega_0) = -\angle H(-\omega_0)$, we have

$$y(t) = \frac{A}{2} |H(\omega_0)| e^{j(\omega_0 t + \theta + \angle H(\omega_0))} + \frac{A}{2} |H(\omega_0)| e^{-j(\omega_0 t + \theta - \angle H(\omega_0))}$$

$$\begin{aligned} y(t) &= A |H(\omega_0)| \cos(\omega_0 t + \angle H(\omega_0) + \theta) \\ &= A |H(\omega_0)| \cos\left(\omega_0 \left(t + \frac{\angle H(\omega_0)}{\omega_0}\right) + \theta\right) \end{aligned}$$

$$y(t) = A |H(\omega_0)| \cos(\omega_0(t - \tau_p(\omega_0)) + \theta)$$

where

$$\tau_p(\omega_0) = -\frac{\angle H(\omega_0)}{\omega_0} \quad (5.70)$$

is the time delay experienced by the single-frequency signal with frequency ω_0 when it passes through the system. Thus, $\tau_p(\omega_0)$ is the system phase delay for a signal with frequency ω_0 . A negative system phase response at positive frequencies indicates that a signal is delayed in time when it passes through the system, whereas a positive system phase response at positive frequencies indicates that a signal is advanced in time when it passes through the system.

Group Delay

When the input signal contains many sinusoidal components with different frequencies that are not harmonically related, each component will go through different phase delays when processed by a frequency-selective LTI system, and the signal delay is determined using a different parameter called the group delay:

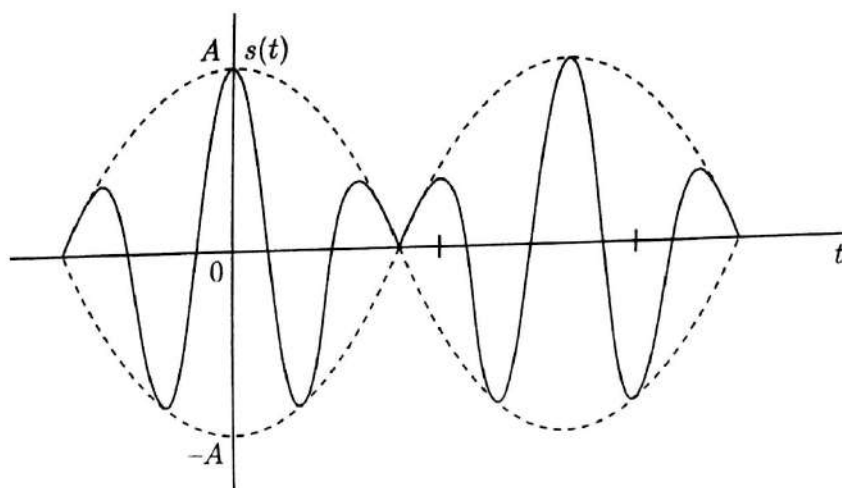
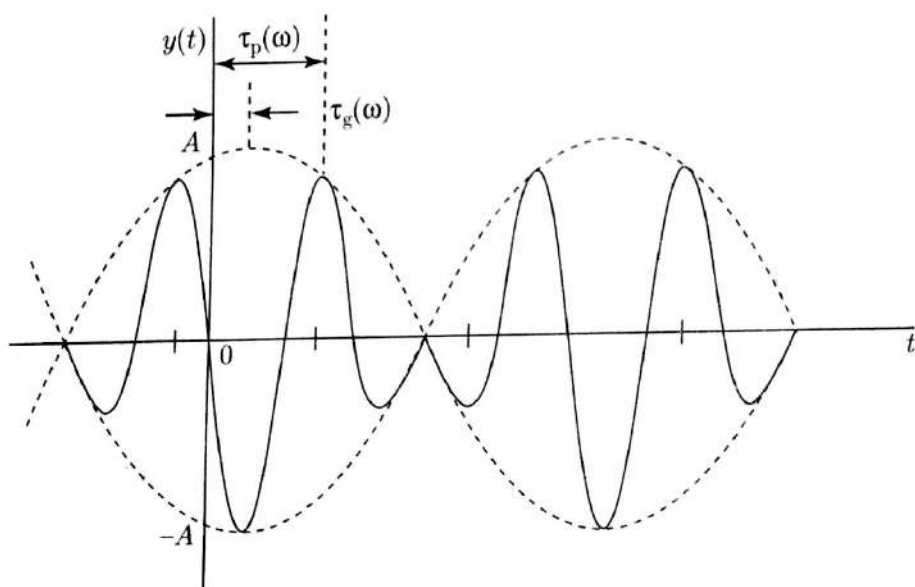
$$\tau_g(\omega) = -\frac{d\angle H(\omega)}{d\omega} \quad (5.71)$$

We now derive Eq. (5.71) that defines the group delay by using a single-frequency modulating and carrier signals with zero phase for simplicity.

The assumed input signal (double-side band-suppressed carrier, i.e., DSB-SC modulated signal) is

$$s(t) = A \cos(\omega_m t) \cos(\omega_c t) \quad (5.72)$$

where ω_c is the carrier frequency and ω_m is the frequency of the modulating signal. This input signal is illustrated in Fig. 5.26.

**Fig. 5.26** Input signal**Fig. 5.27** Output signal

We use the cosine-product trigonometric identity to rewrite the input signal as

$$s(t) = \frac{A}{2} \cos[(\omega_c + \omega_m)t] + \frac{A}{2} \cos[(\omega_c - \omega_m)t]$$

$$s(t) = \frac{A}{2} \cos(\omega_1 t) + \frac{A}{2} \cos(\omega_2 t)$$

where

$$\omega_1 = \omega_c + \omega_m \quad (5.73)$$

and

$$\omega_2 = \omega_c - \omega_m \quad (5.74)$$

Now let the signal $s(t)$ passed through the system with frequency response $H(\omega)$. We assume that the system magnitude response and phase response are $|H(\omega)| = 1$ and $\angle H(\omega) = \phi(\omega)$, respectively. The system output signal is

$$y(t) = \frac{A}{2}|H(\omega_1)| \cos[\omega_1 t + \angle H(\omega_1)] + \frac{A}{2}|H(\omega_2)| \cos[\omega_2 t + \angle H(\omega_2)]$$

$$y(t) = \frac{A}{2} \cos[\omega_1 t + \phi(\omega_1)] + \frac{A}{2} \cos[\omega_2 t + \phi(\omega_2)]$$

where $\phi(\omega_1)$ and $\phi(\omega_2)$ are the phase shifts produced by the system at frequencies ω_1 and ω_2 , respectively. Equivalently, we may express $y(t)$ as

$$\begin{aligned} y(t) &= \frac{A}{2} \cos \left(\omega_c t + \omega_m t + \frac{\phi(\omega_1) + \phi(\omega_2)}{2} + \frac{\phi(\omega_1) - \phi(\omega_2)}{2} \right) \\ &\quad + \frac{A}{2} \cos \left(\omega_c t - \omega_m t + \frac{\phi(\omega_1) + \phi(\omega_2)}{2} - \frac{\phi(\omega_1) - \phi(\omega_2)}{2} \right) \\ &= \frac{A}{2} \cos \left(\left[\omega_c t + \frac{\phi(\omega_1) + \phi(\omega_2)}{2} \right] + \left[\omega_m t + \frac{\phi(\omega_1) - \phi(\omega_2)}{2} \right] \right) \\ &\quad + \frac{A}{2} \cos \left(\left[\omega_c t + \frac{\phi(\omega_1) + \phi(\omega_2)}{2} \right] - \left[\omega_m t + \frac{\phi(\omega_1) - \phi(\omega_2)}{2} \right] \right) \\ y(t) &= A \cos \left(\omega_c t + \frac{\phi(\omega_1) + \phi(\omega_2)}{2} \right) \cos \left(\omega_m t + \frac{\phi(\omega_1) - \phi(\omega_2)}{2} \right) \\ y(t) &= A \cos \left(\omega_c \left[t + \frac{\phi(\omega_1) + \phi(\omega_2)}{2\omega_c} \right] \right) \cos \left(\omega_m \left[t + \frac{\phi(\omega_1) - \phi(\omega_2)}{2\omega_m} \right] \right) \\ y(t) &= A \cos \left(\omega_c [t - \tau_p(\omega_c)] \right) \cos \left(\omega_m [t - \tau_g(\omega_c)] \right) \end{aligned} \quad (5.75)$$

Comparing the output signal $y(t)$ in Eq. (5.75) with the input signal $s(t)$ in Eq. (5.72), we make the following two observations:

1. The carrier component at frequency ω_c in $y(t)$ lags its counterpart in $s(t)$ by $(\phi(\omega_1) + \phi(\omega_2))/2$, which represents a time delay

$$\tau_p(\omega_c) = -\frac{\phi(\omega_1) + \phi(\omega_2)}{2\omega_c} = -\frac{\phi(\omega_1) + \phi(\omega_2)}{\omega_1 + \omega_2} \quad (5.76)$$

2. The modulating signal component at frequency ω_m in $y(t)$ lags its counterpart in $s(t)$ by $(\phi(\omega_1) - \phi(\omega_2))/2$, which represents a time delay

$$\tau_g(\omega_c) = -\frac{\phi(\omega_1) - \phi(\omega_2)}{2\omega_m} = -\frac{\phi(\omega_1) - \phi(\omega_2)}{\omega_1 - \omega_2} \quad (5.77)$$

Suppose that the modulating signal frequency ω_m is small compared with the carrier frequency ω_c , which implies that the frequencies ω_1 and ω_2 are close together, with ω_c

between them. Such a modulated signal is known as narrowband signal. Then we may approximate the phase response $\phi(\omega)$ in the vicinity of $\omega = \omega_c$ by the two-term Taylor expansion

$$\phi(\omega) = \phi(\omega_c) + \left. \frac{d\phi(\omega)}{d\omega} \right|_{\omega=\omega_c} (\omega - \omega_c) \quad (5.78)$$

Evaluating $\phi(\omega_1)$ and $\phi(\omega_2)$ using Eq. (5.78), we get

$$\phi(\omega_1) = \phi(\omega_c) + \left. \frac{d\phi(\omega)}{d\omega} \right|_{\omega=\omega_c} (\omega_1 - \omega_c) \quad (5.79)$$

and

$$\phi(\omega_2) = \phi(\omega_c) + \left. \frac{d\phi(\omega)}{d\omega} \right|_{\omega=\omega_c} (\omega_2 - \omega_c) \quad (5.80)$$

Now, from Eqs (5.73), (5.74), (5.76), (5.79), and (5.80), we get

$$\tau_p(\omega_c) = -\frac{\phi(\omega_c)}{\omega_c} \quad (5.81)$$

Similarly using the Eqs (5.73), (5.74), (5.77), (5.79), and (5.80), we get

$$\tau_g(\omega_c) = -\left. \frac{d\phi(\omega)}{d\omega} \right|_{\omega=\omega_c} = -\left. \frac{d\angle H(\omega)}{d\omega} \right|_{\omega=\omega_c} \quad (5.82)$$

The time delay $\tau_g(\omega)$ is called the group delay or envelope delay. Thus, the group delay at each frequency equals the negative of the slope of the phase at that frequency. If $\tau_g(\omega)$ is constant, all the components are delayed by the same interval. The group delay and phase delay are illustrated by the system output signal shown in Fig. 5.27.

Example 5.33 Find the impulse response of the system whose frequency response is given by

$$|H(\omega)| = \begin{cases} 1, & -\omega_c < \omega < \omega_c \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad \angle H(\omega) = \begin{cases} \frac{\pi}{2}, & \omega > 0 \\ -\frac{\pi}{2}, & \omega < 0 \end{cases}$$

Solution

Using the magnitude response $|H(\omega)|$ and phase response $\angle H(\omega)$, the frequency response of the system is given by

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$$

Substituting the values of $|H(\omega)|$ and $\angle H(\omega)$ into the above equation yields

$$H(\omega) = \begin{cases} e^{j(\pi/2)}, & 0 < \omega < \omega_c \\ e^{-j(\pi/2)}, & -\omega_c < \omega < 0 \end{cases}$$

$$H(\omega) = \begin{cases} j, & 0 < \omega < \omega_c \\ -j, & -\omega_c < \omega < 0 \end{cases}$$

$$H(\omega) = j \left[\text{rect} \left(\frac{\omega}{\omega_c} - \frac{1}{2} \right) - \text{rect} \left(\frac{\omega}{\omega_c} + \frac{1}{2} \right) \right] \quad (5.83)$$

We know that [Eq. (5.49)]

$$A \text{sinc} \left(\frac{t}{2\pi} \right) \longleftrightarrow 2\pi A \text{rect}(\omega)$$

$$\frac{\sin(t/2)}{(t/2)} \longleftrightarrow 2\pi \text{rect}(\omega)$$

$$\frac{\sin(t/2)}{\pi t} \longleftrightarrow \text{rect}(\omega)$$

Using frequency-shifting property, we have

$$\frac{\sin(t/2)}{\pi t} e^{j(1/2)t} \longleftrightarrow \text{rect} \left(\omega - \frac{1}{2} \right)$$

Now, using the time-scaling property, we have

$$\frac{\sin(\omega_c t/2)}{\pi \omega_c t} e^{j(\omega_c/2)t} \longleftrightarrow \frac{1}{\omega_c} \text{rect} \left(\frac{\omega}{\omega_c} - \frac{1}{2} \right)$$

$$\frac{\sin(\omega_c t/2)}{\pi t} e^{j(\omega_c/2)t} \longleftrightarrow \text{rect} \left(\frac{\omega}{\omega_c} - \frac{1}{2} \right)$$

Similarly, we have

$$\frac{\sin(\omega_c t/2)}{\pi t} e^{-j(\omega_c/2)t} \longleftrightarrow \text{rect} \left(\frac{\omega}{\omega_c} + \frac{1}{2} \right)$$

Now, taking the inverse Fourier transform of Eq. (5.83) and using the above results gives

$$h(t) = j \left[\frac{\sin(\omega_c t/2)}{\pi t} e^{j(\omega_c/2)t} - \frac{\sin(\omega_c t/2)}{\pi t} e^{-j(\omega_c/2)t} \right]$$

$$= 2j^2 \frac{\sin(\omega_c t/2)}{\pi t} \left[\frac{e^{j(\omega_c/2)t} - e^{-j\omega_c/2t}}{2j} \right]$$

$$h(t) = -2 \frac{\sin(\omega_c t/2)}{\pi t} \sin \left(\frac{\omega_c t}{2} \right) = -2 \frac{\sin^2(\omega_c t/2)}{\pi t}$$

Example 5.34 Consider the following frequency response for a causal and stable LTI system:

$$H(\omega) = \frac{1 - j\omega}{1 + j\omega}$$

- (a) Show that $|H(\omega)| = A$, and determine the value of A .
 (b) Determine which of the following statements is true about $\tau_g(\omega)$, the group delay.
- (i) $\tau_g(\omega) = 0$, for $\omega > 0$
 - (ii) $\tau_g(\omega) > 0$, for $\omega > 0$
 - (iii) $\tau_g(\omega) < 0$, for $\omega > 0$

Solution

- (a) Given that

$$H(\omega) = \frac{1 - j\omega}{1 + j\omega}$$

$$|H(\omega)| = \frac{\sqrt{1 + \omega^2}}{\sqrt{1 + \omega^2}} = 1$$

Therefore $A = 1$.

- (b) The phase response of the system is

$$\begin{aligned}\angle H(\omega) &= \tan^{-1}(-\omega) - \tan^{-1}(\omega) \\ &= -2 \tan^{-1}(\omega)\end{aligned}$$

Therefore, the group delay is

$$\tau_g(\omega) = -\frac{d\angle(H(\omega))}{d\omega} = \frac{d}{d\omega} 2 \tan^{-1}(\omega) = \frac{2}{1 + \omega^2}$$

Since $\tau_g(\omega) > 0$ for $\omega > 0$, statement (ii) is true.

Example 5.35 A causal LTI system has the frequency response $H(\omega)$ shown in Fig. 5.28. For each of the input signals given below, determine the filtered output signal $y(t)$.

- (a) $x(t) = e^{jt}$
- (b) $x(t) = \sin(\omega_0 t)u(t)$
- (c) $X(\omega) = \frac{1}{j\omega(6 + j\omega)}$
- (d) $X(\omega) = \frac{1}{2 + j\omega}$

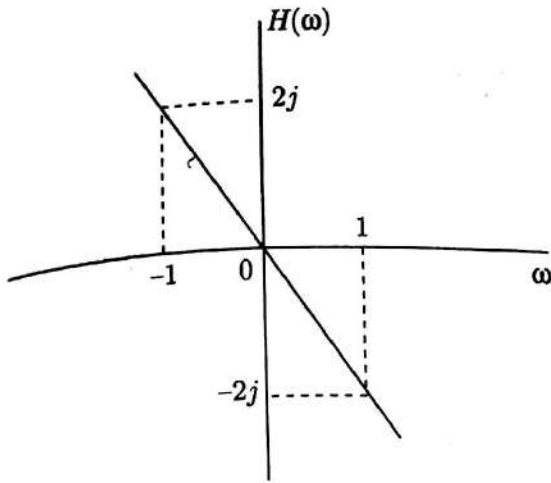


Fig. 5.28 Frequency response

Solution

From Fig. 5.28, the frequency response of the filter can be expressed as

$$H(\omega) = -2j\omega$$

We know that the filtered output signal is given by [Eq. (5.63)]

$$Y(\omega) = H(\omega)X(\omega)$$

$$Y(\omega) = -2j\omega X(\omega)$$

Taking the inverse Fourier transform gives

$$y(t) = -2 \frac{dx(t)}{dt}$$

@sTc

(a) Given that

$$x(t) = e^{jt}$$

Therefore,

$$y(t) = -2 \frac{dx(t)}{dt} = -2 \frac{d}{dt} e^{jt} = -2j e^{jt}$$

(b) Given that

$$x(t) = \sin(\omega_0 t)u(t)$$

Therefore,

$$\begin{aligned} y(t) &= -2 \frac{dx(t)}{dt} = -2 \frac{d}{dt} \sin(\omega_0 t)u(t) \\ &= -2[\omega_0 \cos(\omega_0 t)u(t) + \sin(\omega_0 t)\delta(t)] \\ &= -2\omega_0 \cos(\omega_0 t)u(t) \end{aligned}$$

(c) We know that

$$\begin{aligned} Y(\omega) &= H(\omega)X(\omega) \\ &= \frac{-2j\omega}{j\omega(6+j\omega)} \\ Y(\omega) &= \frac{-2}{6+j\omega} \end{aligned}$$

Taking the inverse Fourier transform, we obtain

$$y(t) = -2e^{-6t}u(t)$$

(d) Given that

$$X(\omega) = \frac{1}{2+j\omega}$$

Taking the inverse Fourier transform, we obtain

$$x(t) = e^{-2t}u(t)$$

The filtered output signal is given by

$$\begin{aligned} y(t) &= -2 \frac{dx(t)}{dt} = -2 \frac{d}{dt} e^{-2t}u(t) \\ &= 4e^{-2t}u(t) - 2e^{-2t}\delta(t) \\ &= 4e^{-2t}u(t) - 2\delta(t) \end{aligned}$$

Example 5.36 Consider a continuous-time low-pass filter whose impulse response $h(t)$ is known to be real and whose frequency response magnitude is given as

$$|H(\omega)| = \begin{cases} 1, & |\omega| \leq 200\pi \\ 0, & \text{otherwise} \end{cases}$$

- (a) Determine and sketch the real-valued impulse response $h(t)$ for this filter when the corresponding group delay function is specified as
 (i) $\tau_g(\omega) = 5$ (ii) $\tau_g(\omega) = 5/2$ (iii) $\tau_g(\omega) = -5/2$
 (b) If the impulse response $h(t)$ would had not been specified to be real, would knowledge of $|H(\omega)|$ and $\tau_g(\omega)$ be sufficient to determine $h(t)$ uniquely? Justify your answer.

Solution

We know that

$$\tau_g(\omega) = -\frac{d\angle H(\omega)}{d\omega}$$

If $\tau_g(\omega) = \alpha$, where α is a constant, then

$$\alpha = -\frac{d\angle H(\omega)}{d\omega}$$

$$\angle H(\omega) = -\alpha\omega + \beta$$

where β is another constant.

(a) We know that if $h(t)$ is real, then $\angle H(\omega)$ has to be an odd function. Therefore, the value of $\beta = 0$. Let us define $H_0(\omega) = |H(\omega)|$. Then

$$|H(\omega)| = H_0(\omega) = \begin{cases} 1, & |\omega| \leq 200\pi \\ 0, & \text{otherwise} \end{cases}$$

Taking the inverse Fourier transform, we have

$$h_0(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H_0(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-200\pi}^{200\pi} e^{j\omega t} d\omega = \frac{\sin(200\pi t)}{\pi t}$$

(i) Given that $\tau_g(\omega) = \alpha = 5$. Hence, $\angle H(\omega) = -5\omega$. Then

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$$

$$= H_0(\omega) e^{-5j\omega}$$

Taking the inverse Fourier transform, we get

$$h(t) = h_0(t - 5) = \frac{\sin(200\pi(t - 5))}{\pi(t - 5)}$$

(ii) Given that $\tau_g(\omega) = \alpha = 5/2$. Hence, $\angle H(\omega) = -(5/2)\omega$. Then

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$$

$$= H_0(\omega) e^{-j(5/2)\omega}$$

Taking the inverse Fourier transform, we get

$$h(t) = h_0\left(t - \frac{5}{2}\right) = \frac{\sin(200\pi(t - 5/2))}{\pi(t - 5/2)}$$

(iii) Given that $\tau_g(\omega) = \alpha = 5/2$. Hence, $\angle H(\omega) = (5/2)\omega$. Then

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$$

$$= H_0(\omega) e^{j(5/2)\omega}$$

Taking the inverse Fourier transform, we get

$$h(t) = h_0\left(t + \frac{5}{2}\right) = \frac{\sin(200\pi(t + 5/2))}{\pi(t + 5/2)}$$

(b) If the impulse response $h(t)$ is not specified to be real, then $\angle H(\omega)$ does not have to be odd function. Therefore, $\beta \neq 0$. Knowledge of $|H(\omega)|$ and $\tau_g(\omega)$ would not be sufficient to determine β uniquely. Therefore, $h(t)$ cannot be determined uniquely.

5.7 IDEAL AND PRACTICAL FILTERS

Filtering is a process by which the essential and useful part of a signal is separated from the extraneous and undesirable components. The idea of filtering using LTI system is based on the convolution property of the Fourier transform, which states that for LTI systems, the Fourier transform of the output $Y(\omega)$ is the product of the Fourier transform of the input $X(\omega)$ and the frequency response $H(\omega)$, i.e., $Y(\omega) = X(\omega)H(\omega)$. In a sense, $H(\omega)$ acts as a weighting function or spectral-shaping function to the different frequency components in the input signal. Ideal filters allow distortionless transmission of a certain band of frequencies and completely suppress the remaining frequencies. The passband of a filter is the band of frequencies that are passed by the system, while the stopband refers to the range of frequencies that are attenuated by the system. In the ideal case, $|H(\omega)| = 1$ in a passband, while $|H(\omega)| = 0$ in a stopband. Filters are usually classified according to their frequency-domain characteristics as low-pass, high-pass, bandpass, bandstop or band elimination, and all-pass filters. The following lists definitions for each of these:

1. low-pass filters are those characterized by a passband that extends from $\omega = 0$ to $\omega = \omega_c$, where ω_c is called the cutoff frequency of the filter [Fig. 5.29(a)]. A low-pass filter attenuates high-frequency components of the input and passes the lower frequency components.
2. Highpass filters are characterized by a stopband that extends from $\omega = 0$ to $\omega = \omega_c$ and a passband that extends from $\omega = \omega_c$ to infinity [Fig. 5.29(b)]. A high-pass filter attenuates low frequencies and passes the high frequencies.

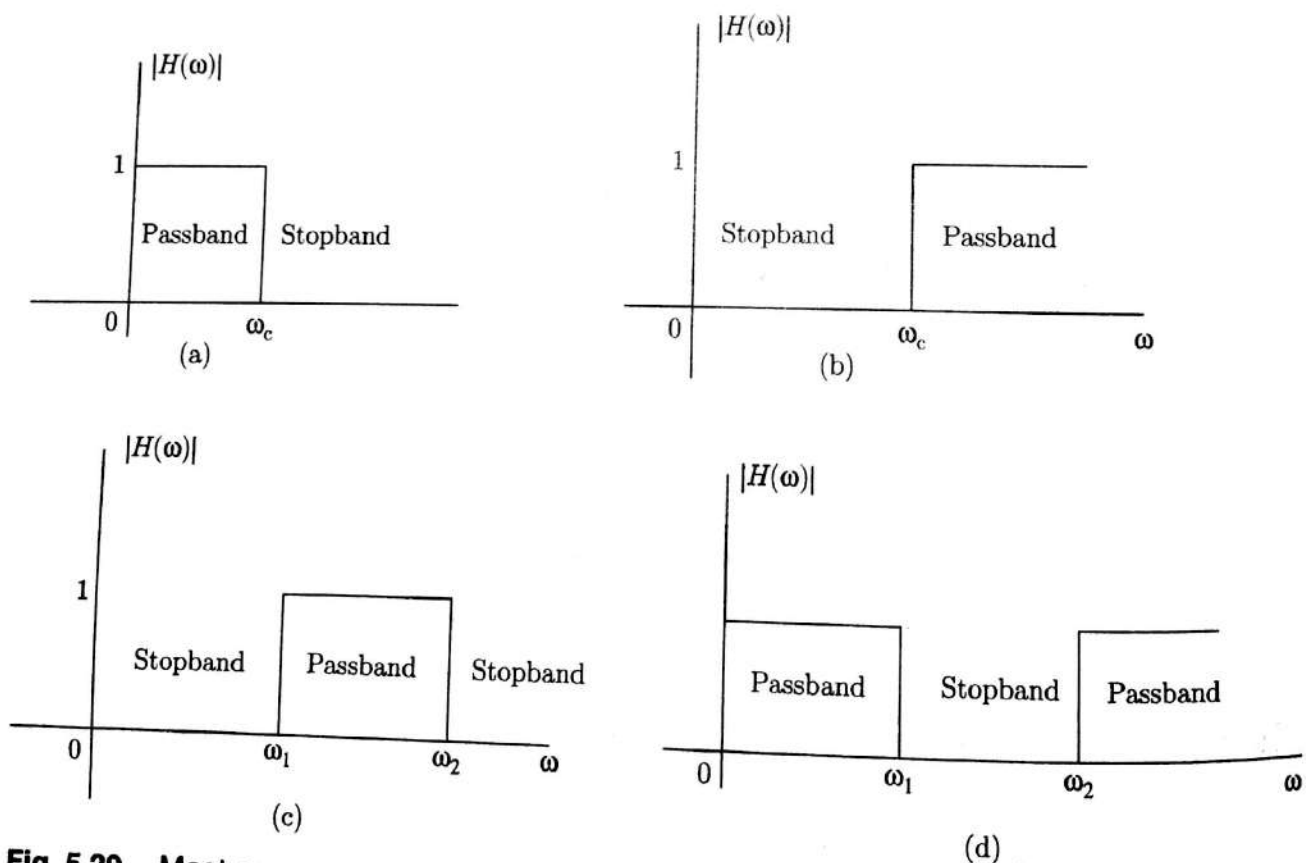


Fig. 5.29 Most common classes of filters: (a) LPF, (b) HPF, (c) BPF, and (d) BSF

3. Bandpass filters are characterized by a passband that extends from $\omega = \omega_1$ to $\omega = \omega_2$ and all other frequencies are stopped [Fig. 5.29(c)]. A bandpass filter passes signals within a certain frequency band and attenuates signals outside that band.
4. Bandstop filters stop frequencies extending from $\omega = \omega_1$ to $\omega = \omega_2$ and pass all other frequencies [Fig. 5.29(d)]. A bandstop filter attenuates signals within a certain frequency band and passes signals outside that band.
5. An allpass filter is characterized by a magnitude response that is constant for all frequencies. The characteristics of an allpass filter are completely determined by its phase-shift characteristics. An allpass filter has nonconstant group delay and different frequencies in the input are delayed by different amounts.

The filters described so far are referred to as ideal filters because they pass one set of frequencies without any change and completely stop others. Since it is impossible to realize filters with characteristic like those shown in Fig. 5.29, with abrupt changes from passband to stopband and vice versa, most of the filters we deal with in practice have some transition band. Ideal filters are noncausal filters and therefore physically unrealizable. For example, consider an ideal low-pass filter whose magnitude response and phase response are (as shown in Fig. 5.30) given by

$$|H(\omega)| = \text{rect}(\omega/2\omega_c) = \begin{cases} 1, & -\omega_c < \omega < \omega_c \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad \angle H(\omega) = -\omega t_d$$

Therefore,

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)} = \text{rect}\left(\frac{\omega}{2\omega_c}\right) e^{-j\omega t_d}$$

Let us define $H_0(\omega) = |H(\omega)|$. Then

$$|H(\omega)| = H_0(\omega) = \begin{cases} 1, & -\omega_c < \omega < \omega_c \\ 0, & \text{otherwise} \end{cases}$$

Taking the inverse Fourier transform, yields

$$h_0(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H_0(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega t} d\omega = \frac{1}{2\pi} \frac{e^{j\omega_c t}}{jt} \Big|_{-\omega_c}^{\omega_c} = \frac{\sin(\omega_c t)}{\pi t}$$

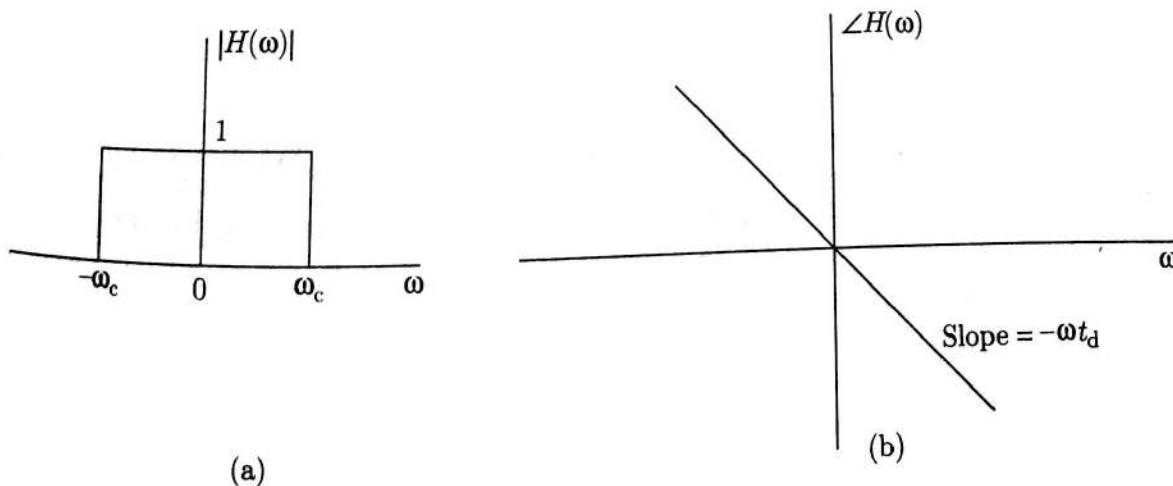


Fig. 5.30 (a) Magnitude response and (b) phase response

Now, consider

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)} = H_0(\omega) e^{-j\omega t_d}$$

Taking the inverse Fourier transform gives

$$h(t) = h_0(t - t_d)$$

$$h(t) = \frac{\sin(\omega_c(t - t_d))}{\pi(t - t_d)}$$

Recall that $h(t)$ is the system response to impulse input $\delta(t)$, which is applied at $t = 0$. Figure 5.31 depicts a curious fact: the response $h(t)$ begins even before the input is applied (at $t = 0$). Clearly, the filter is noncausal and therefore physically unrealizable. Similarly, one can show that other ideal filters are also physically unrealizable.

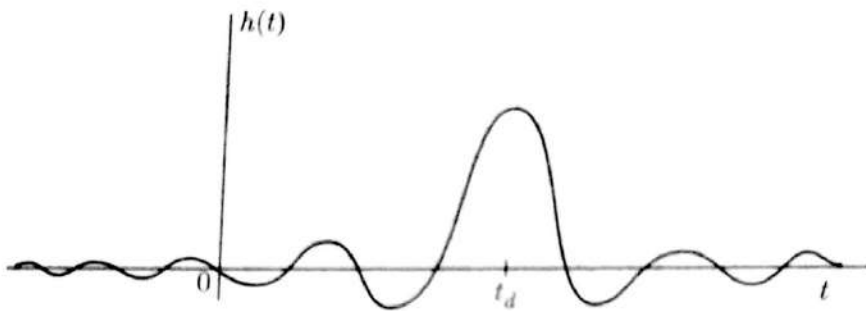


Fig. 5.31 Impulse response of an ideal LPF

5.7.1 Paley-Wiener Criterion

For a physically realizable system, the impulse response $h(t)$ must be causal, i.e.,

$$h(t) = 0, \quad t < 0$$

In the frequency domain, this condition is equivalent to the well-known Paley-Wiener criterion, which states that the necessary and sufficient condition for the magnitude response $|H(\omega)|$ to be realizable is

$$\int_{-\infty}^{\infty} \frac{|\ln|H(\omega)||}{1 + \omega^2} d\omega < \infty \quad (5.84)$$

If $H(\omega)$ does not satisfy this condition, it is unrealizable.

Two important conclusion that we draw from the Paley-Wiener theorem are as follows:

1. The magnitude function $|H(\omega)|$ of a realizable system can be zero at some discrete frequencies, but it cannot be zero over any finite band of frequencies. This means that if $|H(\omega)| = 0$ over any finite band, then $|\ln|H(\omega)|| = \infty$ over that band, and consequently $H(\omega)$ is unrealizable.
2. If $|H(\omega)|$ decays exponentially (or at a higher rate) with ω , the integral in Eq. (5.84) goes to infinity and $|H(\omega)|$ cannot be realized. Clearly, $|H(\omega)|$ cannot decay too fast with ω .

Apparently, causality imposes some tight constraints on a LTI system. In addition to the Paley-Wiener condition, causality also implies a strong relationship between $H_R(\omega) = \text{Re}\{H(\omega)\}$ and $H_I(\omega) = \text{Im}\{H(\omega)\}$, the real and imaginary components of the frequency response $H(\omega)$. To illustrate this dependence, we decompose $h(t)$ into its even and odd parts, i.e.,

$$h(t) = h_e(t) + h_o(t)$$

where

$$h_e(t) = \frac{1}{2}[h(t) + h(-t)]$$

and

$$h_o(t) = \frac{1}{2}[h(t) - h(-t)]$$

Now, if $h(t)$ is causal, it is possible to recover $h(t)$ from its even part $h_e(t)$ or from its odd part $h_o(t)$. Indeed, it can be easily seen that

$$h(t) = 2h_e(t)u(t) = 2h_o(t)u(t) \quad (5.85)$$

$$h_e(t) = h_o(t)\text{sgn}(t) \quad (5.86)$$

and

$$h_o(t) = h_e(t)\text{sgn}(t) \quad (5.87)$$

Since $h_o(t) = h_e(t)$ for $t > 0$, there is a strong relationship between $h_o(t)$ and $h_e(t)$.

If $h(t)$ is absolutely integrable (i.e., BIBO stable), the frequency response $H(\omega)$ exists, and

$$H(\omega) = H_R(\omega) + jH_I(\omega)$$

In addition if $h(t)$ is real valued and causal, the conjugate property of the Fourier transform implies that

$$h_e(t) \longleftrightarrow H_R(\omega) \quad \text{and} \quad h_o(t) \longleftrightarrow jH_I(\omega)$$

Since $h(t)$ is completely specified by $h_e(t)$, it follows that $H(\omega)$ is completely determined if we know $H_R(\omega)$. Alternatively, $H(\omega)$ is completely determined from $H_I(\omega)$. In short, $H_R(\omega)$ and $H_I(\omega)$ are interdependent and cannot be specified independently if the system is causal. Equivalently, the magnitude and phase responses of a causal filter are interdependent and hence cannot be specified independently.

Taking the Fourier transform of Eq. (5.86) gives

$$\begin{aligned} \mathcal{F}[h_e(t)] &= \frac{1}{2\pi} [\mathcal{F}[h_o(t)] * \mathcal{F}[\text{sgn}(t)]] \\ H_R(\omega) &= \frac{1}{2\pi} [jH_I(\omega) * \frac{2}{j\omega}] \end{aligned}$$

$$\begin{aligned}
 H_R(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} jH_I(\theta) \frac{2}{j(\omega - \theta)} d\theta \\
 H_R(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{H_I(\theta)}{\omega - \theta} d\theta
 \end{aligned} \tag{5.88}$$

Similarly, taking the Fourier transform of Eq. (5.87) gives

$$\begin{aligned}
 \mathcal{F}[h_o(t)] &= \frac{1}{2\pi} [\mathcal{F}[h_e(t)] * \mathcal{F}[\text{sgn}(t)]] \\
 jH_I(\omega) &= \frac{1}{2\pi} [H_R(\omega) * \frac{2}{j\omega}] \\
 jH_I(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H_R(\theta) \frac{2}{j(\omega - \theta)} d\theta \\
 H_I(\omega) &= -\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{H_R(\theta)}{\omega - \theta} d\theta
 \end{aligned} \tag{5.89}$$

Equations (5.88) and (5.89) are known as Hilbert transform pair.

To summarize, causality has very important implications in the design of frequency-selective filters. These are as follows:

1. The frequency response $H(\omega)$ cannot be zero except at a finite set of points in frequency.
2. The magnitude $|H(\omega)|$ cannot be constant in any finite range of frequencies and the transition from passband to stopband cannot be infinitely sharp.
3. The real and imaginary parts of $H(\omega)$ are interdependent and are related by the Hilbert transform. As a consequence, the magnitude $|H(\omega)|$ and phase $\angle H(\omega)$ cannot be chosen arbitrarily.

Example 5.37 Using Paley-Wiener criterion, prove that $|H(\omega)| = e^{-\omega^2}$ is not a suitable magnitude response for a causal LTI system.

Solution

According to the Paley-Wiener criterion, the condition for the magnitude response $|H(\omega)|$ to be realizable is

$$\int_{-\infty}^{\infty} \frac{|\ln |H(\omega)||}{1 + \omega^2} d\omega < \infty$$

where

$$|H(\omega)| = e^{-\omega^2}$$

Therefore, we have

$$\begin{aligned}
 \int_{-\infty}^{\infty} \frac{|\ln|e^{-\omega^2}||}{1+\omega^2} d\omega &= \int_{-\infty}^{\infty} \frac{\omega^2}{1+\omega^2} d\omega \\
 &= 2 \int_0^{\infty} \frac{\omega^2}{1+\omega^2} d\omega \\
 &= 2 \int_0^{\infty} \left(1 - \frac{1}{1+\omega^2}\right) d\omega \\
 &= 2[\omega - \tan^{-1}\omega]_0^{\infty} \\
 &= \infty
 \end{aligned}$$

Since the Paley-Wiener criterion is not satisfied, the given magnitude function is not realizable.

5.8 ENERGY SPECTRAL DENSITY (ESD)

Parseval's theorem relates the total signal energy in a signal $x(t)$ to its Fourier transform through

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = \int_{-\infty}^{\infty} |X(\omega)|^2 df$$

Parseval's theorem states that the total energy E_x may be determined either by computing the energy per unit time ($|x(t)|^2$) and integrating over all time or by computing the energy per unit frequency $|X(\omega)|^2$ and integrating over all frequencies. For this reason, $|X(\omega)|^2$ represents energy per unit bandwidth and is often referred to as the *energy spectral density* (per unit bandwidth in hertz) or *energy density spectrum* of the signal $x(t)$ and is denoted by $\Psi_x(\omega)$. Hence,

$$\Psi_x(\omega) = |X(\omega)|^2 \quad (5.90)$$

The units of ESD depend on the units of the underlying signal $x(t)$. For example, if the signal unit is volts (V), its Fourier transform has units of V/Hz and its ESD has units of (V/Hz)² or (V s)².

5.8.1 Relationship Between Input and Output Energy Spectral Densities of an LTI System

Consider an LTI system with frequency response $H(\omega)$, input $x(t)$, and output $y(t)$. If $x(t)$ and $y(t)$ are energy signals, then their energy spectral densities are $\Psi_x(\omega) = |X(\omega)|^2$ and $\Psi_y(\omega) = |Y(\omega)|^2$, respectively. Since, we know that

$$Y(\omega) = H(\omega)X(\omega)$$

it follows that

$$\begin{aligned}
 |Y(\omega)|^2 &= |H(\omega)X(\omega)|^2 \\
 |Y(\omega)|^2 &= |H(\omega)|^2 |X(\omega)|^2 \\
 \Psi_y(\omega) &= |H(\omega)|^2 \Psi_x(\omega)
 \end{aligned} \tag{5.91}$$

5.8.2 Relation of ESD to Autocorrelation

The autocorrelation function $R_{xx}(\tau)$ of an energy signal is defined as [Eq. (2.44)]:

$$\begin{aligned}
 R_{xx}(\tau) &= \int_{-\infty}^{\infty} x(t)x(t-\tau) dt \\
 R_{xx}(\tau) &= x(\tau) * x(-\tau)
 \end{aligned}$$

Taking the Fourier transform of the above equation gives

$$\begin{aligned}
 \mathcal{F}[R_{xx}(\tau)] &= X(\omega)X(-\omega) \\
 &= X(\omega)X^*(\omega) \\
 \mathcal{F}[R_{xx}(\tau)] &= |X(\omega)|^2 \\
 \mathcal{F}[R_{xx}(\tau)] &= \Psi_x(\omega) \\
 R_{xx}(\tau) &\longleftrightarrow \Psi_x(\omega)
 \end{aligned} \tag{5.92}$$

Thus, the autocorrelation function $R_{xx}(\tau)$ and ESD makes a Fourier transform pair.

5.9 POWER SPECTRAL DENSITY (PSD)

PSD has the same relation to power signals as ESD has to energy signals. The expression of PSD may be obtained by assuming the power signal as a limiting case of an energy signal.

Consider a power signal of infinite duration. Since the total energy of a power signal cannot be found, let us first find the ESD of a truncated version $x_T(t)$ of the infinite duration signal $x(t)$. The signal $x_T(t)$ may be expressed as

$$x_T(t) = \begin{cases} x(t), & -\frac{T}{2} < t < \frac{T}{2} \\ 0, & \text{otherwise} \end{cases} \tag{5.93}$$

$$x_T(t) = \text{rect}\left(\frac{t}{T}\right) x(t)$$

The truncated signal $x_T(t)$ is of finite duration therefore it is an energy signal. Now, if $x_T(t) \longleftrightarrow X_T(\omega)$, then using the Parseval's theorem, we have

$$E_{x_T} = \int_{-\infty}^{\infty} |x_T(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X_T(\omega)|^2 d\omega$$

$$\int_{-\infty}^{\infty} |x_T(t)|^2 dt = \int_{-\infty}^{\infty} |X_T(\omega)|^2 df$$

Substituting the value of $x_T(t)$ from Eq. (5.93) into the above equation yields

$$\int_{-T/2}^{T/2} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X_T(\omega)|^2 df$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} |X_T(\omega)|^2 df$$

The LHS of the above equation represents the average power P_x of the signal $x(t)$. Therefore,

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} |X_T(\omega)|^2 df$$

$$= \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{|X_T(\omega)|^2}{T} df$$

$$P_x = \int_{-\infty}^{\infty} G_x(\omega) df \quad (5.94)$$

where

$$G_x(\omega) = \lim_{T \rightarrow \infty} \frac{|X_T(\omega)|^2}{T} \quad (5.95)$$

is the PSD (average power per unit bandwidth). The units of PSD depend on the units of the underlying signal $x(t)$. If the signal unit is amperes (A), the units of PSD are A^2/Hz . If the signal unit is volts (V), the units of PSD are V^2/Hz .

5.9.1 Relationship Between Input and Output Power Spectral Densities of an LTI System

Consider an LTI system with frequency response $H(\omega)$, input $x_T(t)$, and output $y_T(t)$. If $x(t)$ and $y(t)$ are power signals, then their power spectral densities are $G_x(\omega) = \lim_{T \rightarrow \infty} \frac{|X_T(\omega)|^2}{T}$ and $G_y(\omega) = \lim_{T \rightarrow \infty} \frac{|Y_T(\omega)|^2}{T}$, respectively. Since, we know that

$$Y_T(\omega) = H(\omega)X_T(\omega)$$

it follows that

$$\begin{aligned} |Y_T(\omega)|^2 &= |H(\omega)X_T(\omega)|^2 \\ |Y_T(\omega)|^2 &= |H(\omega)|^2 |X_T(\omega)|^2 \\ \lim_{T \rightarrow \infty} \frac{|Y_T(\omega)|^2}{T} &= |H(\omega)|^2 \lim_{T \rightarrow \infty} \frac{|X_T(\omega)|^2}{T} \\ G_y(\omega) &= |H(\omega)|^2 G_x(\omega) \end{aligned}$$

(5.96)

5.9.2 Relation of PSD to Autocorrelation

The autocorrelation function $R_{xx}(\tau)$ of a power signal is defined as [Eq. (2.45)]:

$$\begin{aligned} R_{xx}(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} x_T(t)x_T(t-\tau) dt \\ R_{xx}(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{T} [x_T(\tau) * x_T(-\tau)] \end{aligned}$$

Taking the Fourier transform of the above equation, we get

$$\begin{aligned} \mathcal{F}[R_{xx}(\tau)] &= \lim_{T \rightarrow \infty} \frac{1}{T} X_T(\omega)X_T(-\omega) \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} X_T(\omega)X_T^*(\omega) \\ \mathcal{F}[R_{xx}(\tau)] &= \lim_{T \rightarrow \infty} \frac{1}{T} |X_T(\omega)|^2 \\ \mathcal{F}[R_{xx}(\tau)] &= G_x(\omega) \end{aligned}$$

$$R_{xx}(\tau) \longleftrightarrow G_x(\omega)$$

(5.97)

Thus, the autocorrelation function $R_{xx}(\tau)$ and PSD $G_x(\omega)$ makes a Fourier transform pair.

5.10 PSD OF PERIODIC SIGNALS

The PSD of the periodic signal $x(t)$ with period T is

$$G_x(\omega) = 2\pi \sum_{n=-\infty}^{\infty} |X_n|^2 \delta(\omega - n\omega_0) \quad (5.98)$$

where X_n are the Fourier series coefficients of $x(t)$ and $\omega_0 = 2\pi/T$.

Proof We begin by defining the truncated signals $x_T(t)$ as the product $\text{rect}\left(\frac{t}{T}\right) x(t)$.

$$x_T(t) = \text{rect}\left(\frac{t}{T}\right) x(t)$$

Using the multiplication property, we have

$$\begin{aligned} X_T(\omega) &= \frac{1}{2\pi} \left[\mathcal{F}\left[\text{rect}\left(\frac{t}{T}\right)\right] * X(\omega) \right] \\ &= \frac{1}{2\pi} \left[T \text{sinc}\left(\frac{\omega T}{2\pi}\right) * 2\pi \sum_{n=-\infty}^{\infty} X_n \delta(\omega - n\omega_0) \right] \end{aligned}$$

$$X_T(\omega) = \sum_{n=-\infty}^{\infty} X_n T \text{sinc}\left(\frac{T}{2\pi}(\omega - n\omega_0)\right) \quad (5.99)$$

From Eq. (5.99), we have

$$X_T^*(\omega) = \sum_{m=-\infty}^{\infty} X_m^* T \text{sinc}\left(\frac{T}{2\pi}(\omega - m\omega_0)\right) \quad (5.100)$$

From Eqs (5.99) and (5.100), we get

$$\begin{aligned} X_T(\omega) X_T^*(\omega) &= \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} X_n X_m^* T \text{sinc}\left(\frac{T}{2\pi}(\omega - n\omega_0)\right) T \text{sinc}\left(\frac{T}{2\pi}(\omega - m\omega_0)\right) \\ \frac{|X_T(\omega)|^2}{T} &= \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} X_n X_m^* \left[T \text{sinc}\left(\frac{T}{2\pi}(\omega - n\omega_0)\right) \text{sinc}\left(\frac{T}{2\pi}(\omega - m\omega_0)\right) \right] \end{aligned}$$

Taking the limit of the above equation as $T \rightarrow \infty$, we have

$$\lim_{T \rightarrow \infty} \frac{|X_T(\omega)|^2}{T} = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} X_n X_m^* \lim_{T \rightarrow \infty} \left[T \text{sinc}\left(\frac{T}{2\pi}(\omega - n\omega_0)\right) \text{sinc}\left(\frac{T}{2\pi}(\omega - m\omega_0)\right) \right] \quad (5.101)$$

It has been observed earlier that as $T \rightarrow \infty$, the transform of the rectangular signal approaches to an impulse function and therefore

$$\lim_{T \rightarrow \infty} \left[T \text{sinc}\left(\frac{T}{2\pi}(\omega - n\omega_0)\right) \text{sinc}\left(\frac{T}{2\pi}(\omega - m\omega_0)\right) \right] = \begin{cases} 2\pi \delta(\omega - n\omega_0), & n = m \\ 0, & \text{otherwise} \end{cases}$$

@sTc(small team creation)

Substituting the above value in Eq. (5.101) gives

$$\lim_{T \rightarrow \infty} \frac{|X_T(\omega)|^2}{T} = 2\pi \sum_{n=-\infty}^{\infty} |X_n|^2 \delta(\omega - n\omega_0)$$

$$G_x(\omega) = 2\pi \sum_{n=-\infty}^{\infty} |X_n|^2 \delta(\omega - n\omega_0)$$

SOLVED EXAMPLES

Example 5.38 Given that $x(t)$ has the Fourier transform $X(\omega)$, express the Fourier transforms of the signal listed below in terms of $X(\omega)$.

(a) $x_1(t) = x(1-t) + x(-1-t)$

(b) $x_2(t) = x(3t-6)$

(c) $x_3(t) = \frac{d^2}{dt^2} x(t-1)$

Solution

Given that

$$x(t) \longleftrightarrow X(\omega)$$

(a) Using the time-shifting property, we have

$$x(t+1) \longleftrightarrow X(\omega) e^{j\omega}$$

and

$$x(t-1) \longleftrightarrow X(\omega) e^{-j\omega}$$

Now, using the time-reversal property, we have

$$x(-t+1) \longleftrightarrow X(-\omega) e^{-j\omega}$$

and

$$x(-t-1) \longleftrightarrow X(-\omega) e^{j\omega}$$

Therefore,

$$x_1(t) \longleftrightarrow X_1(\omega)$$

$$x(1-t) + x(-1-t) \longleftrightarrow X(-\omega) e^{-j\omega} + X(-\omega) e^{j\omega}$$

$$x(1-t) + x(-1-t) \longleftrightarrow 2X(-\omega) \cos\omega$$

$$\mathcal{F}[x_1(t)] = X_1(\omega) = 2X(-\omega) \cos\omega$$

(b) Using the time-shifting property, we have

$$x(t-6) \longleftrightarrow X(\omega) e^{-j6\omega}$$

Now, using the time-scaling property, we get

$$x_2(t) = x(3t-6) \longleftrightarrow X_2(\omega) = \frac{1}{3} X\left(\frac{\omega}{3}\right) e^{-j6\omega}$$

$$X_2(\omega) = \frac{1}{3} X\left(\frac{\omega}{3}\right) e^{-j2\omega}$$

(c) Using the differentiation in time-domain property, we get

$$\frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega)$$

Applying this property again, we get

$$\frac{d^2 x(t)}{dt^2} \longleftrightarrow (j\omega)^2 X(\omega)$$

Now applying the time-shifting property, we get

$$x_3(t) = \frac{d^2 x(t-1)}{dt^2} \longleftrightarrow X_3(\omega) = (j\omega)^2 X(\omega) e^{-j\omega}$$

$$X_3(\omega) = -\omega^2 X(\omega) e^{-j\omega}$$

Example 5.39 Find the Fourier transform of the trapezoidal pulse shown in Fig. 5.32(a).

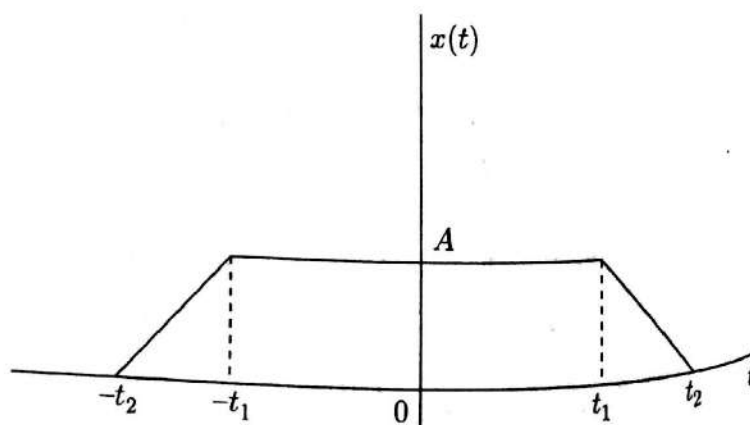
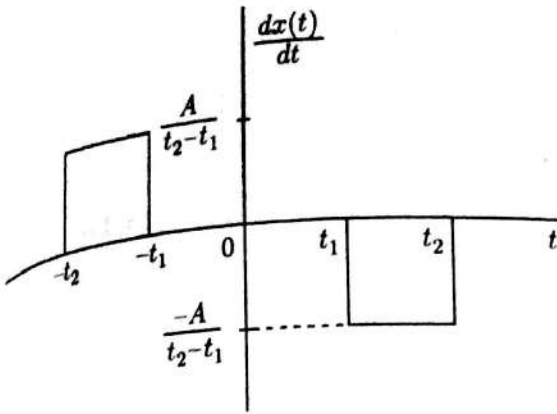
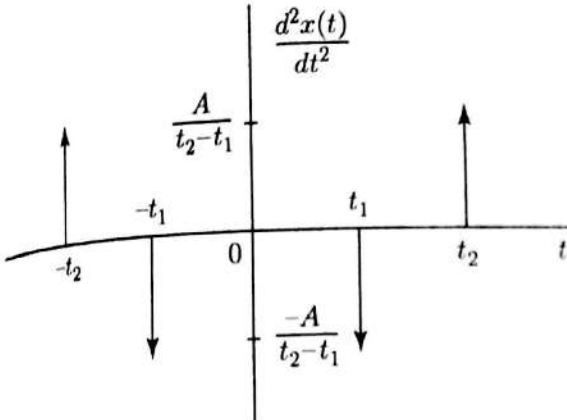


Fig. 5.32(a) Trapezoidal pulse



(b)



(c)

Fig. 5.32(b), (c) (b) First derivative and (c) second derivative

Solution

Using the ramp function $r(t)$, the given trapezoidal pulse can be written as

$$x(t) = \frac{A}{t_2 - t_1} r(t + t_2) - \frac{A}{t_2 - t_1} r(t + t_1) - \frac{A}{t_2 - t_1} r(t - t_1) + \frac{A}{t_2 - t_1} r(t - t_2)$$

The first derivative of the given trapezoidal pulse is given by [Fig. 5.32(b)]

$$\frac{dx(t)}{dt} = \frac{A}{t_2 - t_1} [u(t + t_2) - u(t + t_1) - u(t - t_1) + u(t - t_2)]$$

The second derivative is given by [Fig. 5.32(c)]

$$\frac{d^2x(t)}{dt^2} = \frac{A}{t_2 - t_1} [\delta(t + t_2) - \delta(t + t_1) - \delta(t - t_1) + \delta(t - t_2)]$$

The Fourier transform of the second derivative is

$$\begin{aligned} \mathcal{F} \left[\frac{d^2x(t)}{dt^2} \right] &= \frac{A}{t_2 - t_1} \mathcal{F} [\delta(t + t_2) - \delta(t + t_1) - \delta(t - t_1) + \delta(t - t_2)] \\ \mathcal{F} \left[\frac{d^2x(t)}{dt^2} \right] &= \frac{A}{t_2 - t_1} (\mathcal{F}[\delta(t + t_2)] - \mathcal{F}[\delta(t + t_1)] - \mathcal{F}[\delta(t - t_1)] + \mathcal{F}[\delta(t - t_2)]) \end{aligned}$$

Now, taking the Fourier transform of the above equation and using the differentiation in time domain and the time-shifting property, we get

$$\begin{aligned} (j\omega)^2 X(\omega) &= \frac{A}{t_2 - t_1} [e^{j\omega t_2} - e^{j\omega t_1} - e^{-j\omega t_1} + e^{-j\omega t_2}] \\ &= \frac{2A}{t_2 - t_1} \left[\frac{e^{j\omega t_2} + e^{-j\omega t_2}}{2} - \frac{e^{j\omega t_1} + e^{-j\omega t_1}}{2} \right] \\ (j\omega)^2 X(\omega) &= \frac{2A}{t_2 - t_1} [\cos(\omega t_2) - \cos(\omega t_1)] \\ X(\omega) &= \frac{2A}{\omega^2(t_2 - t_1)} [\cos(\omega t_1) - \cos(\omega t_2)] \end{aligned}$$

Example 5.40 Find the Fourier transform of the sinusoidal pulse shown in Fig 5.33(a).

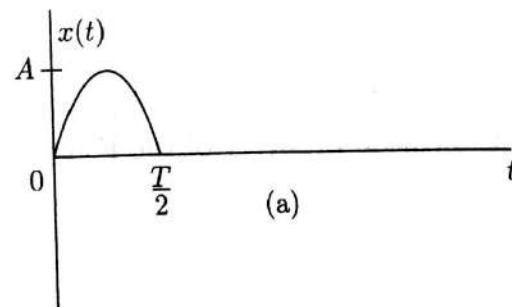


Fig. 5.33(a) Sinusoidal pulse

Solution

Defining a signal $g(t)$ of frequency $\omega_0 = 2\pi/T$ [as shown in Fig. 5.33(b)]

$$g(t) = A \sin(\omega_0 t) u(t)$$

$$g\left(t - \frac{T}{2}\right) = A \sin\left(\omega_0\left(t - \frac{T}{2}\right)\right) u\left(t - \frac{T}{2}\right)$$

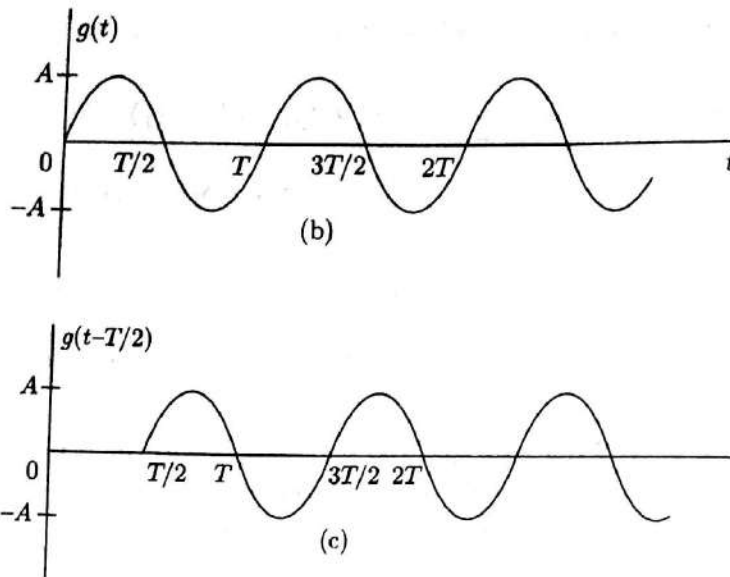


Fig. 5.33(b), (c) (b) Signal $g(t)$ and (c) time-shifted signal

The time-shifted signal $g(t-T/2)$ is shown in Fig. 5.33(c). The given sinusoidal pulse can be expressed as

$$\begin{aligned} x(t) &= g(t) + g(t-T/2) \\ X(\omega) &= G(\omega) + G(\omega)e^{-j\omega(T/2)} \\ &= G(\omega)(1 + e^{-j\omega(T/2)}) \\ &= 2G(\omega) \left(\frac{e^{j\omega(T/4)} - e^{-j\omega(T/4)}}{2} \right) e^{-j\omega(T/4)} \\ X(\omega) &= 2G(\omega) \cos(\omega T/4) e^{-j\omega(T/4)} \quad (5.102) \end{aligned}$$

Now, consider

$$g(t) = A \sin(\omega_0 t) u(t)$$

Applying the multiplication property of the Fourier transform gives

$$\begin{aligned} \mathcal{F}[g(t)] &= \frac{1}{2\pi} \left(\mathcal{F}[A \sin(\omega_0 t)] * \mathcal{F}[u(t)] \right) \\ G(\omega) &= \frac{1}{2\pi} \left(\frac{A\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] \right. \\ &\quad \left. * \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] \right) \\ G(\omega) &= \frac{1}{2\pi} \frac{A\pi^2}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] \\ &\quad + \frac{1}{2\pi} \frac{A\pi}{j} \left[\frac{1}{j(\omega - \omega_0)} - \frac{1}{j(\omega + \omega_0)} \right] \end{aligned}$$

$$\begin{aligned} G(\omega) &= \frac{A\pi}{2j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] \\ &\quad + A \left[\frac{\omega_0}{\omega_0^2 - \omega^2} \right] \end{aligned}$$

Substituting this value of $G(\omega)$ into Eq. (5.102) yields

$$\begin{aligned} X(\omega) &= 2 \frac{A\pi}{2j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] \cos(\omega T/4) e^{-j\omega(T/4)} \\ &\quad + 2A \left[\frac{\omega_0}{\omega_0^2 - \omega^2} \right] \cos(\omega T/4) e^{-j\omega(T/4)} \\ &= \frac{A\pi}{j} [\delta(\omega - \omega_0) \cos(\omega_0 T/4) e^{-j\omega_0(T/4)} \\ &\quad - \delta(\omega + \omega_0) \cos(\omega_0 T/4) e^{-j\omega_0(T/4)}] \\ &\quad + 2A \left[\frac{\omega_0}{\omega_0^2 - \omega^2} \right] \cos(\omega T/4) e^{-j\omega(T/4)} \\ &= \frac{A\pi}{j} [\delta(\omega - \omega_0) \cos(\pi/2) e^{-j\pi/2} \\ &\quad - \delta(\omega + \omega_0) \cos(\pi/2) e^{-j\pi/2}] \\ &\quad + 2A \left[\frac{\omega_0}{\omega_0^2 - \omega^2} \right] \cos(\omega T/4) e^{-j\omega(T/4)} \\ &= 0 + 2A \left[\frac{\omega_0}{\omega_0^2 - \omega^2} \right] \cos(\omega T/4) e^{-j\omega(T/4)} \end{aligned}$$

Therefore,

$$X(\omega) = 2A \left[\frac{\omega_0}{\omega_0^2 - \omega^2} \right] \cos(\omega T/4) e^{-j\omega(T/4)}$$

Example 5.41 Find the magnitude and phase spectrum of the pulse shown in Fig 5.34(a).

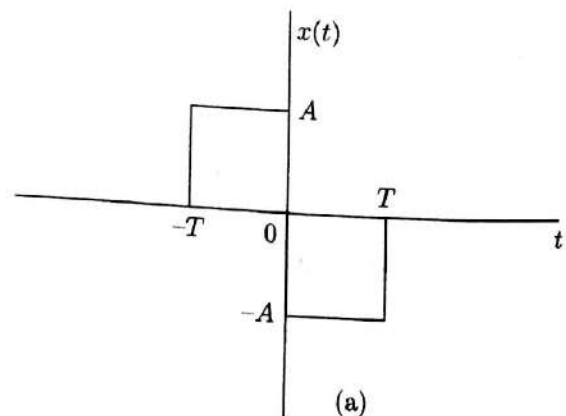


Fig. 5.34(a)

Solution
Analytically, the given pulse can be written as

$$x(t) = \begin{cases} A, & -T < t < 0 \\ -A, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$

The Fourier transform $X(\omega)$ of $x(t)$ is given by

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ &= \int_{-T}^0 A e^{-j\omega t} dt - \int_0^T A e^{-j\omega t} dt \\ &= A \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T}^0 - A \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^T \\ &= \frac{A}{-j\omega} [1 - e^{j\omega T}] + \frac{jA}{\omega} [e^{-j\omega T} - 1] \\ &= \frac{jA}{\omega} [2 - e^{j\omega T} - e^{-j\omega T}] \\ &= \frac{jA}{\omega} \left[2 - 2 \frac{e^{j\omega T} + e^{-j\omega T}}{2} \right] \\ &= \frac{2jA}{\omega} [1 - \cos(\omega T)] \end{aligned}$$

Therefore, the magnitude spectrum $|X(\omega)|$ of $x(t)$ is given by

$$|X(\omega)| = \left| \frac{2A}{\omega} [1 - \cos(\omega T)] \right|$$

Thus, the magnitude spectrum is zero when $1 - \cos(\omega T) = 0$, i.e.,

$$1 - \cos(\omega T) = 0$$

$$\cos(\omega T) = 1 = \cos(2n\pi)$$

or

$$\omega T = 2n\pi$$

$$\omega = \frac{2n\pi}{T}$$

The phase spectrum of $x(t)$ is

$$\angle X(\omega) = \tan^{-1} \left[\frac{X_I(\omega)}{X_R(\omega)} \right]$$

$$\angle X(\omega) = \tan^{-1} \left[\frac{2A/\omega [1 - \cos(\omega T)]}{0} \right]$$

$$\angle X(\omega) = \begin{cases} \frac{\pi}{2}, & \omega > 0 \\ -\frac{\pi}{2}, & \omega < 0 \end{cases}$$

The magnitude and phase spectrum are shown in Figs 5.34(b) and (c).

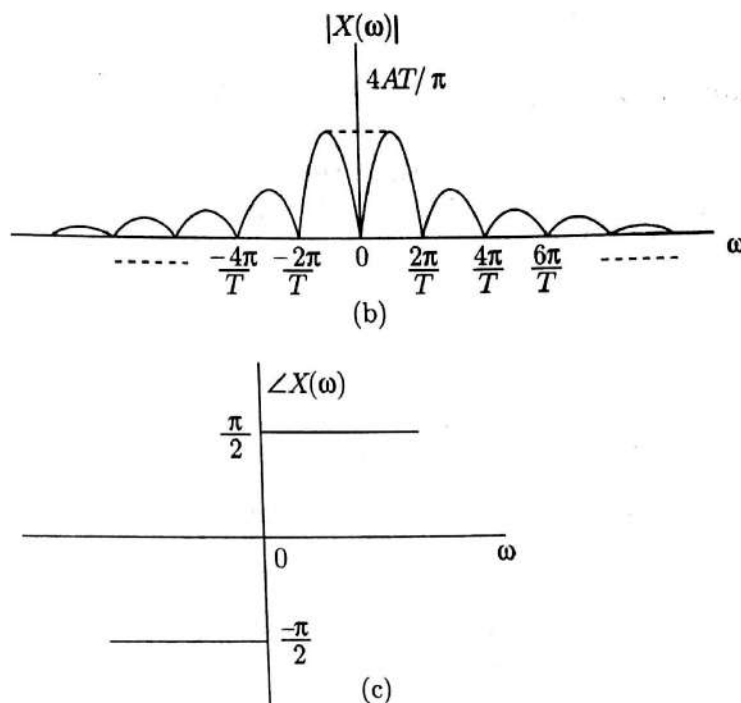


Fig. 5.34(b) and (c)

Example 5.42 For each of the following Fourier transform, use Fourier transform properties to determine whether the corresponding time-domain signal is (i) real, imaginary, or neither and (ii) even, odd, or neither. Do this without evaluating the inverse of any of the given transforms.

- $X_1(\omega) = u(\omega) - u(\omega - 2)$
- $X_2(\omega) = \cos(2\omega) \sin(\omega/2)$
- $X_3(\omega) = A(\omega) e^{jB(\omega)}$, where $A(\omega) = \sin(2\omega)/\omega$ and $B(\omega) = 2\omega + \pi/2$
- $X_4(\omega) = \sum_{k=-\infty}^{\infty} (1/2)^{|k|} \delta(\omega - k\pi/4)$

Solution

(a) We know that the Fourier transform of a real, signal is conjugate symmetric. Since $X_1(\omega)$ is not conjugate symmetric, the corresponding signal $x_1(t)$ is not real. Since $X_1(\omega)$ is neither even nor odd, the corresponding signal $x_1(t)$ is neither even nor odd.

(b) We know that the Fourier transform of a real and odd signal is purely imaginary and odd. Therefore, we may conclude that the Fourier transform of a purely imaginary and odd signal is real and

odd. The given transform $X_2(\omega)$ is real and odd, we may therefore conclude that the corresponding signal $x_2(t)$ is purely imaginary and odd.

(c) We know that if $x(t)$ is real, then its $|X(\omega)| = |X(-\omega)|$ and $\angle X(\omega) = -\angle X(-\omega)$. Here

$$|X(\omega)| = A(\omega) = \frac{\sin(2\omega)}{\omega} = |X(-\omega)|$$

$$\text{and } \angle X(\omega) = B(\omega) = 2\omega = -\angle X(-\omega)$$

so, we may conclude that the signal $x(t)$ is real. Now, consider the signal $x_3(t)$ with Fourier transform $X_3(\omega) = X(\omega)e^{j(\pi/2)} = jX(\omega)$. We may conclude that $x_3(t)$ is imaginary. Since, the Fourier transform $X_3(\omega)$ is neither purely imaginary nor purely real, the signal $x_3(t)$ is neither even nor odd.

(d) Since $X_4(\omega)$ is real and even, the corresponding signal $x_4(t)$ is real and even.

Example 5.43 Consider the signal

$$x(t) = \begin{cases} 0, & t < -\frac{1}{2} \\ t + \frac{1}{2}, & -\frac{1}{2} \leq t \leq \frac{1}{2} \\ 1, & t > \frac{1}{2} \end{cases}$$

- Use the differentiation and integration properties and the Fourier transform pairs to find a closed form expression for $X(\omega)$.
- What is the Fourier transform of $g(t) = x(t) - 1/2$

Solution

The given signal $x(t)$ is as shown in the Fig 5.35(a). We may express $x(t)$ using ramp function as

$$x(t) = r\left(t + \frac{1}{2}\right) - r\left(t - \frac{1}{2}\right)$$

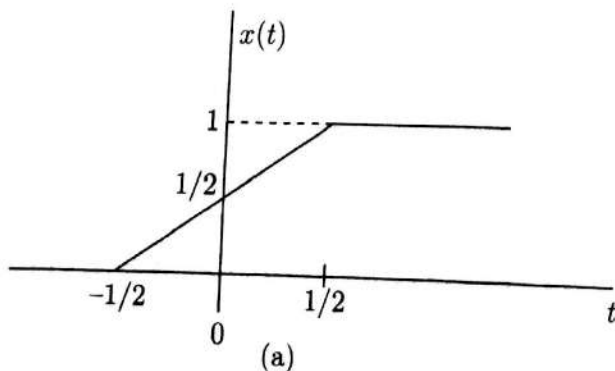


Fig. 5.35(a)

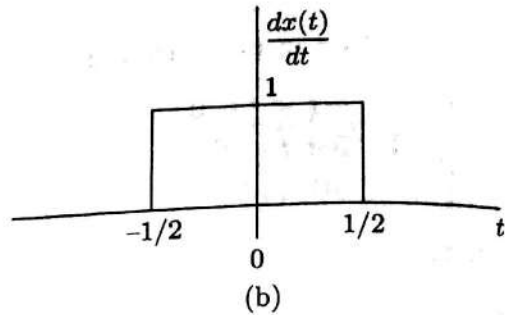


Fig. 5.35(b)

Taking the derivative of $x(t)$ [as shown in Fig. 5.35(b)], we get

$$\frac{dx(t)}{dt} = u\left(t + \frac{1}{2}\right) - u\left(t - \frac{1}{2}\right)$$

Taking the Fourier transform of the above equation and using the time-shifting property, we have

$$\begin{aligned} j\omega X(\omega) &= U(\omega)e^{j(\omega/2)} - U(\omega)e^{-j(\omega/2)} \\ &= 2jU(\omega) \frac{e^{j(\omega/2)} - e^{-j(\omega/2)}}{2j} \end{aligned}$$

$$j\omega X(\omega) = 2jU(\omega) \sin\left(\frac{\omega}{2}\right)$$

$$X(\omega) = \frac{2j}{j\omega} \sin\left(\frac{\omega}{2}\right) \left[\pi\delta(\omega) + \frac{1}{j\omega} \right]$$

$$\begin{aligned} X(\omega) &= \frac{2}{\omega} \sin\left(\frac{\omega}{2}\right) \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] \\ &= \frac{\sin(\omega/2)}{\omega/2} \pi\delta(\omega) + 2 \frac{\sin(\omega/2)}{j\omega^2} \end{aligned}$$

$$X(\omega) = \pi\delta(\omega) + 2 \frac{\sin(\omega/2)}{j\omega^2}$$

(b) Given that

$$g(t) = x(t) - \frac{1}{2}$$

Taking the Fourier transform of the above equation, we get

$$\begin{aligned} G(\omega) &= X(\omega) - \frac{1}{2} 2\pi\delta(\omega) \\ &= \pi\delta(\omega) + 2 \frac{\sin(\omega/2)}{j\omega^2} - \pi\delta(\omega) \end{aligned}$$

$$G(\omega) = 2 \frac{\sin(\omega/2)}{j\omega^2}$$

Example 5.44 (a) Use the Fourier transform properties to determine the Fourier transform of the following signal:

$$x(t) = t \left(\frac{\sin(t)}{\pi t} \right)^2 = t g^2(t)$$

where $g(t) = \sin(t)/\pi t$.

(b) Use Parseval's relation and the result of the previous part to determine the numerical value of

$$A = \int_{-\infty}^{\infty} t^2 \left(\frac{\sin(t)}{\pi t} \right)^4 dt$$

Solution
We know that [Eq. (5.49)]

$$A \operatorname{sinc} \left(\frac{t}{2\pi} \right) \longleftrightarrow 2\pi A \operatorname{rect}(\omega)$$

$$\frac{\sin(t/2)}{t/2} \longleftrightarrow 2\pi \operatorname{rect}(\omega)$$

Using the time-scaling property, we have

$$\frac{\sin(t)}{t} \longleftrightarrow \pi \operatorname{rect} \left(\frac{\omega}{2} \right)$$

$$g(t) = \frac{\sin(t)}{\pi t} \longleftrightarrow G(\omega) = \operatorname{rect} \left(\frac{\omega}{2} \right)$$

$G(\omega)$ is shown in Fig. 5.36(a), which is a rectangle function.

Now, using the multiplication property, we have

$$g^2(t) = g(t)g(t) \longleftrightarrow \frac{1}{2\pi} [G(\omega) * G(\omega)]$$

$$\underbrace{\left(\frac{\sin(t)}{\pi t} \right)^2}_{g_1(t)} \longleftrightarrow \underbrace{\frac{1}{2\pi} \left[\operatorname{rect} \left(\frac{\omega}{2} \right) * \operatorname{rect} \left(\frac{\omega}{2} \right) \right]}_{G_1(\omega)}$$

We know that the convolution of a rectangle function with itself is a triangle function [$G_1(\omega)$ is shown in Fig. 5.36(b)]. Therefore, we have

$$\left(\frac{\sin(t)}{\pi t} \right)^2 \longleftrightarrow \frac{1}{\pi} \Delta \left(\frac{\omega}{2} \right)$$

where

$$G_1(\omega) = \frac{1}{\pi} \Delta \left(\frac{\omega}{2} \right) = \begin{cases} \frac{1}{\pi} \left(1 + \frac{\omega}{2} \right), & -2 \leq \omega < 0 \\ \frac{1}{\pi} \left(1 - \frac{\omega}{2} \right), & 0 \leq \omega \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Using the differentiation in frequency domain property, we have

$$t g_1(t) \longleftrightarrow j \frac{dG_1(\omega)}{d\omega}$$

$$\underbrace{t \left(\frac{\sin(t)}{\pi t} \right)^2}_{x(t)} \longleftrightarrow X(\omega) = \begin{cases} \frac{j}{2\pi}, & -2 \leq \omega < 0 \\ -\frac{j}{2\pi}, & 0 \leq \omega \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$X(\omega)$ is shown in Fig. 5.36(c).

(b) Using Parseval's theorem, we have

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

$$\int_{-\infty}^{\infty} t^2 \left(\frac{\sin(t)}{\pi t} \right)^4 dt = \frac{1}{2\pi} \left[\int_{-2}^0 \frac{1}{4\pi^2} d\omega + \int_0^2 \frac{1}{4\pi^2} d\omega \right]$$

$$\int_{-\infty}^{\infty} t^2 \left(\frac{\sin(t)}{\pi t} \right)^4 dt = \frac{1}{2\pi^3}$$

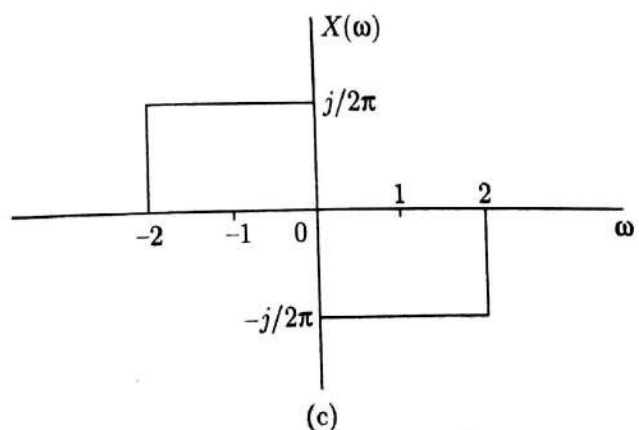
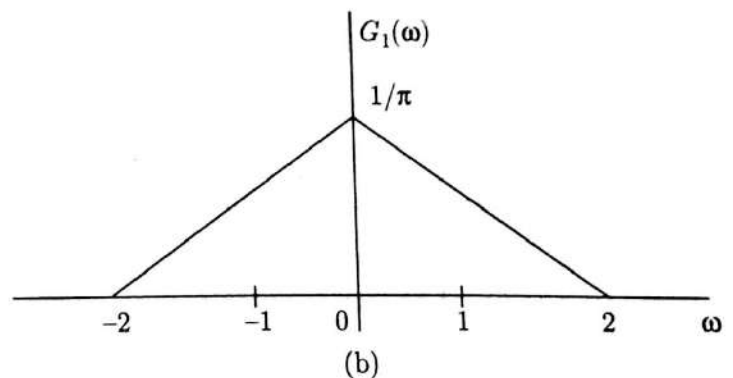
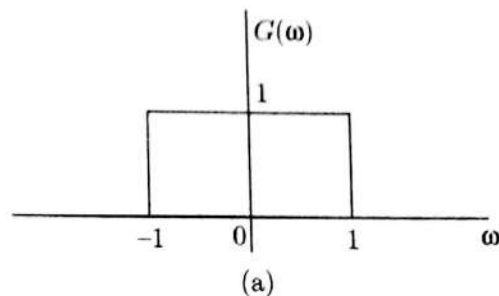


Fig. 5.36

Example 5.45 Find the Fourier transform of the triangular pulse $x(t)$ shown in Fig. 5.37(a).

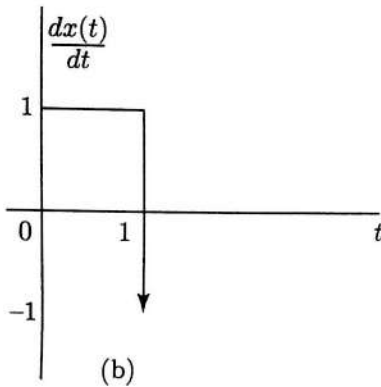
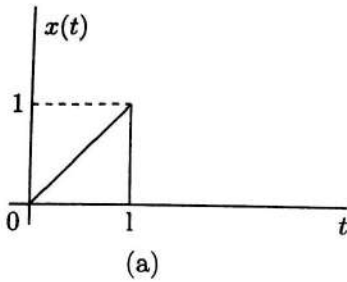


Fig. 5.37

Solution

Using the ramp and step function, the given triangular pulse can be written as

$$x(t) = r(t) - r(t-1) - u(t-1)$$

The first derivative of the triangular pulse is [Fig. 5.37(b)]

$$\frac{dx(t)}{dt} = u(t) - u(t-1) - \delta(t-1)$$

Taking the Fourier transform of the above equation, we have

$$\begin{aligned} j\omega X(\omega) &= \left(\pi\delta(\omega) + \frac{1}{j\omega} \right) \\ &\quad - \left(\pi\delta(\omega) + \frac{1}{j\omega} \right) e^{-j\omega} - e^{-j\omega} \end{aligned}$$

$$\begin{aligned} j\omega X(\omega) &= \pi\delta(\omega)(1 - e^{-j\omega}) + \frac{1}{j\omega} \\ &\quad - \frac{1}{j\omega} e^{-j\omega} - e^{-j\omega} \end{aligned}$$

$$X(\omega) = 0 - \frac{1}{\omega^2} + \frac{1}{\omega^2} e^{-j\omega} - \frac{1}{j\omega} e^{-j\omega}$$

$$X(\omega) = \frac{1}{\omega^2} (e^{-j\omega} + j\omega e^{-j\omega} - 1)$$

Example 5.46 The Fourier transform of the triangular pulse $x(t)$ in Fig. 5.38(a) is expressed as

$$X(\omega) = \frac{1}{\omega^2} (e^{-j\omega} + j\omega e^{-j\omega} - 1)$$

Use this information and the time-shifting and time-scaling properties to find the Fourier transforms of the signals (a) $x_1(t)$ and (b) $x_2(t)$ shown in Figs 5.38(b) and (c), respectively.

Solution

(a) The relationship between $x_1(t)$ and $x(t)$ can be described as

$$x_1(t) = x(t+1) + x(-t+1)$$

Taking the Fourier transform of the above equation, we have

$$\begin{aligned} X_1(\omega) &= X(\omega)e^{j\omega} + X(-\omega)e^{-j\omega} \\ &= \frac{1}{\omega^2} (e^{-j\omega} + j\omega e^{-j\omega} - 1)e^{j\omega} \\ &\quad + \frac{1}{\omega^2} (e^{j\omega} - j\omega e^{j\omega} - 1)e^{-j\omega} \\ &= \frac{1}{\omega^2} (1 + j\omega - e^{-j\omega}) + \frac{1}{\omega^2} (1 - j\omega - e^{j\omega}) \\ &= \frac{1}{\omega^2} (2 - 2\cos(\omega)) \\ &= \frac{2}{\omega^2} (1 - \cos(\omega)) \end{aligned}$$

$$X_1(\omega) = \frac{2}{\omega^2} 2\sin^2(\omega/2) = \frac{\sin^2(\omega/2)}{(\omega/2)^2}$$

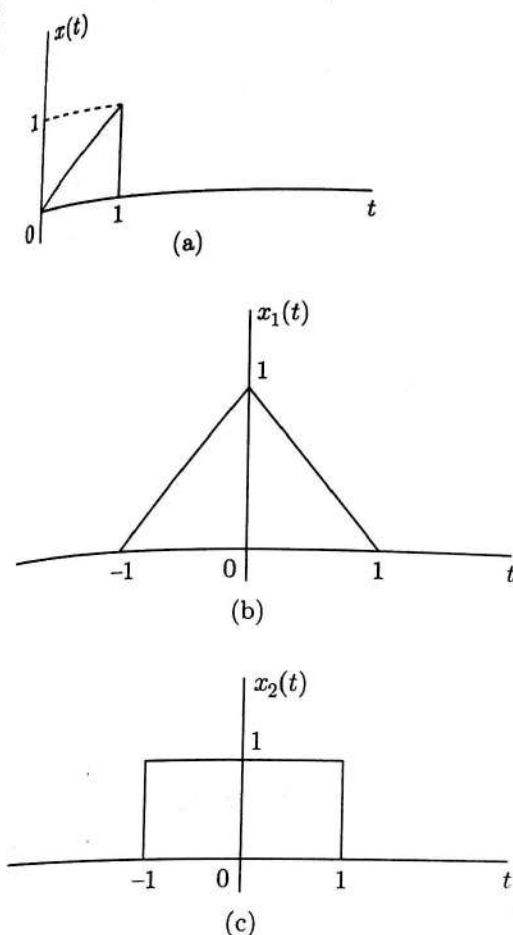


Fig. 5.38

(b) First, we express $x_2(t)$ in terms of $x(t)$. The derivative of $x(t)$ and its folded version is shown in Figs 5.39(a) and (b), respectively. Therefore, the relationship between $x_2(t)$ and $x(t)$ can be expressed as

$$x_2(t) = \frac{dx(t)}{dt} + \frac{dx(-t)}{dt} + \delta(t+1) + \delta(t-1)$$

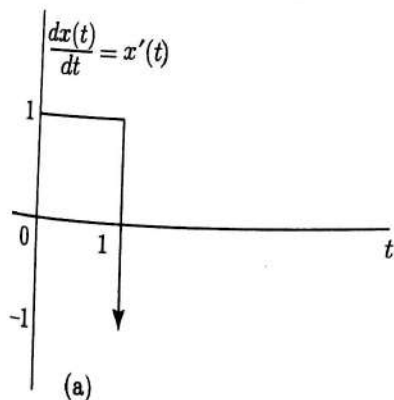


Fig. 5.39(a)

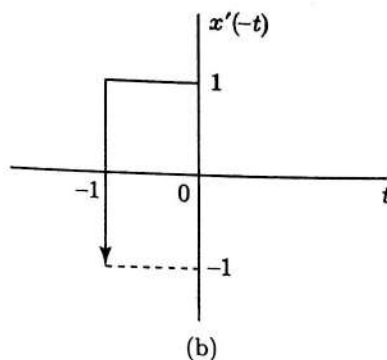


Fig. 5.39(b)

Taking the Fourier transform of the above equation, we have

$$\begin{aligned} X_2(\omega) &= j\omega X(\omega) - j\omega X(-\omega) + e^{j\omega} + e^{-j\omega} \\ &= j\omega[X(\omega) - X(-\omega)] + 2 \frac{e^{j\omega} + e^{-j\omega}}{2} \\ &= \frac{j\omega}{\omega^2} [e^{-j\omega} + j\omega e^{-j\omega} - 1 - e^{j\omega} \\ &\quad + j\omega e^{j\omega} + 1] + 2 \cos(\omega) \\ &= \frac{j}{\omega} \left[-2j \frac{e^{j\omega} - e^{-j\omega}}{2j} + j2\omega \frac{e^{j\omega} + e^{-j\omega}}{2} \right] \\ &\quad + 2 \cos(\omega) \\ &= \frac{j}{\omega} [-2j \sin(\omega) + j2\omega \cos(\omega)] + 2 \cos(\omega) \\ &= 2 \frac{\sin(\omega)}{\omega} - 2 \cos(\omega) + 2 \cos(\omega) \\ X_2(\omega) &= 2 \frac{\sin(\omega)}{\omega} \end{aligned}$$

Example 5.47 Given the relationships

$$y(t) = x(t) * h(t)$$

and

$$g(t) = x(3t) * h(3t)$$

and given that $x(t)$ has Fourier transform $X(\omega)$ and $h(t)$ has Fourier transform $H(\omega)$, use Fourier transform properties to show that $g(t)$ has the form

$$g(t) = Ay(Bt)$$

Determine the values of A and B .

Solution

Given that

$$y(t) = x(t) * h(t)$$

Taking Fourier transform, we get

$$Y(\omega) = X(\omega)H(\omega)$$

Using the time-scaling property, we have

$$x(3t) \longleftrightarrow \frac{1}{3}X\left(\frac{\omega}{3}\right) \quad \text{and} \quad h(3t) \longleftrightarrow \frac{1}{3}H\left(\frac{\omega}{3}\right)$$

Taking the Fourier transform of $g(t) = x(3t) * h(3t)$, we get

$$G(\omega) = \mathcal{F}[x(3t)]\mathcal{F}[h(3t)]$$

$$= \frac{1}{9}X\left(\frac{\omega}{3}\right)H\left(\frac{\omega}{3}\right)$$

$$G(\omega) = \frac{1}{9}Y\left(\frac{\omega}{3}\right)$$

Taking the inverse Fourier transform, we obtain

$$g(t) = \frac{1}{3}y(3t) = Ay(Bt)$$

Therefore, $A = 1/3$ and $B = 3$.

Example 5.48 (a) Use the appropriate Fourier transform properties to find the Fourier transform of $g(t) = te^{-|t|}$.

(b) Use the result of part (a), along with the duality property, to determine the Fourier transform of

$$y(t) = \frac{4t}{(1+t^2)^2}$$

Solution

(a) We know that

$$e^{-a|t|} \longleftrightarrow \frac{2a}{a^2 + \omega^2}$$

and substituting $a = 1$ gives

$$e^{-|t|} \longleftrightarrow \frac{2}{1 + \omega^2}$$

Using the differentiation in frequency domain property, we have

$$te^{-|t|} \longleftrightarrow j \frac{d}{d\omega} \left(\frac{2}{1 + \omega^2} \right) = -\frac{4j\omega}{(1 + \omega^2)^2}$$

(b) We have

$$te^{-|t|} \longleftrightarrow -\frac{4j\omega}{(1 + \omega^2)^2}$$

$$g(t) \longleftrightarrow G(\omega)$$

$$g(t) = te^{-|t|} \quad \text{and} \quad G(\omega) = -\frac{4j\omega}{(1 + \omega^2)^2}$$

$$g(-\omega) = \omega e^{-|\omega|} \quad \text{and} \quad G(t) = -\frac{4jt}{(1 + t^2)^2}$$

The duality property of the Fourier transform states that if

$$g(t) \longleftrightarrow G(\omega)$$

then

$$G(t) \longleftrightarrow 2\pi g(-\omega)$$

$$-\frac{4jt}{(1 + t^2)^2} \longleftrightarrow 2\pi\omega e^{-|\omega|}$$

$$\frac{4t}{(1 + t^2)^2} \longleftrightarrow j2\pi\omega e^{-|\omega|}$$

$$y(t) = \frac{4t}{(1 + t^2)^2} \longleftrightarrow Y(\omega) = j2\pi\omega e^{-|\omega|}$$

Example 5.49 Find the Fourier transform of the periodic signal $x(t)$ as shown in Fig. 5.40.

Solution

The waveform is periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$. The given waveform for one period may be written as

$$x(t) = \begin{cases} A, & |t| < \frac{a}{2} \\ 0, & |t| > \frac{a}{2} \end{cases}$$

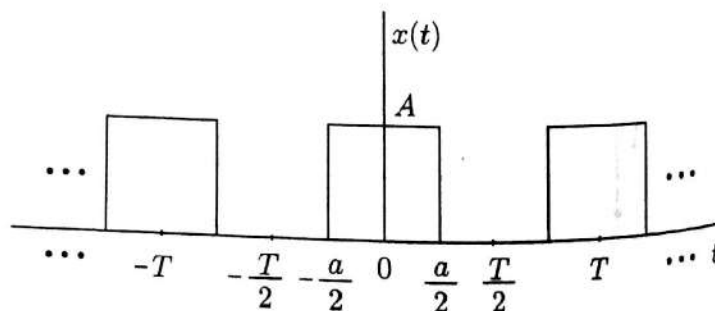


Fig. 5.40

The exponential Fourier series coefficient is

$$\begin{aligned} X_n &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T} \int_{-a/2}^{a/2} A e^{-jn\omega_0 t} dt \\ &= \frac{A}{T} \left[\frac{e^{-jn\omega_0 t}}{-jn\omega_0} \right]_{-a/2}^{a/2} \\ X_n &= \frac{Aa \sin(n\omega_0 a/2)}{T n\omega_0 a/2} \end{aligned}$$

From Eq. (5.61), the Fourier transform of a periodic signal $x(t)$ is given by

$$\begin{aligned} X(\omega) &= 2\pi \sum_{n=-\infty}^{\infty} X_n \delta(\omega - n\omega_0) \\ X(\omega) &= 2\pi \sum_{n=-\infty}^{\infty} \frac{Aa \sin(n\omega_0 a/2)}{T n\omega_0 a/2} \delta(\omega - n\omega_0) \end{aligned}$$

Example 5.50 Let $x(t)$ be a signal whose Fourier transform is

$$X(\omega) = \delta(\omega) + \delta(\omega - \pi) + \delta(\omega - 5)$$

and let

$$h(t) = u(t) - u(t - 2)$$

- Is $x(t)$ periodic?
- Is $y(t) = x(t) * h(t)$ periodic?
- Can the convolution of two aperiodic signals be periodic?

Solution

Given that

$$X(\omega) = \delta(\omega) + \delta(\omega - \pi) + \delta(\omega - 5)$$

Taking the inverse Fourier transform of $X(\omega)$, we have

$$x(t) = \frac{1}{2\pi} + \frac{1}{2\pi} e^{j\pi t} + \frac{1}{2\pi} e^{j5t}$$

The signal $x(t)$ is therefore a constant summed with two complex exponentials whose fundamental fre-

quencies are

$$\begin{aligned} \omega_1 &= \pi \text{ rad/s} \quad \text{and} \quad \omega_2 = 5 \text{ rad/s} \\ T_1 &= \frac{2\pi}{\omega_1} = \frac{2\pi}{\pi} = 2 \quad \text{and} \quad T_2 = \frac{2\pi}{\omega_2} = \frac{2\pi}{5} \end{aligned}$$

The ratio $T_1/T_2 = 5/\pi$ is not a rational number and so $x(t)$ is not periodic.

(b) Given that

$$h(t) = u(t) - u(t - 2)$$

Taking the Fourier transform, we have

$$\begin{aligned} H(\omega) &= U(\omega) - U(\omega) e^{-j2\omega} \\ &= 2jU(\omega) e^{-j\omega} \frac{e^{j\omega} - e^{-j\omega}}{2j} \\ &= 2j e^{-j\omega} \sin(\omega) U(\omega) \\ &= 2j e^{-j\omega} \sin(\omega) \left[\pi \delta(\omega) + \frac{1}{j\omega} \right] \\ H(\omega) &= \frac{2 \sin(\omega)}{\omega} e^{-j\omega} \end{aligned}$$

Consider the signal $y(t) = x(t) * h(t)$. From the convolution property, we have

$$\begin{aligned} Y(\omega) &= X(\omega)H(\omega) \\ &= [\delta(\omega) + \delta(\omega - \pi) + \delta(\omega - 5)]H(\omega) \\ Y(\omega) &= H(0)\delta(\omega) + H(\pi)\delta(\omega - \pi) + H(5)\delta(\omega - 5) \end{aligned}$$

Since $H(\pi) = 0$ and $H(0) = 1$, we get

$$Y(\omega) = \delta(\omega) + H(5)\delta(\omega - 5)$$

Taking the inverse Fourier transform, we get

$$y(t) = \frac{1}{2\pi} + \frac{1}{2\pi} H(5) e^{j5t}$$

Therefore, $y(t)$ is a constant summed with a complex exponential whose fundamental frequency is $\omega = 5 \text{ rad/s}$ and fundamental period $T = 2\pi/5$. We know that adding a constant to a complex exponential does not affect its periodicity. Therefore, $y(t) = x(t) * h(t)$ is periodic.

(c) Yes, it is obvious from parts (a) and (b) that the convolution of two aperiodic signals can be periodic.

Example 5.51 Consider a signal $x(t)$ with Fourier transform $X(\omega)$. Suppose we are given the following facts:

1. $x(t)$ is real and nonnegative.
2. $\mathcal{F}^{-1}[(1 + j\omega)X(\omega)] = Ae^{-2t}u(t)$, where A is independent of t .
3. $\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = 2\pi$.

Determine the closed-form expression for $x(t)$.

Solution

From fact no. 2, we have

$$\mathcal{F}^{-1}[(1 + j\omega)X(\omega)] = Ae^{-2t}u(t)$$

Taking the Fourier transform of the above equation, we obtain

$$(1 + j\omega)X(\omega) = \frac{A}{2 + j\omega}$$

$$X(\omega) = \frac{A}{(1 + j\omega)(2 + j\omega)}$$

Using partial fraction expansion, we have

$$X(\omega) = A \left(\frac{1}{1 + j\omega} - \frac{1}{2 + j\omega} \right)$$

Taking the inverse Fourier transform of the above equation, we get

$$x(t) = Ae^{-t}u(t) - Ae^{-2t}u(t)$$

Using Parseval's theorem, we have

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} 2\pi$$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = 1$$

Substituting the expression of $x(t)$ in the above equation gives

$$\int_{-\infty}^{\infty} |Ae^{-t}u(t) - Ae^{-2t}u(t)|^2 dt = 1$$

$$\int_{-\infty}^{\infty} [A^2 e^{-2t} - 2A^2 e^{-3t} + A^2 e^{-4t}]u(t) dt = 1$$

$$\int_0^{\infty} [A^2 e^{-2t} - 2A^2 e^{-3t} + A^2 e^{-4t}]u(t) dt = 1$$

$$\frac{A^2}{12} = 1$$

$$A = \pm\sqrt{12}$$

Since $x(t)$ is nonnegative, we choose $A = \sqrt{12}$. Therefore,

$$x(t) = \sqrt{12}[e^{-t} - e^{-2t}]u(t)$$

Example 5.52 Let $x(t)$ be a signal with Fourier transform $X(\omega)$. Suppose we are given the following facts:

1. $x(t)$ is real.
2. $x(t) = 0$ for $t \leq 0$.
3. $\frac{1}{2\pi} \int_{-\infty}^{\infty} \Re\{X(\omega)\} e^{j\omega t} d\omega = |t|e^{-|t|}$.

Determine the closed-form expression for $x(t)$.

Solution

From fact no. 3, we have

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \Re\{X(\omega)\} e^{j\omega t} d\omega = |t|e^{-|t|}$$

$$\mathcal{F}^{-1}[\Re\{X(\omega)\}] = |t|e^{-|t|}$$

We know that $\mathcal{E}\{x(t)\} = x_e(t) \longleftrightarrow \Re\{X(\omega)\}$, therefore, we have

$$x_e(t) = |t|e^{-|t|}$$

$$\frac{x(t) + x(-t)}{2} = |t|e^{-|t|}$$

Given that $x(t) = 0$ for $t < 0$. This implies that $x(-t) = 0$ for $t > 0$. So, we may conclude that

$$x(t) = 2|t|e^{-|t|}, \quad t \geq 0$$

Therefore,

$$x(t) = 2te^{-t}u(t)$$

Example 5.53 Consider the signal

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{\sin(k\pi/4)}{(k\pi/4)} \delta\left(t - k\frac{\pi}{4}\right)$$

(a) Determine $g(t)$ such that

$$x(t) = \left(\frac{\sin(t)}{\pi t}\right) g(t)$$

(b) Use the multiplication property of the Fourier transform to argue that $X(\omega)$ is periodic. Specify $X(\omega)$ over one period.

Solution

(a) Consider the given signal

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{\sin(k\pi/4)}{(k\pi/4)} \delta\left(t - k\frac{\pi}{4}\right)$$

$$x(t) = \frac{\sin(t)}{\pi t} \sum_{k=-\infty}^{\infty} \pi \delta\left(t - k\frac{\pi}{4}\right)$$

$$x(t) = \left(\frac{\sin(t)}{\pi t}\right) g(t)$$

Therefore,

$$g(t) = \sum_{k=-\infty}^{\infty} \pi \delta\left(t - k\frac{\pi}{4}\right)$$

(b) From part (a), we have

$$x(t) = \left(\frac{\sin(t)}{\pi t}\right) g(t)$$

Taking the Fourier transform, we get

$$X(\omega) = \frac{1}{2\pi} \left[\mathcal{F}\left(\frac{\sin(t)}{\pi t}\right) * G(\omega) \right]$$

We know that

$$\frac{\sin(t)}{\pi t} \longleftrightarrow \text{rect}\left(\frac{\omega}{2}\right)$$

Since $g(t)$ is an impulse train, its Fourier transform $G(\omega)$ is also an impulse train:

$$G(\omega) = \pi \frac{2\pi}{T_0} \sum_{k=-\infty}^{\infty} \delta\left(\omega - k\frac{2\pi}{T_0}\right)$$

Here, $T_0 = \pi/4$. Therefore,

$$G(\omega) = \pi \frac{2\pi}{\pi/4} \sum_{k=-\infty}^{\infty} \delta\left(\omega - k\frac{2\pi}{\pi/4}\right)$$

$$G(\omega) = 8\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 8k)$$

We see that $G(\omega)$ is periodic with period 8. Now, consider

$$\begin{aligned} X(\omega) &= \frac{1}{2\pi} \left[\mathcal{F}\left(\frac{\sin(t)}{\pi t}\right) * G(\omega) \right] \\ &= \frac{1}{2\pi} \left[\text{rect}\left(\frac{\omega}{2}\right) * 8\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 8k) \right] \end{aligned}$$

$$X(\omega) = 4 \sum_{k=-\infty}^{\infty} \text{rect}\left(\frac{\omega - 8k}{2}\right)$$

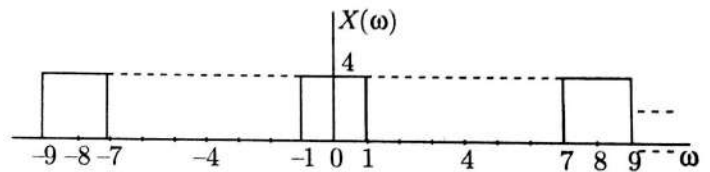


Fig. 5.41

$X(\omega)$ is shown in Fig. 5.41. Therefore, we may specify $X(\omega)$ over one period as

$$X(\omega) = \begin{cases} 1 & |\omega| \leq 1 \\ 0 & 1 < |\omega| \leq 4 \end{cases}$$

Example 5.54 Let $X(\omega)$ denote the Fourier transform of the signal $x(t)$ depicted in Fig. 5.42.

(a) Find $\angle X(\omega)$.

(b) Find $X(0)$.

(c) Find $\int_{-\infty}^{\infty} X(\omega) d\omega$.

(d) Evaluate $\int_{-\infty}^{\infty} X(\omega) \frac{2\sin(\omega)}{\omega} e^{j2\omega} d\omega$.

(e) Evaluate $\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$.

(f) Sketch the inverse Fourier transform of $\Re\{X(\omega)\}$.

Perform all these calculations without explicitly evaluating $X(\omega)$.

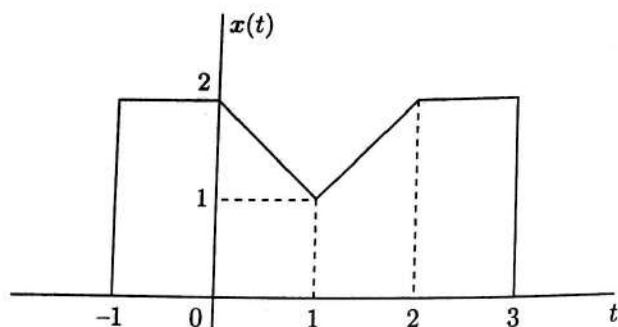


Fig. 5.42

Solution

(a) Note that $g(t) = x(t+1)$ is a real and even signal. Therefore, its Fourier transform $G(\omega)$ is also real and even. This implies that $\angle G(\omega) = 0$. Taking the Fourier transform of $g(t)$, we have

$$\begin{aligned} G(\omega) &= X(\omega) e^{j\omega} \\ X(\omega) &= G(\omega) e^{-j\omega} \\ &= |X(\omega)| e^{j\angle X(\omega)} \end{aligned}$$

Therefore,

$$\angle X(\omega) = -\omega$$

(b) From Eq. (5.58) [area under $x(t)$ property], we have

$$\begin{aligned} X(0) &= \int_{-\infty}^{\infty} x(t) dt \\ &= \left(\int_{-1}^0 2 dt + \int_0^1 (2-t) dt + \int_1^2 t dt + \int_2^3 2 dt \right) \\ &= 2t \Big|_{-1}^0 + \left[2t - \frac{t^2}{2} \right] \Big|_0^1 + \frac{t^2}{2} \Big|_1^2 + 2t \Big|_2^3 \\ X(0) &= 7 \end{aligned}$$

(c) From Eq. (5.59) [area under $X(\omega)$ property], we have

$$\begin{aligned} \int_{-\infty}^{\infty} X(\omega) d\omega &= 2\pi x(0) \\ &= 4\pi \end{aligned}$$

(d) Assume $Y(\omega) = \frac{2 \sin(\omega)}{\omega} e^{j2\omega}$. We know that

$$\begin{aligned} \text{rect}\left(\frac{t}{2}\right) &\longleftrightarrow 2 \frac{\sin(\omega)}{\omega} \\ \text{rect}\left(\frac{t+2}{2}\right) &\longleftrightarrow 2 \frac{\sin(\omega)}{\omega} e^{j2\omega} \end{aligned}$$

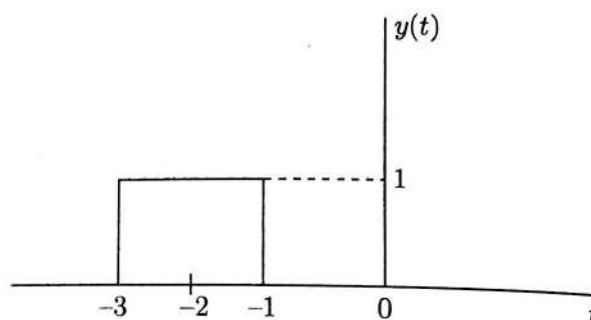


Fig. 5.43

The signal $y(t)$ is as shown in Fig. 5.43.

$$y(t) = \text{rect}\left(\frac{t+2}{2}\right) = \begin{cases} 1, & -3 < t < -1 \\ 0, & \text{otherwise} \end{cases}$$

Consider the given integral

$$\begin{aligned} &\int_{-\infty}^{\infty} X(\omega) \frac{2 \sin(\omega)}{\omega} e^{j2\omega} d\omega \\ &= \int_{-\infty}^{\infty} X(\omega) Y(\omega) d\omega \\ &= 2\pi [x(t) * y(t)]_{t=0} \\ &= 2\pi \left[\int_{-\infty}^{\infty} x(\tau) y(t-\tau) d\tau \right] \Big|_{t=0} \\ &= 2\pi \left[\int_{-\infty}^{\infty} x(\tau) y(-\tau) d\tau \right] \end{aligned}$$

Since the product $x(\tau)y(-\tau)$ is nonzero only for the range $1 < t < 3$. Therefore,

$$\begin{aligned} \int_{-\infty}^{\infty} X(\omega) \frac{2 \sin(\omega)}{\omega} e^{j2\omega} d\omega \\ = 2\pi \left[\int_1^3 x(\tau)y(-\tau) d\tau \right] \\ = 2\pi \left[\int_1^2 \tau d\tau + \int_2^3 2d\tau \right] \\ = 7\pi \end{aligned}$$

(e) Using Parseval's theorem, we have

$$\begin{aligned} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega &= 2\pi \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= \left(\int_{-1}^0 4 dt + \int_0^1 (2-t)^2 dt \right. \\ &\quad \left. + \int_1^2 t^2 dt + \int_2^3 4 dt \right) \\ &= 26\pi \end{aligned}$$

(f) We know that

$$x_e(t) \longleftrightarrow \Re\{X(\omega)\}$$

Therefore,

$$\mathcal{F}^{-1}[\Re\{X(\omega)\}] = x_e(t) = \frac{x(t) + x(-t)}{2}$$

Using step and ramp functions, $x(t)$ can be written as

$$\begin{aligned} x(t) &= 2u(t+1) - r(t) + 2r(t-1) \\ &\quad - r(t-2) - 2u(t-3) \end{aligned}$$

The time-reversal signal is shown in Fig. 5.44(a) and can be written as

$$\begin{aligned} x(-t) &= 2u(t+3) - r(t+2) + 2r(t+1) \\ &\quad - r(t) - 2u(t-1) \end{aligned}$$

Therefore,

$$\begin{aligned} x_e(t) &= \frac{x(t) + x(-t)}{2} \\ &= u(t+3) - \frac{1}{2}r(t+2) + r(t+1) \\ &\quad + u(t+1) - r(t) + r(t-1) - u(t-1) \\ &\quad - \frac{1}{2}r(t-2) - u(t-3) \end{aligned}$$

This is as shown in Fig. 5.44(b).

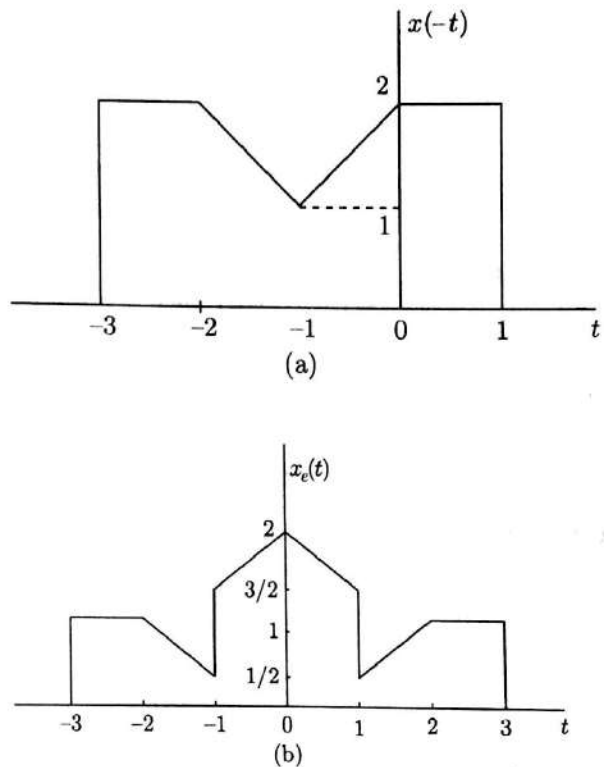


Fig. 5.44

Example 5.55 The output $y(t)$ of a causal LTI system is related to the input $x(t)$ by the equation

$$\frac{dy(t)}{dt} + 10y(t) = \int_{-\infty}^{\infty} x(\tau)z(t-\tau) d\tau - x(t)$$

where $z(t) = e^{-t}u(t) + 3\delta(t)$.

(a) Find the frequency response $H(\omega)$ of this system.

(b) Determine the impulse response of the system.

Solution

(a) Consider the equation

$$\frac{dy(t)}{dt} + 10y(t) = \int_{-\infty}^{\infty} x(\tau)z(t-\tau) d\tau - x(t)$$

$$\frac{dy(t)}{dt} + 10y(t) = [x(t) * z(t)] - x(t)$$

Taking the Fourier transform of the above equation, we have

$$j\omega Y(\omega) + 10Y(\omega) = X(\omega)Z(\omega) - X(\omega)$$

$$Y(\omega)[10 + j\omega] = X(\omega)[Z(\omega) - 1]$$

Given that $z(t) = e^{-t}u(t) + 3\delta(t)$, this implies that $Z(\omega) = \frac{1}{1+j\omega} + 3$. Substituting $Z(\omega)$ in the above equation, we get

$$Y(\omega)[10 + j\omega] = X(\omega) \left[\frac{1}{1+j\omega} + 3 - 1 \right]$$

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = \frac{3 + 2j\omega}{(1+j\omega)(10+j\omega)}$$

(b) Finding the partial fraction expansion of $H(\omega)$, we obtain

$$H(\omega) = \frac{1}{9} \frac{1}{1+j\omega} + \frac{17}{9} \frac{1}{10+j\omega}$$

Taking the inverse Fourier transform, we obtain

$$h(t) = \frac{1}{9} e^{-t}u(t) + \frac{17}{9} e^{-10t}u(t)$$

Example 5.56 For the LTI system described by the impulse response $h(t) = \delta(t) - 2e^{-2t}u(t)$. Determine and sketch the frequency response. Name the type of filter the system represents.

Solution

Given that

$$h(t) = \delta(t) - 2e^{-2t}u(t)$$

Taking the Fourier transform of $h(t)$, we obtain

$$H(\omega) = 1 - \frac{2}{2+j\omega}$$

$$H(\omega) = \frac{j\omega}{2+j\omega}$$

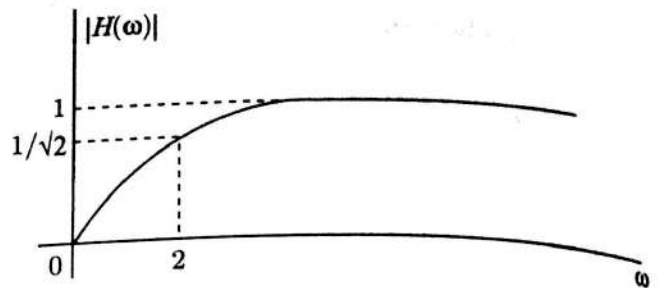


Fig. 5.45 Magnitude response of HPF

Taking the magnitude of $H(\omega)$, we obtain

$$|H(\omega)| = \frac{\omega}{\sqrt{4 + \omega^2}}$$

$$|H(\omega)| = \frac{1}{\sqrt{1 + \frac{4}{\omega^2}}}$$

The magnitude response is plotted in Fig. 5.45. At lower frequencies, the magnitude response is also low, and at higher frequencies the magnitude response is also high and so this filter blocks low frequencies and passes high frequencies. This is a high-pass filter.

Example 5.57 Determine and sketch the magnitude and phase response of the LTI causal system described by the differential equation

$$\frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} - x(t)$$

Solution

Consider the differential equation

$$\frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} - x(t)$$

Taking the Fourier transform of the above equation, we obtain

$$j\omega Y(\omega) + Y(\omega) = j\omega X(\omega) - X(\omega)$$

$$Y(\omega)[j\omega + 1] = X(\omega)[j\omega - 1]$$

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = \frac{j\omega + 1}{j\omega - 1}$$

The magnitude response is given by

$$|H(\omega)| = \frac{\sqrt{\omega^2 + 1}}{\sqrt{\omega^2 + 1}} = 1$$

and the phase response is given by

$$\begin{aligned}\angle H(\omega) &= \tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{-1}\right) \\ &= \tan^{-1}(\omega) + \tan^{-1}(\omega) \\ \angle H(\omega) &= 2\tan^{-1}(\omega)\end{aligned}$$

Both magnitude and phase response are shown in Fig. 5.46.

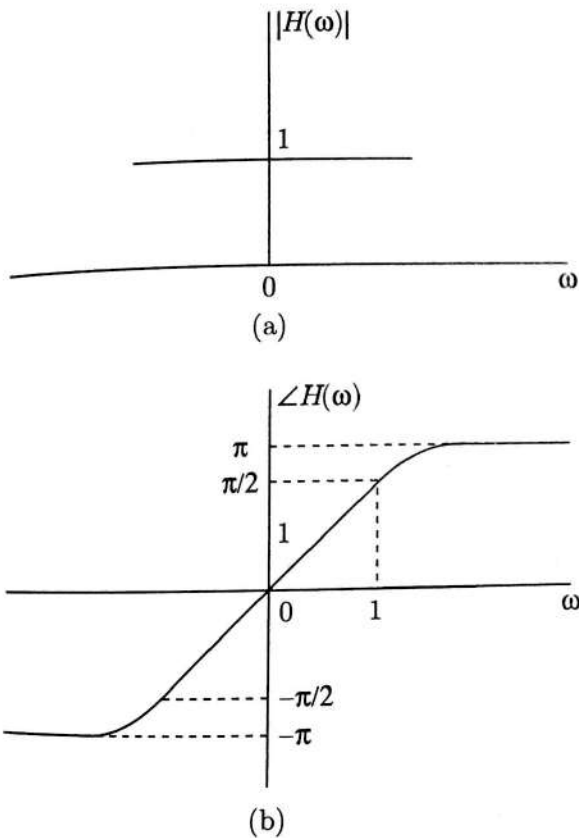


Fig. 5.46 (a) Magnitude response and (b) phase response

Example 5.58 Let

$$g(t) = x(t)\cos^2(t) * \frac{\sin(t)}{\pi t}$$

Assuming that $x(t)$ is real and $X(\omega) = 0$ for $|\omega| \geq 1$, show that there exists an LTI system S such that

$$x(t) \longrightarrow g(t)$$

Solution

Assume

$$y_1(t) = \cos^2(t) = \frac{1 + \cos(2t)}{2}$$

Taking its Fourier transform, we obtain

$$Y_1(\omega) = \frac{1}{2}2\pi\delta(\omega) + \frac{1}{2}\pi[\delta(\omega - 2) + \delta(\omega + 2)]$$

$$Y_1(\omega) = \pi\delta(\omega) + \frac{\pi}{2}\delta(\omega - 2) + \frac{\pi}{2}\delta(\omega + 2)$$

Now, let

$$y_2(t) = x(t)y_1(t) = x(t)\cos^2(t)$$

Taking the Fourier transform, we obtain

$$\begin{aligned}Y_2(\omega) &= \frac{1}{2\pi}[X(\omega) * Y_1(\omega)] \\ &= \frac{1}{2\pi}\left[X(\omega) * \left(\pi\delta(\omega) + \frac{\pi}{2}\delta(\omega - 2) + \frac{\pi}{2}\delta(\omega + 2)\right)\right] \\ Y_2(\omega) &= \frac{1}{2}X(\omega) + \frac{1}{4}X(\omega - 2) + \frac{1}{4}X(\omega + 2)\end{aligned}$$

Now consider

$$\underbrace{\frac{\sin(t)}{\pi t}}_{y_3(t)} \longleftrightarrow \underbrace{\text{rect}\left(\frac{\omega}{2}\right)}_{Y_3(\omega)} = \begin{cases} 1, & |\omega| < 1 \\ 0, & \text{otherwise} \end{cases}$$

$X(\omega)$, $Y_2(\omega)$, and $Y_3(\omega)$ are as shown in Figs 5.47(a), (b), and (c) respectively.

Now, consider

$$g(t) = y_2(t) * y_3(t)$$

Taking the Fourier transform, we obtain

$$G(\omega) = Y_2(\omega)Y_3(\omega)$$

From Figs 5.47(b) and (c), it is evident that

$$G(\omega) = \frac{1}{2}X(\omega)$$

$$\frac{G(\omega)}{X(\omega)} = H(\omega) = \frac{1}{2}$$

Taking its inverse Fourier transform, we have

$$h(t) = \frac{1}{2}\delta(t)$$

A change of variables is performed by letting $\omega = \tan(\alpha)$, which also yields $d\omega = \sec^2(\alpha) d\alpha$, $\alpha \rightarrow -\pi/2$ as $\omega \rightarrow -\infty$, and $\alpha \rightarrow \pi/2$ as $\omega \rightarrow \infty$. Therefore,

$$\begin{aligned}
 E_y &= \frac{4}{2\pi} \int_{-\pi/2}^{\pi/2} \frac{\sec^2(\alpha)}{\sec^4(\alpha)} d\alpha \\
 &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \cos^2(\alpha) d\alpha \\
 &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \frac{1}{2} [1 + \cos(2\alpha)] d\alpha \\
 &= \frac{1}{\pi} \left[\alpha + \frac{\sin(2\alpha)}{2} \right]_{-\pi/2}^{\pi/2} \\
 E_y &= \frac{1}{\pi} \left[\frac{\pi}{2} + \frac{\pi}{2} + 0 \right] \\
 E_y &= 1
 \end{aligned}$$

Since the cutoff frequency of the LPF is finite, therefore all the frequency components greater than this cutoff frequency will not be allowed to pass through the LPF and the energy associated with such frequency components will be attenuated. So, the output energy will always be less than the input energy.

Example 5.60 Find the power contained at the output of an RC LPF [shown in Fig. 5.50(a)] if the input voltage has the PSD:

- $G_x(\omega) = k$ for $-\infty < \omega < \infty$
- $G_x(\omega)$ is a gate function of height 1 and width 2 placed symmetrically about the vertical axis.
- $G_x(\omega) = \pi[\delta(\omega + 1) + \delta(\omega - 1)]$.

In each case also find the power contained in the input signal and find whether power at the input and output are equal or not?

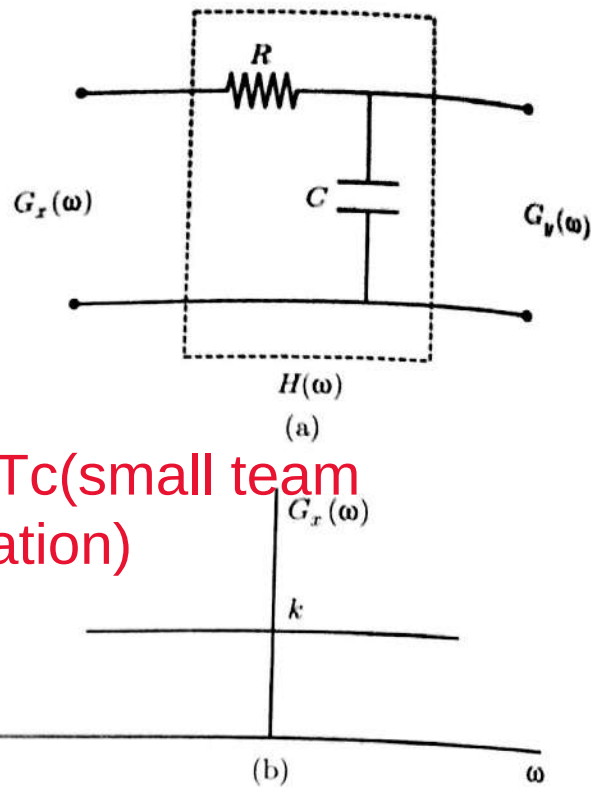


Fig. 5.50 (a) RC LPF and (b) input PSD

Solution

(a) Given that $G_x(\omega) = k$ for $-\infty < \omega < \infty$ [shown in Fig. 5.50(b)]. The power contained in the input signal is given by

$$\begin{aligned}
 P_x &= \frac{1}{2\pi} \int_{-\infty}^{\infty} G_x(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} k d\omega \\
 &= \frac{k}{2\pi} \omega \Big|_{-\infty}^{\infty} \\
 P_x &= \infty
 \end{aligned}$$

The frequency response $H(\omega)$ of the RC LPF of cut-off frequency $\omega_c = 1/RC = 1$ rad/s is given by

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{1 + j\omega}$$

The output PSD is given by

$$\begin{aligned}
 G_y(\omega) &= |H(\omega)|^2 G_x(\omega) \\
 &= \frac{k}{1 + \omega^2}
 \end{aligned}$$

@sTc(small team creation)

The output power is given by

$$\begin{aligned}
 P_y &= \frac{1}{2\pi} \int_{-\infty}^{\infty} G_y(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{k}{1+\omega^2} d\omega \\
 &= \frac{k}{2\pi} \tan^{-1}(\omega) \Big|_{-\infty}^{\infty} \\
 &= \frac{k}{2\pi} \left[\frac{\pi}{2} + \frac{\pi}{2} \right] \\
 P_y &= \frac{k}{2}
 \end{aligned}$$

The output power is less than the input power.

(b) The input PSD is defined as

$$G_x(\omega) = \begin{cases} 1, & -1 < \omega < 1 \\ 0, & \text{otherwise} \end{cases}$$

The power contained in the input signal is given by

$$\begin{aligned}
 P_x &= \frac{1}{2\pi} \int_{-\infty}^{\infty} G_x(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-1}^1 1 d\omega \\
 &= \frac{1}{2\pi} \omega \Big|_{-1}^1 \\
 P_x &= \frac{1}{\pi}
 \end{aligned}$$

The output PSD is given by

$$\begin{aligned}
 G_y(\omega) &= |H(\omega)|^2 G_x(\omega) \\
 G_y(\omega) &= \frac{1}{1+\omega^2}, \quad -1 < \omega < 1
 \end{aligned}$$

The output power is given by

$$\begin{aligned}
 P_y &= \frac{1}{2\pi} \int_{-\infty}^{\infty} G_y(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-1}^1 \frac{1}{1+\omega^2} d\omega
 \end{aligned}$$

$$\begin{aligned}
 P_y &= \frac{1}{2\pi} \tan^{-1}(\omega) \Big|_{-1}^1 \\
 &= \frac{1}{2\pi} \left[\frac{\pi}{4} + \frac{\pi}{4} \right] \\
 P_y &= \frac{1}{4}
 \end{aligned}$$

The output power is less than the input power.

(c) The input PSD is given by

$$G_x(\omega) = \pi[\delta(\omega+1) + \delta(\omega-1)]$$

The power contained in the input signal is given by

$$\begin{aligned}
 P_x &= \frac{1}{2\pi} \int_{-\infty}^{\infty} G_x(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \pi[\delta(\omega+1) + \delta(\omega-1)] d\omega \\
 &= \frac{\pi}{2\pi} \left[\int_{-\infty}^{\infty} \delta(\omega+1) d\omega + \int_{-\infty}^{\infty} \delta(\omega-1) d\omega \right] \\
 P_x &= \frac{1}{2} [1+1] \\
 P_x &= 1
 \end{aligned}$$

The output PSD is given by

$$\begin{aligned}
 G_y(\omega) &= |H(\omega)|^2 G_x(\omega) \\
 G_y(\omega) &= \frac{\pi}{1+\omega^2} [\delta(\omega+1) + \delta(\omega-1)]
 \end{aligned}$$

The output power is given by

$$\begin{aligned}
 P_y &= \frac{1}{2\pi} \int_{-\infty}^{\infty} G_y(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\pi}{1+\omega^2} [\delta(\omega+1) + \delta(\omega-1)] d\omega \\
 &= \frac{1}{2} \left[\int_{-\infty}^{\infty} \frac{1}{1+\omega^2} \delta(\omega+1) d\omega \right. \\
 &\quad \left. + \int_{-\infty}^{\infty} \frac{1}{1+\omega^2} \delta(\omega-1) d\omega \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \left[\frac{1}{1+\omega^2} \Big|_{\omega=-1} + \frac{1}{1+\omega^2} \Big|_{\omega=1} \right] \\
 &= \frac{1}{2} \left[\frac{1}{2} + \frac{1}{2} \right]
 \end{aligned}$$

$$P_v = \frac{1}{2}$$

The output power is less than the input power.

SUMMARY

1. The Fourier transform of $x(t)$ is defined by

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

2. The inverse Fourier transform of $X(\omega)$ is defined by

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

3. The magnitude of $X(\omega)$ plotted against ω is called the magnitude spectrum of $x(t)$ and $|X(\omega)|^2$ is called the energy spectrum.
 4. The angle of $X(\omega)$ plotted versus ω is called the phase spectrum.
 5. Parseval's theorem states that

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

6. The PSD of $x(t)$ is defined by

$$G_x(\omega) = \lim_{T \rightarrow \infty} \frac{|X_T(\omega)|^2}{T}$$

where

$$x_T(t) \longleftrightarrow X_T(\omega)$$

and

$$x_T(t) = \begin{cases} x(t), & -T/2 < t < T/2 \\ 0, & \text{otherwise} \end{cases}$$

7. The more a signal is localized in one domain (time or frequency), the less it is localized in the other domain.
 8. Convolution and multiplication of functions are dual in the time and frequency domains.
 9. The Fourier transform of a periodic signal consists only of impulses.
 10. Signal energy is conserved in the Fourier transformation process.
 11. The correlation function indicates how correlated two signals are as a function of how much one of them is shifted in time.
 12. Correlation and convolution are closely related mathematical processes.
 13. The correlation of a signal with a shifted version of itself is called autocorrelation.
 14. Autocorrelation is closely related to signal energy or power and contains important information about how rapidly a signal varies in time.
 15. ESD and PSD are the frequency-domain counterparts of autocorrelation, related through the Fourier transform.
 16. ESD and PSD indicated how the energy or power of a signal varies with frequency.

MULTIPLE-CHOICE QUESTIONS

1. The Fourier transform of a signal $x(t) = e^{2t}u(-t)$ is given by

- (a) $\frac{1}{2-j\omega}$
 (b) $\frac{2}{1-j\omega}$
 (c) $\frac{1}{j2-\omega}$

(d) $\frac{2}{j2-\omega}$

2. The Fourier transform of a unit step function

- (a) does not exist
 (b) is another unit step
 (c) contains impulse functions
 (d) is $\frac{1}{j\omega}$

3. If the notation $*$ is used to denote the convolution, and $x(t) \longleftrightarrow X(\omega)$, $y(t) \longleftrightarrow Y(\omega)$ makes the Fourier transform pair, then $x(t)y(t) \longleftrightarrow F(\omega)$ is given by

- (a) $X(\omega) * Y(\omega)$
- (b) $X(\omega)Y(\omega)$
- (c) $\frac{1}{2\pi}X(\omega)Y(\omega)$
- (d) $\frac{1}{2\pi}[X(\omega) * Y(\omega)]$

4. The Fourier transform of a rectangular pulse is

- (a) another rectangular pulse
- (b) triangular pulse
- (c) sinc function
- (d) impulse.

5. The property of Fourier transform that states that the compression in time domain is equivalent to expansion in the frequency domain is

- (a) duality
- (b) time shifting
- (c) time scaling
- (d) frequency shifting

6. The spectrum of $x(t)$ extends from $-\omega_0$ to $+\omega_0$, while that of $h(t)$ extends from $-2\omega_0$ to $+2\omega_0$. The spectrum of

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau \text{ extends from}$$

- (a) $-4\omega_0$ to $+4\omega_0$
- (b) $3\omega_0$ to $+3\omega_0$
- (c) $-2\omega_0$ to $+2\omega_0$
- (d) $-\omega_0$ to $+\omega_0$

7. Inverse Fourier transform of $U(\omega)$ is

- (a) $\frac{1}{2}\delta(t) + \frac{j}{2\pi t}$
- (b) $\frac{1}{2}\delta(t)$
- (c) $2\delta(t) + \frac{j}{\pi t}$
- (d) $\delta(t) + \text{sgn}(t)$

8. The Fourier transform of a function $x(t)$ is $X(f)$. The Fourier transform of $\frac{dx(t)}{dt}$ will be

- (a) $\frac{dX(f)}{df}$
- (b) $j2\pi fX(f)$

- (c) $jfX(f)$
- (d) $\frac{X(f)}{jf}$

9. The frequency of a continuous-time signal $x(t)$ changes on transformation from $x(t)$ to $x(\alpha t)$, $\alpha > 0$, by a factor

- (a) α
- (b) $\frac{1}{\alpha}$
- (c) α^2
- (d) $\sqrt{\alpha}$

10. The Fourier transform of the exponential signal $e^{j\omega_0 t}$ is

- (a) a constant
- (b) a rectangular pulse
- (c) an impulse
- (d) a series of impulses

11. The Fourier transform of $u(t)$ is

- (a) $\frac{1}{j2\pi f}$
- (b) $j2\pi f$
- (c) $\frac{1}{1+j2\pi f}$
- (d) none of these

12. The Fourier transform of a real and even signal $x(t)$ is

- (a) real and even function of frequency.
- (b) real and odd function of frequency.
- (c) complex, in general.
- (d) purely imaginary.

13. The Fourier transform of a real and odd signal $x(t)$ is

- (a) imaginary and odd function of frequency.
- (b) real and odd function of frequency.
- (c) complex, in general.
- (d) purely imaginary.

14. Fourier transform of a d.c. signal with unity strength is

- (a) zero
- (b) 1
- (c) $2\pi\delta(\omega)$
- (d) $2\delta(\omega)$

15. For distortionless transmission through an LTI system, amplitude of $H(\omega)$ is

- (a) constant
- (b) one

- (c) zero
(d) linearly dependent on ω
16. For distortionless transmission through an LTI system, phase of $H(\omega)$ is
- (a) constant
(b) one
(c) zero

- (d) linearly dependent on ω
17. The Fourier transform of $\text{sgn}(t)$ is
- (a) $\frac{2}{j\omega}$
(b) $-\frac{2}{j\omega}$
(c) 1
(d) zero

PROBLEMS

5.1 If the unit impulse response of a causal LTI system contains no impulses at the origin, then show that with

$$H(\omega) = H_R(\omega) + jH_I(\omega)$$

$H_R(\omega)$ and $H_I(\omega)$ satisfy the following conditions:

$$H_R(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{H_I(\theta)}{\omega - \theta} d\theta$$

$$H_I(\omega) = -\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{H_R(\theta)}{\omega - \theta} d\theta$$

5.2 The input and the output of a causal LTI system are related by the differential equation

$$\frac{d^2 y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 8y(t) = 2x(t)$$

- (a) Find the impulse response of this system.
(b) What is the response of this system if $x(t) = te^{-2t}u(t)$?
(c) Repeat part (a) for the causal LTI system described by the equation

$$\frac{d^2 y(t)}{dt^2} + \sqrt{2} \frac{dy(t)}{dt} + y(t) = 2 \frac{d^2 x(t)}{dt^2} - 2x(t)$$

5.3 Consider an LTI system whose response to the input

$$x(t) = [e^{-t} + e^{-3t}]u(t)$$

is

$$y(t) = [2e^{-t} - 2e^{-4t}]u(t)$$

- (a) Find the frequency response of this system.
(b) Determine the systems' impulse response.
(c) Find the differential equation relating the input and output of this system.

5.4 Consider the signal

$$x(t) = \begin{cases} 0, & |t| > 1 \\ \frac{t+1}{2}, & -1 \leq t \leq 1 \end{cases}$$

- (a) Determine the closed form expression for $X(\omega)$.
(b) Take the real part of your answer to part (a), and verify that it is the Fourier transform of the even part of $x(t)$.
(c) What is the Fourier transform of the odd part of $x(t)$.

5.5 Find the impulse response of a system with the frequency response

$$H(\omega) = \frac{\sin^2(3\omega) \cos(\omega)}{\omega^2}$$

5.6 Determine the inverse Fourier transform of $X(\omega) = (4/\omega^2) \sin^2(\omega)$, using the convolution property of the Fourier transform.

5.7 Let $X(\omega) = \text{rect}(\omega - 1/2)$. Using the Fourier transform properties, find the Fourier transform of the following functions:

- (a) $x(-t)$
(b) $tx(t)$
(c) $x(t+1)$
(d) $x(-2t+4)$
(e) $(t-1)x(t+1)$
(f) $\frac{dx(t)}{dt}$
(g) $t \frac{dx(t)}{dt}$
(h) $x(2t-1)e^{-j2t}$
(i) $x(t)e^{-j2t}$

5.8 Let $x(t) = e^{-2t}u(t)$ and let $y(t) = x(t+1) + x(t-1)$. Find $Y(\omega)$.

5.9 Find the energy of the following signals, using Parseval's theorem.

(a) $x(t) = e^{-2t}u(t)$

(b) $x(t) = u(t) - u(t - 5)$

(c) $x(t) = \frac{\sin(\pi t)}{\pi t}$

5.10 A Gaussian-shaped signal

$$x(t) = A e^{-at^2}$$

is applied to a system whose input/output relationship is

$$y(t) = x^2(t)$$

Find the Fourier transform of the output $y(t)$.

5.11 The signal $x(t) = e^{-at}u(t)$ is input into a system with impulse response $h(t) = \frac{\sin(2t)}{\pi t}$.

(a) Find the Fourier transform $Y(\omega)$ of the output.

(b) For what value of a does the energy in the output equal one-half the input signal energy.

5.12 A signal $x(t)$ can be expressed as the sum of even and odd components:

$$x(t) = x_e(t) + x_o(t)$$

(a) If $x(t) \longleftrightarrow X(\omega)$, show that for real $x(t)$,

$$x_e(t) \longleftrightarrow \Re\{X(\omega)\}$$

and

$$x_o(t) \longleftrightarrow j\Im\{X(\omega)\}$$

(b) Verify these results by finding the Fourier transform of the even and odd components of the following signals: (i) $u(t)$ and (ii) $e^{-at}u(t)$.

5.13 Show that the energy of the Gaussian pulse

$$x(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}}$$

is $\frac{1}{2\sigma\sqrt{\pi}}$. Verify this result by using Parseval's theorem to derive the energy E_x from $X(\omega)$.

5.14 If $x(t) = 25 \operatorname{rect}\left(\frac{t-4}{10}\right)$, find the Fourier transform of $x(t)$.

5.15 What is the total area under the function, $x(t) = 100 \operatorname{sinc}\left(\frac{t-8}{30}\right)$.

5.16 Find the Fourier transform of the signal $x(t) = e^{-a|t|} \operatorname{sgn}(t)$.

ANSWERS TO MULTIPLE-CHOICE QUESTIONS

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|--------|
| 1. (a) | 2. (c) | 3. (d) | 4. (c) | 5. (b) | 6. (d) | 7. (a) | 8. (b) | 9. (b) |
| 10. (c) | 11. (d) | 12. (a) | 13. (a) | 14. (c) | 15. (a) | 16. (d) | 17. (a) | |